COUNTING FROM THE FIRST TICK

Life, the Universe, and God

A Software Engineer's Instrument for the Immeasurable

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2026 · built against the ucal source tree

Contents

Why this might be worth your time

There is a distinction that people have been trying to hold steady for about two and a half thousand years, and losing.

It is the line between what a measurement **establishes** and what it merely **points at**. Everyone agrees it exists. Aristotle has a version of it, Basil of Caesarea has a version of it, Kant spent a large part of the **Critique** on it, and every one of them observed that it does not stay where you put it. Kant was the bluntest: the illusion that erodes the line is natural, unavoidable, and **survives being diagnosed**. You can see exactly why the moon is not larger at the horizon and go on seeing it larger.

The usual remedy is vigilance — argue carefully, mark the boundary, and hope the next reader is paying attention. It works for as long as attention lasts, which is not long. In 1922 a first-rate mathematician who had the distinction fully available to him, in his own tradition, going back fifteen centuries, crossed it in a single paragraph and did not notice.

This book is about trying something else: holding the line with a **compiler**.

The project is a working piece of software — a calendar that counts time in Planck units from an origin it declares it cannot measure. Inside it, the uncertain claim about where that origin actually falls is recorded in full, cited, with its exact magnitude — and given a type that has no arithmetic operations at all. You can read it. You can print it. You cannot compute with it, and three tests exist whose job is to fail to build if you ever could.

That is a small technical fact with a large consequence. A philosophical argument reaches people who are willing to follow it. A type refuses people who are not. Someone who thinks the distinction is pedantic sits down, writes the line of code that ignores it, and is stopped in under a second by something with no interest in whether they agree.

The other reason to keep reading is that the machinery answered back.

The specification behind this software claimed, twice, in two published revisions, that its calendar-derivation mechanism reproduces both the Julian and the Gregorian leap rules from first principles. It reproduces the Julian one exactly. The Gregorian rule is not there — not at that depth, not at any depth, and a rule twelve times simpler is more accurate than it.

The author found that out from his own program, about his own published claim, and printed it. Chapter 10 is that story, and there is a check in the build now whose only purpose is to stop anyone quietly reversing the correction.

If a book about a calendar that admits it cannot measure its own zero sounds like it might be an elaborate way of saying nothing, chapter 10 is the answer, and it is the shortest one available.

Some of what is in here

A calendar with no Earth in its arithmetic turns out to be a good instrument for looking at calendars that do have Earth in theirs. Some of what came out:

The Julian leap rule is derivable and the Gregorian is not — Feed the mechanism nothing but Earth's rotation and orbit and it produces `1/4` as its first answer — the Julian calendar, with no knowledge of Rome. It never produces `97/400`. Two simpler fractions beat the Gregorian rule, one of them by a factor of 124.

The Persian calendar of 1079 is derivable — The rule worked out by a commission including Omar Khayyam is the third convergent — exactly where the arithmetic says the good approximation lives. So "derived" and "accurate" are independent properties, and the older calendar is the one the mathematics agrees with.

The Metonic cycle falls out unaided — Nineteen years, 235 lunar months — known to Babylon, named for an Athenian, still fixing the date of Easter and the Hebrew calendar's leap years. It appears as the sixth convergent of two numbers, with nothing supplied but Earth's periods.

Mars has no month, and that is the correct output — Neither of its moons is anything a person would call one. The mechanism returns nothing rather than inventing a Martian month, because "month-like" turns out to be an Earth predicate and there is no way to compute it for somewhere else.

The Hebrew calendar's epoch is the same idea, eleven centuries early — **Molad tohu** — "the new moon of chaos" — is a computational origin, placed before the event it anchors, and named for its own emptiness. Whoever fixed it understood precisely what this project spent four chapters working out, and named it better.

And what kind of book it is

Nine philosophical and religious traditions are read here — Greek, Jewish, Islamic, Patristic, Orthodox, Latter-day Saint, modern European, Russian — and none of them is argued to be true. They are treated as **readers of the artifact**: each chapter says where the tradition and the software agree, and then, at greater length, where they collide.

Every one of those chapters has a section headed **the conflict**, and four of them cut at the project rather than at the tradition. A Baghdad argument from 1095 shows that the software took a metaphysical side without noticing it had one. A Russian philosopher shows that its central prohibition is a contested position rather than a safeguard. Those are in here because a tradition that only agrees has not been read.

The book also marks itself. Where a passage is the author's interpretation rather than a checkable fact, it sits in a ruled block that says so — and there is a script that deletes every one of those blocks, rebuilds the book, and fails if any technical claim stopped standing. The software marks where it stops computing; the book marks where it stops asserting. That symmetry is the whole design.

And there is a chapter reserved for what did not work. Six experiments were run against real material; chapter 28 reports what each of them failed to establish, including one that essentially failed outright and one measurement the author cannot make and has recorded as not made.

Three things are true about this project, and the third only follows if you accept the first two. So they go here, on the first page, before anything has a chance to soften them.

The artifact is real. `ucal` is working Rust — six crates, a library and a command line, published and installable. Time is an unsigned integer count of Planck-time units since a stipulated datum. There is no floating-point value anywhere in the workspace, not in a signature, a field, an intermediate, or the rendering path. Every constant in it is reproducible by two independent derivations that agree bit for bit. At the commit this book is pinned to, 381 tests pass on both integer backends. You can check every sentence in that paragraph; the last page of this book tells you how.

Its practical utility is questionable. No task you have today needs a Planck-tick count since a stipulated datum. It does not compete with `chrono`, `time`, or `hifitime`, and it does not try to. Nothing on its ladder of units is near a second or an hour — a second is 21.4 beats, an hour is 24.6 arcs — so it is not merely unnecessary for ordinary work but actively awkward for it. There is exactly one qualification to this, it concerns timekeeping across more than one planet, it takes up a single page in Part V, and it is not a rescue.

Therefore it is research of another kind. Art, philosophy, theology — conducted in the medium of a working program. That is not a consolation prize awarded after usefulness failed. It is what the thing is.

The rigor is the medium

Here is the part that matters, and it is easy to misread as modesty.

That the code compiles, that the constants reproduce, that the tests pass — these are not preliminaries to the argument. They **are** the argument, in the same way that the metre of a poem is not preliminary to the poem.

An essay can assert that a distinction ought to be respected. It can argue at length, and persuade you, and then it is over, and the discipline it recommended survives exactly as long as the next author's attention does. A type system can do something an essay cannot: it can make violating the distinction **fail to build**. Not discouraged. Not deprecated with a warning. Refused, by a machine, to a person who disagrees.

That difference is the thesis of this book, and the reason it is a book about a program rather than a program with a book attached.

What this is not

Some of these will be obvious. They are listed because a book that surveys nine philosophical and religious traditions alongside a Rust crate will be misread in predictable directions, and it is cheaper to close those doors now than to keep apologising later.

Not an apologetic No tradition is argued true here. The code is evidence for none of them. Where a tradition and the instrument agree, that is reported as a convergence and never as a vindication of either.

Not a proof of usefulness The second fact above stands unretracted.

Not a claim of discovery When a tenth-century argument turns out to match a twenty-first-century type signature, the older text is not thereby shown to have anticipated anything. Convergence is not anticipation.

Not numerology No constant, base, or magnitude in this system acquires meaning by resembling a number in a tradition. Base 5 was chosen because $5^5 = 3125$ gives a tier of exactly five digits. That is the whole reason. The book says so plainly and then reports, as a curiosity and not as evidence, that the first place the research went looking for a five, it found one waiting.

Not a tutorial If you want to use the crate, its documentation is better at that than this book will be.

Not a defence of the datum Tick zero is stipulated. That it is stipulated is the point, and Part IV is four chapters about why.

Two paths through the book

The book serves two readers who are rarely the same person, and it does not ask either of them to pretend to be the other.

The **engineer's path** is Parts I, II, III, and V. It is a complete technical article about an unusual piece of software: what it computes, what its type system refuses, where the design lost arguments with the compiler, and how far the approach generalises beyond Earth. You can stop at the end of Part V and have received the whole of that.

The **reader's path** is Parts I, IV, VI, VII, and VIII. It is a book about what it means to build a measuring instrument that points at something it declares itself unable to measure, and about nine traditions that have thought carefully about origins, number, and time — read here as readers of the artifact rather than as authorities over it.

Part I belongs to both. Nothing later in the book may redefine what Part I establishes.

A NOTE ON MARKING

This book marks its own claims. Where a passage is **interpretation** — the author reading a meaning into the artifact — it sits in a ruled block that says so. Where a coincidence is genuinely striking but proves nothing, it is labelled **resonance**, and it may never appear as a premise for anything.

That apparatus is not decoration. It is the book submitting to the discipline it documents: the software marks where it stops computing, and the book marks where it stops asserting fact. If you delete every marked block, every technical claim in Parts I through III and V must still stand. Making sure that is true is a scheduled step in producing this book, not a hope about it.

A word about being wrong

The most valuable document in this project's history is a correction.

Two revisions of the specification asserted that the continued-fraction machinery reproduces the Julian and Gregorian calendar rules as convergents. The Julian rule does appear — it is the first convergent. The Gregorian rule does not appear at any depth, and a rule twelve times simpler is more accurate than it. The machinery contradicted its author, in writing, on a claim he had published twice.

That correction is in this book, dated, including the revisions that were wrong. It is in here because a system built to make its author's conclusions provable would never have produced it, and because the charge that such systems are **always** built that way is nine hundred years old and deserves an answer made of evidence rather than assurance.

If you find yourself reading a passage that seems to be presenting that correction as foresight rather than as an error published — a demonstration of rigour rather than an argument the author lost — then the book has failed at the thing it cares most about, and you should say so.

The rules, in one page

The software this book is about is specified by twenty-four rules, each named by a letter. The book cites them constantly and from chapter 2 onward, so they are here at the front rather than only in the back.

You do not need to learn these. Read past them on first encounter and come back when a chapter leans on one — that is what this page is for.

rule	name	what it requires	ch.
Q	the datum is stipulated	declared, not measured — and not computable with	3
Z	time is unsigned	nothing precedes the datum; earlier is an error	3
G	the tier grid	units are powers of five; a timestamp is base 5	4
N	names are display only	a tier's identity is its exponent, not its name	4
P	profiles are tagged and type-bound	two profiles' values cannot be compared	6
K	one mechanism; Earth is an instance	every calendar is built by the same path	8
J	the anchor is declared and required	phase is supplied per body; absence is an error	8
C	body parameters carry provenance	epoch, rate, window — outside it, a warning	8
X	certified enclosures	an interval, with its two error sources kept apart	5
Y	metrology	foreign units cross one boundary you can point at	7
B	fixed 64-byte binary	big-endian, never minimal, so the format never changes	6
S	sort order	byte order is chronological order — for binary and UCID	6
M	order is total and monotone	of any two instants, exactly one order holds	5
F	the frame is declared	what it does not model, it says it does not model	16

Ten further rules govern parts of the software the book does not discuss. Appendix D lists all twenty-four, together with the non-goals, failure modes and specification corrections cited in the text. The normative statements are in `spec/RULES.md` in the source tree.

WHY LETTERS

Because the specification names them that way, and renaming them for the book would make every citation in the source tree — there are 538 — resolve to nothing.

The letters are not mnemonic and the specification does not pretend they are. `Q` is the datum rule because it was the seventeenth thing written down.

PART I

Foundations

Definitional. Nothing later in the book may redefine these, and a reader who stops here must not leave with a false impression.

CHAPTER 1

1

Intent

This began as an irritation, and it is worth being honest that it began as an engineer's irritation rather than a philosopher's question.

Read almost any account of the early universe and you will find a sentence like this one: **recombination occurred about 380,000 years after the Big Bang.** It is a good sentence. It is doing real work. And if you stop and ask what its units are, something uncomfortable happens.

A year is the time Earth takes to go around the Sun. Not approximately — definitionally. The Julian year used in astronomy is exactly 365.25 days of exactly 86,400 seconds, and those numbers are what they are because of how fast one particular rock spins and how long it takes to circle one particular star. So **380,000 years after the Big Bang** means 380,000 × 365.25 × 86,400 seconds, which means it is a quantity expressed in units defined by the rotation and orbit of a planet that would not exist for another nine billion years.

The number is correct. It is not wrong in any way that matters to a cosmologist. But it carries a passenger, and the passenger is Earth.

The irritation, stated precisely

The problem is not that the units are arbitrary. All units are arbitrary. The metre is a bar in a vault, then a wavelength, then a fraction of a light-second; none of that makes it a bad metre.

The problem is **provenance leaking into arithmetic**. When you compute with Earth years, Earth's orbital period is inside every intermediate value. Usually that costs nothing. Occasionally it costs you a rounding you did not intend, because 365.25 days is not a whole number of anything and the conversions do not close. And structurally — which is the part that would not let go — it means the description of an event 13.8 billion years old is phrased in terms of a body that had no bearing on it.

> **WHAT THIS IS NOT AN ARGUMENT FOR**
>
> None of this is a criticism of how cosmology is done. Astronomers use Julian years because they are stable, agreed, and unambiguous, which is exactly what a unit should be. The complaint here is narrow and it is aesthetic before it is technical: a quantity ought to be expressible in units that do not smuggle in a planet.

What would have to be true instead

Suppose you wanted a time system with no Earth content in its arithmetic. What would it need?

It would need a **unit** that is not defined by any body's motion — something built out of constants of nature rather than out of a rotation.

It would need an **origin** that is not a calendar event, since every calendar epoch is someone's civil history.

It would need a **notation** in which writing fewer digits means saying less precisely, rather than meaning zero — because at these magnitudes you will constantly be quoting numbers you do not know to full precision, and a system that silently pads them with zeros is lying on your behalf.

And it would need to be honest about the one thing it cannot do: it cannot tell you what time it is on Earth without, at some point, being told about Earth. That contact has to happen somewhere. The question is whether it happens **in the arithmetic** or at a declared boundary you can point to.

That last question turned out to be the whole design.

What was built

`ucal` is the answer to those four requirements, and Part II is the technical account of it. In one paragraph: absolute time is an unsigned integer count of Planck-time units since a stipulated datum; those ticks are grouped in a ladder of powers of five, so a timestamp is the tick count written in base 5 and truncating it **is** rounding it; Earth enters through exactly one declared constant and leaves at the formatting boundary; and no floating-point value appears anywhere in the workspace.

It works. It is also, as the preface said and will keep saying, of questionable use.

Where the trouble started

The requirement that has no clean answer is the second one: the origin.

If time is a count since some zero, the zero has to be somewhere. And the obvious place to put it — the beginning — is not available, for reasons that took four chapters of this book to state properly and that Part IV is entirely about.

The short version, so you can carry it through Parts II and III: **the age of the universe is a measurement, and measurements have error bars, and an error bar cannot be the origin of an exact integer count.** The published figure is 13.787 billion years give or take 0.020 billion. That uncertainty, converted to ticks, is a number with fifty-eight digits. If the datum inherited it, every timestamp in the system would inherit it too, and the exactness the whole design exists to protect would be theatre.

So the datum is **stipulated**. Declared, not discovered. Tick zero is a reference point that the specification says, in as many words, is not a measurement and not an observed event.

And then the interesting problem: how do you say that in a way that a future maintainer — or the author, three years later, in a hurry — cannot quietly forget?

> **INTERPRETATION**
>
> The answer this project arrived at is the reason it turned into a book. The physical claim about the origin is declared, cited, and given an exact magnitude. And then it is made **impossible to compute with**: a type with no arithmetic operations at all, guarded by a test that fails to compile if the type ever reaches an operand position.
>
> You cannot forget a discipline that the compiler enforces. That is a small technical fact with a large philosophical shadow, and the shadow is what Parts IV, VII, and VIII are about.

What this book will and will not claim

Before Part II starts and the prose becomes technical for a hundred pages, here is the calibration.

It **will** claim that a distinction enforced in a type system is a philosophical contribution and not merely an implementation detail. It will claim that the approach generalises to any celestial body, and it will spend a chapter on exactly where it fails. It will claim that the instrument does philosophical work no prose argument can do, and it will demonstrate that rather than assert it.

It will **not** claim that time began, that any tradition is correct, that the base is meaningful, or that anyone should adopt this calendar.

WHAT THIS CHAPTER ESTABLISHED

- Cosmic durations are conventionally expressed in units defined by one planet's motion — correct, useful, and carrying a passenger.
- A time system with no Earth content in its arithmetic needs a physical unit, a non-calendrical origin, a notation where truncation means imprecision, and one declared point of contact with Earth.
- The origin cannot be measured, because a measurement has error bars and an exact integer count cannot inherit one.
- So it is stipulated — and the project's contribution is making that stipulation mechanically impossible to forget.

CHAPTER 2

2

The tick

Everything this system computes is an unsigned integer count of ticks. Nothing else is primitive — not the second, not the beat, not the day. If you understand why the tick was chosen and, more importantly, what choosing it does **not** claim, you have the foundation of the whole instrument.

TERM **Tick**

The Planck time, $t_P = \sqrt{\hbar G/c^5} \approx 5.391 \times 10^{-44}$ seconds. In this system it is the atomic unit of duration: the smallest quantity representable, and the unit in which every other quantity is counted.

Why this unit

The Planck time is composed from three constants: the gravitational constant G, the reduced Planck constant $\hbar$, and the speed of light c. Those three bound gravitation, quantum action, and the propagation of causality respectively — and there is exactly one combination of them with the dimension of time.

That is the appeal, and it is worth stating carefully because it is easy to overstate. The tick is not defined by any body's motion. It does not know about Earth, or about the Solar System, or about matter being organised into planets at all. It is the interval you get when you ask the three limiting constants of physics what a duration would be, and they answer.

For a system whose entire purpose is to have no Earth content in its arithmetic, that is the right floor.

What the tick is not

Here is where a great deal of writing about the Planck time goes wrong, and where this project declines to follow.

The tick is not a quantum of time. It is the resolution floor of an instrument, not a structure discovered in the world. Nothing in this project asserts that time comes in discrete lumps, that there is a smallest possible interval, or that asking about a shorter duration is physically meaningless. The system cannot represent a shorter duration; that is a fact about the system.

The distinction matters more than it looks, because asserting otherwise would silently take a side in an argument that has been running for twenty-four centuries.

> Aristotle holds in **Physics** VI that time is a continuum: divisible without limit, and the instant is a **limit** of an interval rather than a part of it, in the way that a

> point is a limit of a line and not a very short line. On this view a smallest duration is not merely unobserved but incoherent.
>
> Against that, temporal atomism has a long and respectable history. Epicurus held that there are **minima** of time as there are minima of magnitude. The Kalām tradition developed a thoroughgoing atomism in which time is composed of indivisible moments and accidents do not endure from one to the next.

The instrument is discrete. If discreteness were asserted as a claim about the world, this project would have joined Epicurus and the Kalām against Aristotle — and it would have done so not on the strength of an argument but as a side effect of choosing an integer type.

That is not a respectable way to take a metaphysical position. So the position is declined, explicitly, and the declining is written down: the tick is where the instrument's resolution stops, and the instrument makes no claim about where the world's does.

> A RULE YOU WILL SEE AGAIN
>
> This pattern — **the implementation forces a choice; name the choice; refuse the claim that would come free with it** — recurs throughout the design. It is the same move that Part IV makes about the datum, at much greater length and with much higher stakes.

The concession: the tick's length

There is one place where the tick is not free of Earth, and it should be admitted here rather than discovered later.

The Planck time's **numerical value** is known only as well as G is measured, and G is the worst-measured of the fundamental constants by a wide margin. Worse, expressing it at all requires a unit of time to express it **in** — and that unit is the SI second, which is defined by a caesium transition counted by instruments on Earth.

So the tick's length is fixed by convention against the second. The system declares a value, records where the value came from, and uses that declared value everywhere. It does not re-derive it, and it does not pretend the value is more certain than the measurement behind it.

INTERPRETATION

What this concedes is **metrology** and nothing else. The arithmetic contains no Earth content: ticks are counted, not converted. What is Earth-flavoured is the statement of how long a tick is in seconds — which is a sentence about translating between two systems, not a fact used inside either.

Whether that is a satisfying answer is a fair question, and Part III returns to it. It is at least an honest one, and its honesty is structural: the conversion lives at a declared boundary you can point to rather than dissolved into the operations.

Counting, not measuring

One more consequence, which will matter in Part IV.

Because time here is a **count**, the system's exactness is the exactness of integer arithmetic. Tick 8,070,204,002,895,596,515,944,343,085,635,637,180,530,466,139,316,558,837,890,625 is a specific tick, and adding one to it gives the next one, and that is all perfectly exact — in the same way that the number of steps you have walked is exact whether or not you know it.

What is **not** exact, and cannot be made exact by any amount of careful arithmetic, is the correspondence between tick zero and any physical event. That is a separate question, it has a separate answer, and keeping those two things apart is the discipline the rest of this book is about.

WHAT THIS CHAPTER ESTABLISHED

- The tick is the Planck time, built from G, $\hbar$, and c — the one combination of the three limiting constants with the dimension of time.
- It is the resolution floor of an instrument, **not** a quantum of time. Asserting otherwise would take a side in a live argument as a side effect of choosing an integer type.
- Its length in seconds is fixed by convention against SI. This concedes metrology and nothing else, at a boundary you can point to.
- Time is counted, not measured. The count is exact; what tick zero corresponds to is a different question entirely.

CHAPTER 3

3

The zero of time

A count needs somewhere to start counting from. This chapter says where, and says the one thing about it you need in order to read Parts II and III without being misled. The full argument is Part IV, four chapters of it, and the deferral is deliberate — but a reader who stops at the end of Part I must not leave with a false impression, so the essential point is made here and made plainly.

TERM **Datum**

Tick zero. A **stipulated** reference point, conventionally identified with the Friedmann–Lemaître–Robertson–Walker $t \to 0$ limit. It is not a measurement and not an observed event.

The domain is unsigned

Absolute time in this system is an unsigned integer. There is no tick -1.

This is not a storage optimisation. It is a statement about what the system is willing to represent: nothing precedes the datum, because the datum is where the count begins, and a count does not go backwards past its own start.

Ask for an instant before tick zero and you do not get a negative number. You get an error — `UCAL-E0020` — and the operation fails.

INTERPRETATION

There is a difference between answering a question with a negative number and refusing the question, and the system takes the second option deliberately.

A negative tick count would be an **answer**: it would say that the time before the datum exists, is measurable, and happens to lie on the other side of a chosen origin — the way 1 BC lies on the other side of the Gregorian epoch. The refusal says something else: that the question, as posed to this instrument, is malformed.

Augustine gives the classic form of the move in **Confessions** XI, answering what God was doing before creating heaven and earth. His answer is not a description of that interval but a rejection of the question's shape, because time is among the things created and there is no "before" of the sort the question wants. Whether or not you find the theology congenial, the logical structure is exactly `UCAL-E0020`: an error, not a negative number.

Why it is stipulated and not measured

Here is the one-line version. Part IV gives it four chapters and three further reasons; this is the one you can carry.

Exactness cannot come from measurement.

The age of the universe is a measured quantity. The current best figure is 13.787 billion years, plus or minus 0.020 billion. That is an excellent measurement. It is also, like every measurement, an interval rather than a point.

Convert that uncertainty into ticks and it becomes:

ucal datum — the claim's magnitude

```
big_bang_claim:
  half_width_ticks
1170697614114106987000000000000000000000000000000000000000
  half_width_drifts  141.53
```

Fifty-eight digits. About 0.145% of the whole span from the datum to now.

Now suppose the datum were defined **as** the measured age. Every timestamp in the system would inherit that uncertainty, because every timestamp is an offset from the datum. The exact integer arithmetic — the thing this entire design exists to protect — would be computing precise-looking answers on top of a foundation that wobbles by a number with fifty-eight digits in it.

You cannot get an exact origin from an inexact measurement. So the origin is not taken from the measurement. It is **declared**, and the measurement is recorded separately.

What is actually claimed

Precision about this matters, so here is the claim in full, with nothing left implicit.

The system asserts that tick zero is tick zero. That is a definition, and definitions are not the sort of thing that can be wrong.

It asserts, **separately and as metadata**, that tick zero is conventionally identified with the FLRW $t \to 0$ limit, and that the published uncertainty in that identification is ±0.020 Gyr, citing Planck 2018. This is a claim about the world and it may be revised.

And it asserts nothing whatever about whether time began, whether there was a first moment, or whether the FLRW limit corresponds to a physical event at all.

DEFERRAL IS NOT EVASION

You have now been told that the datum is a decision rather than a discovery, and why the short reason forces it. What Part IV adds is three further reasons the specification gives, one more that it does not, and — the part that turned this project into a book — what was **done** about it: how a claim can be declared, cited, given an exact magnitude, and simultaneously made impossible to compute with.

If you are on the engineer's path and skipping Part IV, the sentence to carry forward is this one: the uncertainty in the age of the universe does not make the timestamps uncertain, because it is not an operand.

The datum is in ordinary company

One last thing, because "stipulated origin" can sound exotic and is not.

TAI's zero is 1 January 1958, chosen by agreement. The Julian Day epoch is 1 January 4713 BC, chosen because it made a convenient common multiple of three cycles. The Unix epoch is 1 January 1970, chosen because it was recent and round. None of these is a discovery about the universe; all of them are decisions that turned out to be useful.

INTERPRETATION

What is unusual here is not that the origin is stipulated. It is that the stipulation is **declared as such in the artifact**, with the physical claim held separately and marked non-computable — rather than being a fact about the system's history that you would have to go and read about.

Most epochs are stipulated and silent about it. This one is stipulated and says so, in a place the compiler can reach.

WHAT THIS CHAPTER ESTABLISHED

- Tick zero is the datum: stipulated, conventionally identified with the FLRW $t \to 0$ limit, and explicitly not a measurement.
- The domain is unsigned. A request for a time before the datum is an error, not a negative number — a refusal of the question rather than an answer to it.
- Exactness cannot come from measurement: the ±0.020 Gyr uncertainty is a 58-digit number of ticks, and an exact count cannot inherit it.
- The physical claim is recorded separately as metadata, cited, and — as Part IV shows — made impossible to use in arithmetic.

CHAPTER 4

4

The universe second

A tick is far too small to think in. It is 5.4×10^{-44} seconds; the age of the universe is about 8×10^{60} of them. Nobody reads a 61-digit number.

So there is a ladder of larger units above the tick, and it has an unusual property: every rung is a whole power of five of the one below, and the whole ladder is a whole power of five of the tick. That property is what makes the notation work, and this chapter is about what it buys and what it costs.

TERM Beat

5^{60} ticks, about 46.762 milliseconds. The reference rung of the ladder, and what the specification calls the **universe second** — a unit of human-noticeable size with no Earth content whatever.

Why base five

Because $5^5 = 3125$, and 3125 is a number of exactly five base-5 digits.

That is the entire reason. It is worth being blunt about it, because a book that spends its second half among Pythagoreans and Neoplatonists has an obligation to say clearly where its numbers came from, and this one came from a digit-packing convenience.

RULE N

No constant, base, or magnitude in this system acquires significance by resembling a number in a tradition. Not the five, not the sixty in 5^{60}, not the 3125.

The research did, as it happens, go looking. The first place it looked — Losev on the Neoplatonic tetrad — had a five waiting, and the Pythagorean tetraktys sums to ten and builds from four. Those are interesting facts about those texts. They are not evidence about this software, they were not inputs to it, and the book reports them once, in Part VI, labelled as what they are.

The ladder

Each rung is $5^5 = 3125$ of the rung below, which is to say each rung is exactly five base-5 digits wide. Rung k is 5^{60+5k} ticks, indexed from the beat at $k = 0$.

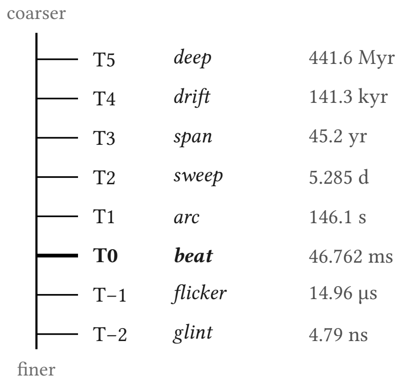

Figure 2 — The named rungs of the tier ladder. Each is 3125 of the one below. The ladder continues in both directions; only the named tiers are shown.

What the ladder buys

Three things, and they are all consequences of the same fact — that a timestamp is the tick count written in base 5 and grouped in fives.

Truncation is rounding. Write fewer groups and you have said the same thing less precisely. There is no separate rounding step, no scaling, no loss of a different kind: the digits you dropped are exactly the precision you gave up. This is the property that makes the uncertainty discipline of Part II possible at all.

Prefix comparison is chronological comparison. Two timestamps compare in the order their digits compare, because base-5 positional notation is monotone. Sorting the text sorts the times.

Writing all the digits pinpoints one tick. There is no accumulated conversion error between the coarse view and the fine one, because they are the same integer read at different widths.

Here is one real instant, at full precision and then at three coarser tiers:

ucal explain — one instant, four precisions

```
ticks
8070205189128471254993117657693008777530466139316558837890625

human     UC1 0031·0687·2481·3000·1638·3018:0779·2671·2006·1837·…
tiers:
  T5 deep   31
  T4 drift  687
  T3 span   2481
  T2 sweep  3000
  T1 arc    1638
  T0 beat   3018
```

The `0031` at the front is deeps since the datum. The `3018` before the colon is beats within the current arc. Every group is a base-5 number written in decimal, which is why none of them exceeds 3124.

> **INTERPRETATION**
>
> There is something worth noticing about that colon. It sits at the beat — at 5^{60} — and it is the only place in the notation where a human convenience has been allowed in. It is there because the beat is roughly the scale at which a person can notice a duration, so it is the natural place to put the decimal point of a time system built for beings who perceive at that scale.
>
> That is a concession to the reader, not to Earth. It changes no arithmetic.

What the ladder costs

Nothing on this ladder is near anything you know.

A second is 21.385 beats — not a whole number, and not close to one. An hour is 24.6 arcs. A day is 0.189 sweeps. A year is 0.699 spans.

> **THE TWO SECONDS DO NOT DIVIDE**
>
> This is the sharpest form of the cost, and the system says so on its own ladder output:
>
> The two seconds are incommensurable above T−6. One bridge second is 21.385061835 beats, not a whole number, because `BEAT` carries 5^{60} while `SECOND` carries only 5^{30}. They share a common measure only at the tick.
>
> So the beat is not a second in disguise, and no amount of rescaling would make it one. They agree at the tick and nowhere above it — which is exactly why the tick is primitive rather than either of them.

That is the honest consequence of leaving the Earth paradigm. If you want units with no planetary content, you do not get to keep the hour. The hour **is** planetary content.

The mark

The project's own mark is a diagram of this chapter, and it is worth reading as one.

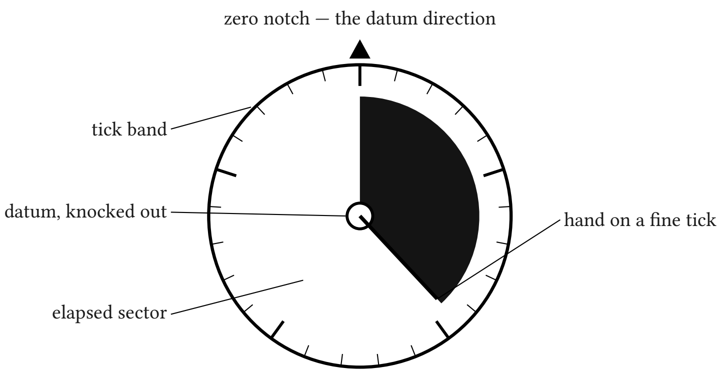

Figure 4 — The mark, read as a diagram. Five heavy divisions on the tick band, not twelve — the base is visible at a glance. The centre is negative space.

The centre dot has its core knocked out. That is deliberate: the datum is stipulated rather than observed, so the mark declines to draw anything there. The hand lands on a fine tick rather than a coarse division, because the claim the instrument makes is that it can pinpoint an arbitrary exact instant — not that it can round to a tidy one.

INTERPRETATION

A mark that showed a filled centre would be making a claim the specification refuses. The identity was drawn after the rule existed, and it obeys it. That is a small thing, and it is the sort of small thing this project is made of.

WHAT THIS CHAPTER ESTABLISHED

- The beat is 5^{60} ticks ≈ 46.762 ms — the **universe second**, a human-scale unit with no Earth content.
- Base five because $5^5 = 3125$ is five base-5 digits wide. Rule N: no further meaning, and the book says so before Part VI can be misread.
- The uniform ladder makes truncation **be** rounding, prefix comparison **be** chronological comparison, and full precision pinpoint one tick.
- The cost is that nothing on the ladder is near a second or an hour — one second is 21.385 beats, and the two share a measure only at the tick.

PART II

What was built

An engineer may stop at the end of this part and have received a complete technical article.

CHAPTER 5

5

The domain

Part I established what is counted. This part is about how the counting is implemented, and it starts with the number that holds the count.

Three properties, and the third is the one that turns out to have consequences all the way into the cosmology: the domain is **unsigned**, it is **closed**, and every operation on it is **checked**.

Unsigned, and why that is not a saving

There is no tick −1. Chapter 3 said why in one line — nothing precedes the datum, because the datum is where the count begins — and the type system is where that line is enforced.

`Instant` has no `Sub` implementation returning a signed value. Subtracting a later instant from an earlier one does not produce a negative `Delta`; it produces `UCAL-E0020` and the operation fails. To get the distance between two instants regardless of order you must ask for it by name, and the name says what you are doing.

> **THE SAME DISCIPLINE, APPLIED TO RATIONALS**
>
> `Ratio` — the exact rational type used throughout the derivations — carries the identical rule. `sub` refuses a negative result and directs the caller to `abs_diff`. That is Rule Z applied one level down, and it means the same class of mistake is impossible in the calendar derivations as in the timestamps.

Closed: 512 bits, and what that buys

The tick count is a 512-bit unsigned integer. The ceiling is $2^{512} - 1$, which the system will print for you:

ucal doctor — the domain

```
domain_max_ticks
13407807929942597099574024998205846127479365820592393377

72356144372176403007354697680187429816690342769003185818
                    6486050853753882811946569946433649006084095
domain_bits         512
```

In years, that ceiling is about 2.29×10^{103}.

To give that a scale: proton decay, if it happens at all, is expected somewhere around 10^{34} years. The last black holes evaporate around 10^{100}. The domain outlasts both, with room left.

The present epoch sits at about 8.07×10^{60} ticks — which is 6.0×10^{-94} of the range. The counter is, for all practical purposes, still at zero.

A NUMBER THIS BOOK HAD TO CORRECT

RFC UCAL-A1 attaches the figure 7×10^{-17} to this paragraph. That number is real but belongs to a different range: it is the present epoch as a fraction of the **UCID's** 256-bit space, not of the 512-bit domain. The domain fraction is 6.0×10^{-94}.

Rule S is why this correction is here rather than the original figure. The article RFC describes what the book intended to say; the source tree is what is true. This is the first place in the book where those two disagreed, and it will not be the last.

INTERPRETATION

The width was not chosen to be impressive. It was chosen so that **the width never has to change**, because under Rule B the value width is a wire-format commitment: the canonical binary encoding is 64 bytes because the domain is 512 bits, and widening the domain later would invalidate every stored timestamp in existence.

A system that might need to grow its integer has a migration in its future. This one bought its way out of that with sixty-four bytes, once.

Checked: overflow is a typed error

No operation on a time type wraps, and none saturates.

Every arithmetic method that could exceed the domain returns a `Result`. Overflow is `UCAL-E0060`, an ordinary error you handle, not a silently wrong answer you ship. The release profile sets `overflow-checks = true`, so the guarantee holds in optimised builds and not only in debug ones — which is where this class of bug normally hides.

There is also a workspace lint that fails the build on `wrapping_*` or `saturating_*` appearing on a time type. That is belt and braces, and it is there because the failure mode it guards is the quietest one available: a wrapped tick count is not an error message, it is a timestamp from the wrong end of eternity that looks entirely plausible.

Two backends, one domain

The tick integer has two implementations. The default is a fixed 512-bit stack integer; the alternative is an arbitrary-precision heap integer. They are selected by feature flag and are mutually exclusive at compile time.

The interesting requirement is that they must accept and reject **exactly the same values**. The arbitrary-precision backend is unbounded and would happily represent 2^{513}; it is made to enforce the same ceiling anyway, and the whole test suite runs against both.

> **INTERPRETATION**
>
> It would have been easier to let the unbounded backend be unbounded. The reason not to is that the two would then be different systems wearing the same name — a timestamp accepted by one and rejected by the other, with no way to tell from the type which you were holding.
>
> The rule that forbids this is one line long, and enforcing it cost a full second copy of the test suite in CI. That ratio — a cheap sentence, an expensive guarantee — is characteristic of the whole design.

No floating point. Anywhere.

This is the constraint with the largest blast radius, so it gets stated in its strongest form: there is no floating-point value in any shipped crate of this workspace. Not in a signature, not in a field, not in a constant, not in an intermediate, and not in the rendering path that produces the human-readable output.

A lint scans the source and fails the build on any float token. It permits exactly one exception — a float reference implementation used as a **test oracle** — and it requires the exception to be marked, and it prints every exemption it honoured on every run, so an escape hatch cannot quietly become a retreat.

What it costs
The cost is real and it is concentrated in one place: cosmology.

Computing the age of the universe at a given redshift means evaluating an integral. In any normal library that is a call to a quadrature routine over `f64`, and it returns a number with an error estimate. Here there is no `f64`, so it is certified interval quadrature over exact rationals — thousands of panels, each bounded above and below, with directed integer square roots underneath.

It is slower by orders of magnitude. The default subdivision depth takes about half a second where a float routine would take microseconds.

What it buys
An **enclosure** rather than an estimate.

The float routine returns a number and a tolerance, and the tolerance is a well-informed guess about accumulated error. The interval routine returns two numbers and a proof that the true value lies between them. Not probably. Provably, given the model.

> INTERPRETATION
>
> For most software the float answer is the better engineering, and it is not close. A guess accurate to twelve digits, computed instantly, beats a proof accurate to four, computed slowly.
>
> This project takes the other option because its subject is a quantity whose uncertainty is the interesting part. Part IV is four chapters about an origin that cannot be measured; a cosmology module that reported a confident point estimate would be undermining the book's central claim in the one place where the claim is most testable.

The two widths

One consequence deserves its own note, because it is where the arithmetic discipline becomes an argument.

Every cosmological result carries **two** uncertainty widths, and they are never merged: the width contributed by the quadrature, and the width contributed by the model's own measured parameters. Asking for a single combined tolerance requires calling a method with a name that says you are combining them.

At recombination the quadrature width is about 251 years and the parameter width is about 10,900 years. Merged into one number, the answer would be "about eleven thousand years of uncertainty" — and the fact that forty-three forty-fourths of it comes from the **measurement** rather than from the computation would be invisible.

That distinction is what tells you more computing power would buy nothing here. A merged tolerance hides it. Keeping them apart is a line in the specification, and it is the reason the specification has that line.

WHAT THIS CHAPTER ESTABLISHED

- The domain is unsigned: a result before the datum is `UCAL-E0020`, not a negative number. The same rule applies one level down, to `Ratio`.
- 512 bits reaches 2.29×10^{103} years, past proton decay and black-hole evaporation. The present epoch is 6.0×10^{-94} of it — and the width never has to change, because it is a wire-format commitment.
- Overflow is a typed error, enforced in release builds and by a lint, because a wrapped tick count looks entirely plausible.
- Two backends, one domain: the unbounded one enforces the same ceiling, verified by running the whole suite twice.
- No floating point anywhere. It costs orders of magnitude in the cosmology and buys a certified enclosure instead of an estimated tolerance.

CHAPTER 6

6

Notation

One value, four ways of writing it down. This chapter is about why that is not redundancy, and about what each form is for.

ucal explain — one instant, four notations

```
ticks
8070205189128471254993117657693008777530466139316558837890625

human     UC1 0031·0687·2481·3000·1638·3018:0779·2671·2006·1837·…

digit5    UC1/5 00000.00000. … .00111.10222.34411.44000.23023. …

ucid      000000000050PM6K45MKCAVY5MPYAMHCJQ142JHE26A2ZAJ9FJ1
```

Two text forms, and why both

The **human form** writes each tier group as a decimal number. The **digit form** writes the underlying base-5 digits directly, five to a group.

They encode the same integer. The difference is who is reading.

A person reading `0031·0687·2481` can see at a glance that this is 31 deeps, 687 drifts, 2481 spans — three quantities in a unit system, each written the way people write numbers. The same prefix in the digit form is `00111.10222.34411`, which is correct, canonical, and unreadable.

> **INTERPRETATION**
>
> The temptation is to call the human form a convenience wrapper and the digit form the real one. That is not quite right, and the distinction matters for Part VIII.
>
> The digit form is canonical **for machines** — parsing, sorting, round-tripping. The human form is canonical **for statements about time**, because tier groups are the units the system actually reasons in. Neither is a rendering of the other; both are renderings of the integer.

Each form carries a tag — `UC1` or `UC1/5` — so a value never travels without declaring which profile and which notation produced it. That is Rule P doing its work at the notation layer: two timestamps from different profiles cannot be compared by accident, because the text itself says they are different things.

Where each form anchors

This is the detail that most repays care, and it is the subject of one of the corrections this project made against its own specification.

A form printed to fewer groups is **less precise**, not **padded with zeros**. Chapter 4 called this "truncation is rounding". But saying that precisely requires knowing which tier each form counts from.

The human form anchors at **T0**, the beat: the group immediately before the colon. The digit form anchors at **T32**, the top of the ladder. So the same act of dropping trailing groups means the same thing in both — you have said less — but the tier a given group sits at is read from a different reference.

D-A8 — AN AMENDMENT THE IMPLEMENTATION FORCED

The specification originally described the anchoring loosely enough that two readers could disagree about what a truncated form denoted. Implementation made the ambiguity concrete: the human form **cannot express a precision coarser than T0**, because it has no groups above the colon to drop.

That is not a defect. It is a consequence of anchoring at the beat, and it is why the command line falls back to a different form when asked to render a timeline at a tier above T0 — rather than printing a blank field, which is what the first implementation did and which looked exactly like a value.

The canonical binary form

Sixty-four bytes, big-endian, fixed width. Never length-prefixed, never minimal, never a varint.

The width is not negotiable and the reason is worth stating: **byte order is numeric order**. A fixed-width big-endian encoding sorts lexicographically into chronological order, so the encoding is directly usable as a database key or a sort key with no comparator at all.

A minimal encoding — the obvious space optimisation — would break that, and it would break something else: the wire format would depend on the **magnitude** of the value rather than on the profile. Two timestamps of different sizes would have different lengths, and a reader would have to know the value to know how to read it.

> **INTERPRETATION**
>
> Sixty-four bytes for a timestamp is extravagant by any ordinary standard. Unix time fits in eight.
>
> What the extra fifty-six buy is that the format never has to change. The domain is 512 bits, the encoding is 512 bits, and there is no future in which a stored timestamp becomes unreadable because the range was widened. That is a real cost paid once against a class of migration that software normally pays repeatedly.

UCID: the transcribable form

The fourth notation is a 52-character identifier in Crockford base-32.

the alphabet

```
0123456789ABCDEFGHJKMNPQRSTVWXYZ
```

`I`, `L` and `O` are absent because they are confusable with `1` and `0`. `U` is absent to avoid accidental obscenity. The alphabet is strictly ascending in ASCII — and there is a test that checks that rather than trusting it — which is what makes lexicographic order equal numeric order here too.

Why 256 bits and not 512

Because the UCID is meant to be read, typed, and quoted by a person, and the domain is not.

At five bits per character, covering the full 512-bit domain would take **103 characters**. At 256 bits it takes 52, which is 260 bits of capacity — so the leading character encodes a single significant bit and is always `0` or `1`.

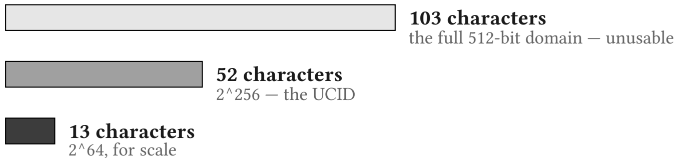

Figure 3 — UCID length against range. Five bits per character, so width is linear in bits — and the domain is twice the bits.

The ceiling that buys is 2^{256} ticks, about 1.98×10^{26} years. That is past the end of the stelliferous era and thirty orders of magnitude beyond the present epoch — and far short of the domain's 2.29×10^{103} years.

Above the ceiling you do not get a truncated identifier. You get `UCAL-E0031`, **instant outside UCID range**. The same refusal-rather-than-approximation that runs through the whole design.

What UCID is not

The source documentation is emphatic here, and so is this book.

UCID contains no randomness. It is a pure function of the instant. Two events at the same tick receive the same identifier, which makes it useless as a unique key for concurrent events.

It is worse than merely non-random. Chapter 7 shows that an instant read from a nanosecond clock has at least twenty-one trailing base-5 zeros by construction, so the low-order characters are not just predictable but **structurally constrained** — consecutive nanoseconds share more than twenty leading characters.

There is a test called `ucid_has_no_entropy` whose entire job is to measure how badly the identifier would fail if someone used it as a UUID.

INTERPRETATION

Writing a test that demonstrates your own type is unsuitable for an obvious misuse is an unusual thing to do. It costs nothing to omit; nobody would notice.

It is here because the failure would be silent. A UCID **looks** like a UUID — same length class, same character set, same shape in a log line — and the moment someone reaches for it as one, the collisions begin and never announce themselves. A test that fails loudly at the boundary is cheaper than an incident.

Four forms, one integer

To close the chapter, the whole picture in one table.

form	for	width	sorts?
`ticks`	the value itself, in decimal	variable	no
human	statements about time, by people	variable	only if padded
`digit5`	parsing, round-tripping, canonical text	variable	only if padded
binary	storage, wire, database keys	64 bytes	yes
UCID	quoting, transcribing, URLs	52 chars	yes

The two that sort are the two that are fixed-width. That is not a coincidence, it is the whole mechanism, and it is why the text forms carry the caveat "only if padded" rather than being quietly assumed to sort.

WHAT THIS CHAPTER ESTABLISHED

- Two text forms encode one integer: the human form in decimal tier groups, the digit form in raw base-5. Neither is a rendering of the other.
- Each form declares its anchor — human at T0, digit at T32 — which is what makes truncation mean **less precise** rather than **padded**.
- Canonical binary is 64 bytes, fixed-width, big-endian, so byte order is chronological order and the format never has to change.
- UCID is 52 characters over 2^{256} because 512 bits would need 103. Beyond the ceiling is `UCAL-E0031`, not a truncation.
- UCID carries no entropy and is not a UUID. A test measures exactly how badly it would fail as one, because that failure would otherwise be silent.

CHAPTER 7

7

The bridge

Chapter 1 said the design's whole question was whether Earth enters **the arithmetic** or at **a declared boundary you can point to**. This chapter is the boundary.

One constant

Earth enters the system through exactly one number.

the bridge constant

```
SECOND = 18 548 584 399 861 000 000 000 000 000 000 000 000 000
000 ticks
```

That is how many ticks are in one SI second. It is an exact integer — declared, not computed — and it is the only place in the workspace where a foreign unit is named at all. A lint fails the build if any identifier in the core crate mentions a foreign unit outside the bridge declaration.

> **INTERPRETATION**
>
> The number of things this constant is **not** doing is the point.
>
> It is not a conversion factor applied throughout the code. It is not a scale on the tick. It does not appear in any arithmetic that produces a timestamp from another timestamp. Every operation the system performs on absolute time is integer addition, subtraction and comparison on tick counts, and none of them has ever heard of a second.
>
> `SECOND` is consulted when a foreign value arrives and when one leaves. In between, it is inert.

Why the direction matters

Converting **into** absolute time is multiplication by an integer. Multiplication of integers is exact. So a whole number of seconds becomes a whole number of ticks with no rounding whatsoever, ever, under any circumstances.

Converting **out** is division, which is where rounding lives — and so the rendering path is the only place a rounding mode is chosen, and it is always chosen explicitly by the caller.

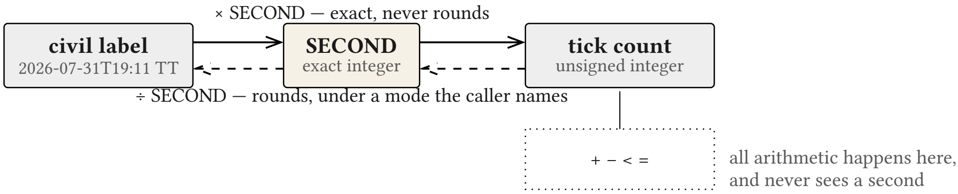

Figure 9 — The bridge. Earth enters and leaves at one declared constant; the arithmetic in between is integer operations on tick counts.

Refusing rather than rounding

Ask the system to accept a time finer than the bridge constant can represent and it does not round. It returns `UCAL-E0043` — **foreign-unit input finer than the bridge constant permits**.

The threshold is 10^{-30} seconds, and it is not arbitrary: it is where the declared constant's own decimal precision runs out. Below it, the system does not know the answer, and it says so instead of inventing one.

> **THE PATTERN, AGAIN**
>
> This is the third time in three chapters. Before the datum: refuse the question (`UCAL-E0020`). Beyond the UCID range: refuse to truncate (`UCAL-E0031`). Finer than the bridge: refuse to round (`UCAL-E0043`).
>
> A system that answered all three would be more convenient and would be lying in three different ways.

The invariants that fall out free

Here is the part that makes the design feel less like a set of choices and more like a consequence.

`SECOND` was set to a multiple of 10^{30}. Because $10^{30} = 2^{30} \cdot 5^{30}$, that means `SECOND` carries thirty factors of five. The nanosecond — a thousandth of a millionth of it — carries twenty-one.

So:

quantity	factors of 5	consequence
`SECOND`	30	a whole second lands with 30 trailing base-5 zeros
`NANOSECOND`	21	a whole nanosecond lands with 21 trailing base-5 zeros
`ORIGIN_OFFSET`	61	the bridge epoch is zero in every tier below T0

Read that in the notation of chapter 6 and it means something concrete: **any instant that came from a wall clock has a long tail of zeros in its digit form**. Six full tier groups of them, for a nanosecond clock.

> **INTERPRETATION**
>
> These are excellent invariant tests, and the project uses them as such — they are cheap to check and they fail loudly if the bridge is ever mis-declared.
>
> They are also the reason chapter 6 could say the UCID is not merely non-random but **structurally constrained**. The zeros are not a coincidence about particular timestamps; they are a theorem about every timestamp that entered through the bridge. An identifier whose low-order digits are provably zero is not a candidate for a unique key, and it is better to know that from the invariant than to discover it from a collision.

One pivot, and where the leap seconds live

Civil time is a mess, and the design's response is to confine the mess.

There is exactly one pivot scale: **Terrestrial Time**. TT is a uniform scale with no leap seconds, which is precisely why it was chosen — a timestamp converted through TT never has to know about the irregular ones.

Leap seconds exist, and the system knows all twenty-seven of them. They are applied at the parse and format boundary and nowhere else. No arithmetic on absolute time ever encounters one.

ucal doctor — the leap table

```
leap_seconds:
  table_version     IERS Bulletin C 70 (no leap second to
2026-06-30)
  entries           27
  complete_through  2026-06-30
  pre_1972          the 1961-1972 rubber-second era is modelled
exactly;
                    UTC before 1961-01-01 is UCAL-E0041
  network           never; the table is bundled and offline
```

Two details in that output are worth pausing on.

The table is bundled and offline. No command in this system performs network access, ever. A time library that phones home is a time library whose answers depend on whether the network was up.

The rubber-second era is modelled exactly. Between 1961 and 1972, UTC did not tick at the same rate as TAI — it ran at declared fractional offsets that changed periodically, and the offsets were not whole seconds. Most software declines to handle this and starts its UTC support in 1972.

INTERPRETATION

It turns out to be exactly representable in this system, and the reason is a small arithmetic accident worth recording. The era's rate coefficients — 0.001296, 0.0011232, 0.002592 seconds per day — are all divisible by 27. A day is 86,400 seconds, which carries $3^3 = 27$. The 27s cancel, and what is left is exact in the rationals the system already uses.

Nobody designed that. It was discovered while implementing, and it meant an era that could have been declared out of scope became eleven years of exactly convertible history instead. Before 1961 there is nothing to convert exactly, and the system returns `UCAL-E0041` rather than guessing.

What the bridge costs

Honesty, since Part III is about what implementation refused.

The bridge is where the claim “no Earth content” is at its weakest. `SECOND` is an Earth-derived constant sitting in the core crate, and the tick’s **length** is fixed by convention against it, as chapter 2 admitted.

What the design achieves is not the elimination of that dependency — that is not available to anyone — but its **localisation**. There is one constant, in one declared place, named by one identifier, with a lint that fails the build if a second one appears. The arithmetic is clean; the boundary is dirty; and the boundary is visible.

> **INTERPRETATION**
>
> Whether that is enough is a fair question and this book does not think it is settled. What can be said is that the alternative designs are worse in a specific way: a system with conversion factors distributed through its operations has the same dependency and no place to point at.
>
> A dependency you can point at can be argued about. One that has been dissolved into the arithmetic cannot.

WHAT THIS CHAPTER ESTABLISHED

- Earth enters through one exact integer constant, `SECOND`, and a lint fails the build if the core names a foreign unit anywhere else.
- Conversion **in** is multiplication and never rounds; conversion **out** is division and is the only place a rounding mode is chosen.
- Input finer than 10^{-30} s is refused, not rounded — the third refusal in three chapters.
- Whole seconds land with 30 trailing base-5 zeros and whole nanoseconds with 21, which makes good invariant tests and explains why UCID has no entropy.
- TT is the only pivot; all 27 leap seconds live at the parse/format boundary, offline. The 1961–1972 rubber-second era turns out exactly representable, by accident.
- The bridge does not eliminate the Earth dependency. It localises it to one place you can point at and argue about.

CHAPTER 8

8

Derived calendars

The ladder of chapter 4 has no days and no years in it, deliberately. But people live on planets, and a calendar that cannot say what day it is locally is an instrument of limited interest.

So there is a second layer: local calendars, derived from a body's own motion. This chapter is how that derivation works. Part V is how far it generalises and where it fails.

Four components

Every calendar in the system is the same shape:

component	what it is	source
Body	rotation, solar day, orbital period — as exact rationals of ticks	cited measurement
Anchor	where the count starts in phase	**declared, cited, empirical**
LeapRule	intercalation: how the fractional day is absorbed	**derived**
Cycles	grouping periods, if the calendar names a satellite	**derived**

Three of the four are computed. One is not, and the one that is not is the interesting one — it gets flagged here and gets a full treatment in chapter 16.

Units, as rationals of ticks

A body's periods enter as exact rational numbers of ticks, each carrying its citation, its epoch, its secular rate, and the window over which it is valid.

That last part matters more than it sounds. Earth's rotation is slowing by roughly 1.8 milliseconds per century; its tropical year is shortening by about half a second per century. A parameter treated as a constant is wrong the moment you leave the epoch it was measured at — and this system is aimed at deep time, where "leaving the epoch" is the normal case.

So evaluating a parameter outside its stated window produces a warning rather than a confident extrapolation. The system will tell you the number is out of its depth.

Intercalation is derived, never declared

Here is the heart of the mechanism.

A body's year is not a whole number of its days. Earth takes about 365.2422 rotations per orbit. Some scheme has to absorb the 0.2422, and every civilisation that has built a calendar has invented one.

This system does not invent one. It **computes** the schemes, by expanding the fractional part as a continued fraction and reading off the convergents — the sequence of best rational approximations, in the precise sense that no fraction with a smaller denominator comes closer.

ucal cal show earth-d — the derivation

```
intercalation:
  whole_days_per_year  365
  rule                 31/128
  bound                1 local day per 10000 local years
  walked:
    1: 1/4        — 1 day slips in 128 local years
    2: 7/29       — 1 day slips in 1234 local years
    3: 8/33       — 1 day slips in 4269 local years
    4: 31/128     — 1 day slips in 400000 local years   <- chosen
    5: 752/3105   — 1 day slips in 62100000 local years
    6: 4543/18758 — 1 day slips in 937900000 local years
```

Read the first line of that walk. Convergent 1 is **1/4** — one leap day every four years. That is the Julian calendar, derived from Earth's rotation and orbit with no knowledge of Rome.

The system then walks the ladder until a convergent meets the requested drift bound, and reports the entire walk rather than only the winner — so you can see what was rejected and why.

> **INTERPRETATION**
>
> The reporting-the-whole-walk decision looks like a nicety and is not. A mechanism that returns only its answer cannot be argued with. One that shows its working can be checked, and — as chapter 10 will describe at length — can turn out to contradict the person who built it.
>
> The single most important output this project ever produced came from reading a walk and noticing what was **not** in it.

Cycles, and the absence of them

Months are grouping periods, and they come from a satellite.

If a calendar names a grouping satellite, the system derives the cycle structure the same way — continued-fraction expansion of the ratio between the orbital period and the satellite's synodic period. Earth names the Moon, and the derivation produces 235/19, the Metonic cycle, known since antiquity and recovered here from Earth's periods alone.

If a calendar names no grouping satellite, it has no months. Not a default month, not a synthesised one, not a fallback to Earth's. None.

ucal cal list — three derived calendars

```
earth-d:
  leap_rule        31/128 (convergent 4)
  cycles           from moon

mars-d:
  leap_rule        45/76 (convergent 6)
  cycles           none — the calendar names no grouping satellite

titan-d:
  status  no anchor: complete in units, intercalation and cycles,
          incomplete in phase. Asking for local fields is UCAL-
E0062.
```

MARS HAS NO MONTH, AND THAT IS THE CORRECT OUTPUT

Mars has two satellites. Phobos orbits in about 0.45 of a Martian day; Deimos in about 5.4. Neither is anything a person would call a month.

The system could have synthesised one — divided the Martian year into twelve, say, or borrowed a period from somewhere. It returns nothing instead, because a month-shaped structure imposed on a body that has no month is Earth structure leaking through a mechanism built specifically to keep it out.

The absence is not a gap in the implementation. It is the implementation working.

The anchor: the one thing that cannot be derived

Everything above came out of arithmetic on cited parameters. The anchor does not, and cannot.

Intercalation tells you how long a year is. It does not tell you **when the year starts** — and no amount of tick counting will, because phase is not a consequence of period. You can know exactly how long Earth takes to rotate and still not know whether it is currently noon.

So the anchor is declared: one cited, interval-valued constant per body, with a revision number and an uncertainty window.

ucal cal show earth-d — the anchor

```
anchor:
  phase         mean solar midnight
  revision      1
  method        mean solar midnight at the prime meridian on
2000-01-01,
                i.e. 00:00:00 UT1, converted through TT = UT1 +
Delta-T
                with Delta-T(2000.0) = 63.8285 s
  uncertainty   dominated by the resolution of the published
Delta-T series
  window_ticks  3709716879972200000000000000000000000000
  citation      IERS Conventions (2010) and the IERS Earth
Orientation
                Centre's published Delta-T series
```

A calendar with no anchor is not an error to construct. It is an error to **ask local fields of**: `UCAL-E0062`. Titan is in exactly that state, and chapter 16 explains why it will stay there.

> **INTERPRETATION**
>
> The anchor is where this mechanism admits it is not self-sufficient, and the admission is structural rather than apologetic. It is one declared constant, no more privileged than the rotation period sitting beside it and no less necessary.
>
> What would have been dishonest is a default — an anchor that quietly fell back to Earth's, or to zero, so that every body appeared to have a working calendar. That would produce confident local dates for a body whose phase nobody knows.

Derived and legacy, held apart

The Gregorian calendar is in this system, and it is not a derived calendar.

ucal cal list — the legacy entries

```
earth-civil:
  kind       legacy — declared tables (§8.6)
  arbitrary  4
  leap_rule  97/400 (NOT a convergent — declared, not derived)

earth-julian:
  kind       legacy — declared tables (§8.6)
  arbitrary  4
```

`earth-civil` has four items of **arbitrary content** — irregular month lengths, the seven-day week (which corresponds to no astronomical period at all), the leap rule, and the epoch. None of them follows from Earth's motion. They are declared tables, and the type system keeps them in a separate trait: a function that wants a derived calendar will not accept a legacy one, and there is no blanket conversion.

Note the parenthesis on the leap rule. `97/400` is marked **NOT a convergent**. That label is the subject of chapter 10, and it is where this book's most uncomfortable finding lives.

> **INTERPRETATION**
>
> It is worth being explicit that "legacy" here is a technical classification and not a judgement. A declared table is not a worse calendar than a derived one; the Gregorian reform solved a real problem and solved it well enough to still be in use.
>
> What the classification records is **where the authority comes from** — arithmetic on cited parameters, or a decision someone made. Part VI applies that same distinction to scriptural chronologies, and it has to apply it even-handedly or not at all. A system that calls the Gregorian derived and Seder Olam declared would be doing something other than classifying.

WHAT THIS CHAPTER ESTABLISHED

- Every calendar is (Body, Anchor, LeapRule, Cycles). Three are computed; the anchor is not.
- Body parameters are exact rationals of ticks carrying epoch, secular rate, and validity window — and evaluating outside the window warns rather than extrapolating.
- Intercalation is derived by continued-fraction expansion, and the whole walk is reported, not just the winner. Convergent 1 for Earth is 1/4 — the Julian rule, derived.
- Cycles come from a named grouping satellite or do not exist. Mars has no month, and the absence is the mechanism working.
- The anchor cannot be derived, because phase is not a consequence of period. It is one declared, cited, interval-valued constant, and its absence is `UCAL-E0062` rather than a default.
- Legacy calendars are held apart by the type system. `97/400` is marked **not a convergent** — the subject of chapter 10.

PART III

What implementation refused

Design as something that loses arguments.

CHAPTER 9

9

Divergence

Parts I and II described a system. This part describes the arguments that system won against the person who specified it.

That framing is deliberate. A specification written before implementation is a set of predictions about what will turn out to be buildable, and some of them are wrong. The usual practice is to quietly fix the specification afterwards so that it appears to have been right all along. This chapter is the record of not doing that.

What was checked, and how

Before a line of the library was written, every constant and derivation in the specification was recomputed independently — twice, by two separate integer routes that share no code, and compared against what the document claimed.

That harness still runs:

cargo run -p xtask

```
UC-P0 constants harness — RFC UCAL-1, profile UC-1

  ok    routes A and B agree on every field
  ok    §3.3 profile constants are const-constructible and correct
  ok    BEAT: 5^60 (computed) == literal
  ok    provenance: AGE_s
  ...
  96 passed, 0 failed
```

It produced fifteen findings. Fourteen stand; one was withdrawn, and the withdrawal is the most instructive of the lot.

The map

Twenty-four named rules. Six were amended by contact with the implementation.

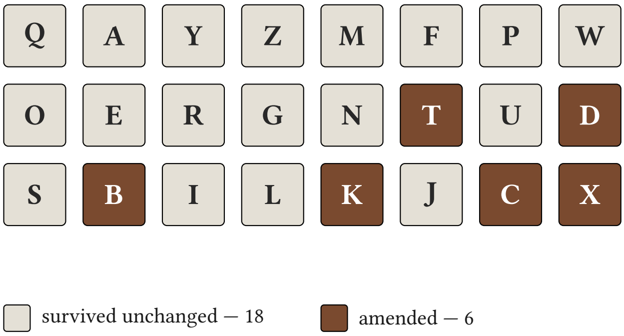

Figure 10 — The twenty-four rules by fate. None was dropped; six changed, and four of the six changed because the implementation proved the original could not be built as written.

rule	delta	what changed
T — truncation is uncertainty	D-A8	what a printed form **means**, and how each form anchors
D — two text forms	D-A8, D-A4	human anchors at T0, digit at T32; the fixtures were truncated
B — canonical binary	D-A7	full-width encode is 45 `divmod` steps, not 44
K — one derivation mechanism	D-A5	grouping cycles are declared per body, not admitted by a global bound
C — parameter provenance	D-A11, D-A13	obliquity cannot be a `RatedParam`; a drift bound is a rate, not a duration
X — certified enclosures	D-A14	the cosmological integral cannot be quadratured as written

Three kinds of losing

The fourteen findings sort into three classes, and the classes matter more than the count.

Editorial — five findings where the specification stated a number or a description inaccurately, with no behavioural consequence. `ORIGIN_OFFSET` has 61 trailing base-5 zeros, not 62. Appendix B's seconds column was chained from neighbouring rows instead of computed independently, and drifts in the fifth significant figure. These are the cheap ones.

Correction — six findings where the specification was wrong and the implementation could not follow it. The synodic-period formula computed the wrong quantity. A drift bound was typed as a duration when it is a rate. The cosmological integral, as written, was improper and could not be certified at all.

Amendment — three findings where the specification was coherent, buildable, and adopted differently on purpose after implementation revealed what it cost.

> INTERPRETATION
>
> The amendments are the interesting category, because nothing forced them.
>
> D-A5 is the clearest. The specification admitted a satellite as a "grouping satellite" — a source of months — if its synodic period fell within a bracket of 5 to 100 solar days. That is a perfectly implementable rule. It was replaced because the bracket is **calibrated on Earth's Moon**, which puts an Earth-derived constant inside the one mechanism whose entire purpose is to keep Earth from being the template.
>
> Nothing broke. The tests passed. The rule was changed because implementing it made visible that "month-like" is not a derivable predicate — **month-like** is an Earth predicate — and a bound tuned until Mars gave the expected answer would have been fitting the constant to a conclusion.

What each change cost

Honesty requires the costs alongside the corrections.

D-A5 cost a calendar the ability to have months by default: a body must now **name** a grouping satellite, with a citation for why. Mars names none, so `mars-d` has no months at all. That is the correct output, and it is also less than the specification promised.

D-A13 cost a type. `max_drift: &Delta` became `DriftBound { days, per_years }` in the body's own local units — which means the bound cannot be stated in seconds, which means a caller who thinks in seconds has to convert first and think about what they are doing.

D-A14 cost the cosmology module its shape. The specification's integral runs to infinity; certified quadrature needs a compact interval. The substitution $u = 1/(1 + z)$ fixes it, and finding that substitution was a day of work that the specification's phrasing implied would not be necessary.

The withdrawal

One finding was raised and retracted, and it stays in the record.

The verification claimed that Appendix C's fixture for 15 March 44 BC was one day late. It was not. The checking oracle was wrong: it applied an era adjustment that exists to make **truncating** integer division behave like flooring, on top of a language whose division already floors. The correction was correct for positive years, which is why seven of the eight fixtures passed and only the one negative year failed.

Appendix C is eight for eight.

> **INTERPRETATION**
>
> A retracted claim is more useful in the record than an absent one, and the reason is not modesty.
>
> A verification process that only ever reports findings is indistinguishable from one that manufactures them. The retraction is the evidence that the oracle was itself checked — that when the specification and the harness disagreed, the harness was not automatically believed.
>
> It is the smallest of the fifteen entries and it does the most work.

Did it diverge enough?

The plan for this book set a condition on Part III existing at all: if the implementation turned out to follow the specification closely, this part would collapse into two pages inside Part II, because a book claiming that implementation talks back has to show that it did.

The answer is yes, and here is the count. Fourteen standing findings. Six of twenty-four rules amended. Six places where the specification was simply wrong. One where its central method could not be executed as written. And separately from the rules, four of the project's six gated experiments hit their kill criteria — including one that fired before implementation started, when the integer library the specification named changed its API mid-build.

THE COUNT IS NOT THE ARGUMENT

Fourteen findings out of a document this size is not a scandal, and it is not presented as one. Specifications of this length routinely contain more.

What Part III claims is narrower: that the findings were **recorded** rather than absorbed, that the record distinguishes what was wrong from what was merely changed, and that one entry is a retraction. Those three properties are what make the next chapter's claim believable, and the next chapter is why this part exists.

WHAT THIS CHAPTER ESTABLISHED

- Every constant and derivation was recomputed by two independent routes **before** the library was written; the harness still runs, 96 checks.
- Fifteen findings: five editorial, six corrections, three amendments — and one withdrawal.
- Six of twenty-four rules were amended. None was dropped.
- The amendments were not forced. D-A5 replaced a working rule because implementing it revealed an Earth constant inside the mechanism built to exclude Earth.
- Each change cost something, and the costs are recorded next to the corrections.
- The withdrawal stays in the record, because a process that only ever finds things is indistinguishable from one that invents them.

CHAPTER 10

10

The 97/400 correction

This is the chapter the book exists for.

Everything else here is a claim that a program can do philosophical work. This chapter is the evidence, and the evidence takes the only form that counts: the program produced a result its author did not want, about a claim its author had published twice, and the author printed it.

The claim that was made

Chapter 8 described the intercalation mechanism. A body's year is not a whole number of its days; the fraction is expanded as a continued fraction; the convergents are the best rational approximations, in the precise sense that no fraction with a smaller denominator comes closer.

The specification, in two successive revisions, said this about it:

> The mechanism reproduces the Julian and Gregorian rules as convergents.

It is a satisfying sentence. It says the machinery, given nothing but Earth's rotation and orbit, rediscovers the two calendars Western civilisation actually built. If true, it would be the strongest possible demonstration that the derivation is not merely consistent but **correct** — that it finds what people found, from first principles.

Half of it is true.

What the mechanism actually returns

ucal cal show earth-d — the full walk

```
intercalation:
  whole_days_per_year  365
  rule                 31/128
  bound                1 local day per 10000 local years
  walked:
    1: 1/4         — 1 day slips in 128 local years
    2: 7/29        — 1 day slips in 1234 local years
    3: 8/33        — 1 day slips in 4269 local years
    4: 31/128      — 1 day slips in 400000 local years   <- chosen
    5: 752/3105    — 1 day slips in 62100000 local years
    6: 4543/18758 — 1 day slips in 937900000 local years
    7: 9838/40621 — 1 day slips in 4062100000 local years
    8: 24219/100000 — 1 day slips in never (exact) local years
```

Convergent 1 is **1/4**. One leap day every four years. That is the Julian calendar, derived from Earth's motion with no knowledge of Rome, and the first half of the claim is confirmed exactly.

Now look for `97/400`.

It is not there. It is not at depth 4, where the chosen rule sits. It is not at depth 8, where the expansion terminates because the parameter is exact. **It is not at any depth**, and there is a check in the constants harness whose only job is to assert its absence.

Why it cannot be there

This is not an accident of where the walk stopped. It is a theorem.

The convergents of a continued fraction have a defining property: convergent p/q is the **best** rational approximation with denominator at most q. Nothing with a smaller or equal denominator gets closer. So if `97/400` were a convergent, no fraction with a denominator below 400 could beat it.

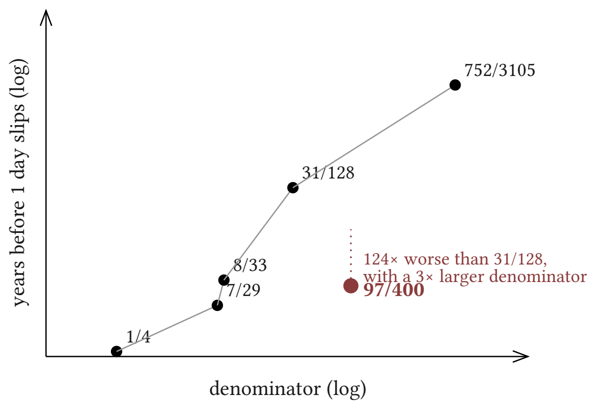

Figure 6 — Earth's convergent ladder, and the Gregorian rule off it. Every point on the line is the best approximation available at its denominator. `97/400` *is not on the line.*

Two fractions with smaller denominators beat it:

rule	denominator	1 day slips in	against 97/400
8/33	33	4,269 yr	1.32× more accurate, denominator 12.1× smaller
31/128	128	400,000 yr	**124× more accurate**, denominator 3.1× smaller

97/400	400	3,226 yr	— the Gregorian rule

`31/128` is a hundred and twenty-four times more accurate than the Gregorian rule while being a simpler fraction. That is the finding, and it is not close.

What that means, and what it does not

It does not mean the Gregorian calendar is bad. The reform of 1582 solved the problem it was aimed at — Easter drifting against the equinox — and it solved it well enough to still be in use four and a half centuries later. Accuracy against the tropical year was one constraint among several, and it was not the binding one.

It does not mean 97/400 was a mistake. A rule of 400 years is expressible as "every four years, except centuries, except every fourth century," which a person can apply without arithmetic. `31/128` cannot be stated that way at all.

What it means is precisely this: **the Gregorian rule is a declared table, not a derivation.** Chapter 8′s classification — legacy versus derived — is not a value judgement dressed up as taxonomy. It is a factual claim about where a rule's authority comes from, and here is the case where it bites.

> **INTERPRETATION**
>
> The interesting thing is not that the specification was wrong about a fraction. It is that the specification was wrong **in the direction of its own thesis**.
>
> A machinery that rediscovers both historical calendars is a better story than one that rediscovers one and quietly demonstrates the other was never derived. The sentence was not a careless error; it was the error the argument wanted to be true. It survived two revisions because nobody checked the half that was flattering.

Maimonides' charge

There is a nine-hundred-year-old objection to systems like this one, and it is worth stating in its own words because the correction above is the only kind of answer it admits.

In the **Guide of the Perplexed** I.73, Maimonides considers the Kalām practice of constructing metaphysical premises and observes — about people he otherwise treats with respect — that their premises were not derived from the world but assembled to make a desired conclusion provable. They knew where they were going and built the road backwards from it.

The charge generalises immediately to any formal system built by someone with a thesis. A calendar that derives calendars, built by a person who believes derivation is superior to declaration, is exactly the sort of thing that could be tuned until it produced the flattering result. The drift bound could be adjusted. The parameter could be chosen. The depth could be stopped where the answer looked good.

> **INTERPRETATION**
>
> You cannot answer that charge with assurance, because assurance is what it is about. You can only answer it with evidence, and the evidence has to be a case where the system contradicted its builder on something he had already committed to in public.
>
> This is that case. The claim was published twice. The mechanism said no. The mechanism's answer was recorded, the two revisions that said otherwise are named as wrong, and the harness now contains an assertion whose only purpose is to keep the correction from being quietly reversed.
>
> That is not proof of good faith — nothing is. It is the strongest available substitute: a system that has demonstrably been permitted to answer back.

What was done about it

Four things, and the fourth is the one that matters in five years.

The specification's sentence was corrected rather than deleted, and the revisions that carried it are named.

The mechanism was left alone. No bound was adjusted, no parameter retuned, no depth extended in the hope that `97/400` would appear further along. It does not appear further along; the expansion terminates at depth 8 because Earth's parameter is exact in the system, and the fraction is absent from all eight.

`earth-civil` was reclassified as a legacy calendar, and its leap rule carries the label in the tooling's own output:

ucal cal list

```
earth-civil:
  kind       legacy — declared tables (§8.6)
  arbitrary  4
  leap_rule  97/400 (NOT a convergent — declared, not derived)
```

And a check was added to the constants harness asserting that `97/400` is absent from Earth's convergents at every depth. It runs on every build. If someone ever changes a parameter in a way that makes the flattering claim true, the build will say so — which is the only form of vigilance that survives the author losing interest.

WHAT THIS CHAPTER ESTABLISHED

- The specification claimed twice that the mechanism reproduces the Julian **and** Gregorian rules as convergents.
- The Julian rule is convergent 1, exactly as claimed. `97/400` is not a convergent at any depth, and it cannot be: two simpler fractions beat it.
- `31/128` is 124× more accurate with a denominator 3.1× smaller; `8/33` beats it with a denominator 12.1× smaller.
- This is not a criticism of the Gregorian reform. It is the factual content of the legacy/derived distinction — where a rule's authority comes from.
- The error ran in the direction of the author's own thesis, which is why it survived two revisions.
- Maimonides' charge in **Guide** I.73 admits only one kind of answer: a case where the system contradicted its builder in public. This is that case, and a build-time assertion now keeps the correction from being reversed.

PART IV

The datum

Why it cannot be measured, and what was done instead.

CHAPTER 11

11

Why it cannot be measured

Chapter 3 gave one reason and promised three more. Here they are, and then a fourth the specification does not state, which is the best of them.

The question this part answers is narrow and it is worth keeping in view: **why is the origin of an exact integer count not simply the measured age of the universe?** The measurement exists. It is very good. Why not use it?

First: exactness cannot come from measurement

The published age is 13.787 billion years, plus or minus 0.020 billion.

The plus-or-minus is not a formality. Converted into ticks it is a number with fifty-eight digits — about 0.145% of the entire span from the datum to now, or 141.53 drifts on the ladder of chapter 4. If tick zero were **defined** as the measured age, every timestamp would be an offset from a foundation that wobbles by that much, and the exactness of the integer arithmetic would be decoration over a guess.

This is the reason that forces the decision. The others explain why the decision is not a workaround.

Second: the limit is not an event

The FLRW $t \to 0$ limit is not an observable event. It is where a mathematical model's coordinate goes when you extrapolate backwards, and the model is known to stop describing anything real well before you get there.

You cannot measure the time of an event that the theory containing it declines to assert exists. What is measured is the **age** — how long the universe has been expanding under the model — which is a different quantity that happens to be numerically what you want.

Third: the extrapolation is model-dependent

The figure 13.787 is not read off an instrument. It is inferred: a set of parameters is fitted to observations, and the age falls out of the fit. Change the model and the number changes.

A datum defined by that inference would inherit not only its error bar but its **dependence on a cosmological model that will be revised**. Timestamps written today would mean something slightly different after the next data release — which is failure mode F1, and the reason profiles exist.

Fourth: Kant, and the question's shape

The specification stops at three. There is a fourth, older than all of them, and it is the one that makes the other three feel less like obstacles and more like symptoms.

> In the **Critique of Pure Reason**, the First Antinomy sets out a thesis — that the world has a beginning in time — and an antithesis — that it does not — and argues that **both proofs are valid**.
>
> Kant's resolution is that both conclusions are false, because both share a premise that cannot be granted: that the world-series is a completed totality, given as a whole, about which a question of finite-or-infinite extent has an answer waiting to be found. On his account the series of past states is not an object of possible experience at all; it is something reason extends indefinitely, never something it is handed.

If that is right, then the search for the true origin is not a hard empirical problem awaiting better instruments. It is a well-formed-looking question with nothing of the appropriate kind to be true **of**.

> **INTERPRETATION**
>
> This is the strongest available support for stipulating the datum, and it is worth being precise about what it does and does not do.
>
> It does not show that the universe has no beginning. Kant is explicit that the antithesis fails too. What it shows is that the **question as posed** — where is the real zero, so that we may count from it — is asking for something that could not be supplied even in principle.
>
> Which means stipulating is not a retreat from a better method. There is no better method. Chapter 3 said the system refuses the question rather than answering it with a negative number; this is the philosophical form of the same refusal, and it arrived two centuries before the measurement did.

The book returns to Kant once more, in its last chapter, for the part of his argument that is less comfortable.

The datum is in ordinary company

All of that can make a stipulated origin sound exotic. It is the normal case.

epoch	zero	chosen because
TAI	1 January 1958	international agreement; atomic clocks were running
Julian Day	1 January 4713 BC	a convenient common multiple of three cycles
Unix	1 January 1970	recent, round, and before the machines that used it
UC-1	the datum	an exact count needs an exact origin

None of the first three is a discovery about the universe. All are decisions that turned out to be useful, and nobody finds them troubling.

THE EXACT PARALLEL: THE SI SECOND

The second is the sharpest case, because it is doing the same thing in the same system.

The SI second is defined as 9,192,631,770 periods of a caesium-133 hyperfine transition. That number is not a fact about caesium that was waiting to be found. It was **chosen**, in 1967, to match the ephemeris second then in use — itself defined from a particular fraction of a particular tropical year.

So the second is a stipulation calibrated against an older stipulation, and the calibration carries an uncertainty that the definition does not inherit. Nobody says the second is uncertain because the 1900 tropical year was imperfectly measured. The definition is exact; the **correspondence** to what it was calibrated against is not.

That is precisely the structure of the datum, and it is why the datum is unremarkable once you see where it already lives.

What is left over

Stipulation solves the exactness problem completely and leaves a different problem entirely intact.

The physical claim is still true or false. Something **is** the case about the relationship between tick zero and the early universe, and the published uncertainty in that relationship is real information that a serious system should not discard.

So there are two things to hold at once: an origin that is exact by definition, and a claim about that origin that is uncertain by measurement. Conflate them and you get either a false precision or a useless one.

Chapter 12 is how they were kept apart — and the answer is the reason this project became a book rather than a library with a good README.

WHAT THIS CHAPTER ESTABLISHED

- The measured age carries ±0.020 Gyr — 141.53 drifts, 0.145% of the span. An exact count cannot inherit that.
- The FLRW $t \to 0$ limit is not an observable event; what is measured is the age under a model, which is a different quantity.
- The inference is model-dependent, so a datum defined from it would shift with each revision — failure mode F1.
- Kant's First Antinomy adds a fourth: the question presupposes a completed totality that is not a possible object of knowledge, so there is nothing of that kind to find.
- Stipulated origins are ordinary — TAI, Julian Day, Unix — and the SI second is the exact parallel: a definition calibrated against something uncertain, without inheriting the uncertainty.
- What remains is a real claim about the world that must be kept, and kept separate.

CHAPTER 12

12

What was done instead

Three moves. Stipulate the datum. Declare the physical claim separately. Make the claim impossible to compute with.

The first two are ordinary good practice. The third is the one this book is about, and it is the only place where a code listing earns its space.

The exhibit

`BIG_BANG_CLAIM` is a value of type `SignedWindow`. Here is the type's entire definition:

```
pub struct SignedWindow {
    lo: Signed,
    hi: Signed,
}
```

Two fields. What matters is everything that is absent, and the source says so explicitly:

> This type deliberately has:
>
> - no arithmetic operators of any kind,
> - no `From<SignedWindow>` for `Delta`, `Instant` or `Window`,
> - no method returning any of those types.
>
> It cannot be added to an `Instant` and cannot be widened into one. Attempting to use it as an operand is `UCAL-E0025`, and the type system is what makes that unreachable rather than a runtime check.

The claim is fully available. You can read its bounds, render it, print its citation, compare it to itself. What you cannot do is **arithmetic** with it. There is no addition, no conversion, no escape hatch — and no runtime guard either, because a runtime guard is a check that can be forgotten, disabled, or bypassed by a code path nobody thought about.

The tests that prove the absence

An absence is hard to test. You cannot write an assertion that a method does not exist, because the file would not compile if you named it.

So the project uses compile-fail tests: programs that are **required to fail to build**, checked on every run.

tests/compile_fail/signed_window_arithmetic.rs

```rust
use ucal_core::{Profile, UC1};

fn main() {
    let a = UC1::big_bang_claim();
    let b = UC1::big_bang_claim();
    let _ = a + b;
}
```

tests/compile_fail/signed_window_as_operand.rs

```rust
use ucal_core::{Instant, Profile, UC1};

fn main() {
    let t: Instant<UC1> = Instant::zero();
    let claim = UC1::big_bang_claim();
    // A SignedWindow is not a Delta and must never become one.
    let _ = t.checked_add(&claim);
}
```

tests/compile_fail/signed_window_into_delta.rs

```rust
use ucal_core::{Delta, Profile, UC1};

fn main() {
    let claim = UC1::big_bang_claim();
    let _d: Delta = claim.into();
}
```

Each of these is a person trying, in the most natural way available, to use the uncertainty in the age of the universe as a number. Each fails to compile. The test suite passes precisely because they do.

INTERPRETATION

This is the whole thesis in nine lines of Rust, and it is worth spelling out why it is different in kind from a comment saying **do not do this**.

A comment addresses a reader who is paying attention. A runtime check addresses a program that reaches a particular line. A type addresses **everyone who ever writes code against this library**, including the author in three years, including someone who has not read a word of this book, including someone who actively disagrees with the distinction being enforced.

They will not be persuaded. They will be refused, by a machine, in under a second, with an error message. That is a different relationship between an argument and its audience than any essay can have.

Why the type is signed at all

A detail that is easy to pass over and is load-bearing.

The window is **signed** — it can express values before the datum — while `Instant` is unsigned and cannot. That is not an inconsistency. It is the reason the type had to be separate.

The FLRW limit may lie before tick zero. The published uncertainty is symmetric: ±0.020 Gyr around the datum, so its lower bound is negative. A type that could not express that would have forced the claim to be truncated at zero, which would have misreported the measurement.

INTERPRETATION

So the design faced a genuine dilemma. The claim needs a representation the timeline cannot have; the timeline needs a discipline the claim would break.

Splitting them into two types with no bridge is not a compromise between those requirements — it satisfies both exactly. The claim keeps its sign and its full symmetric range. The timeline keeps its unsigned domain. And the absence of any conversion is what stops the first from contaminating the second.

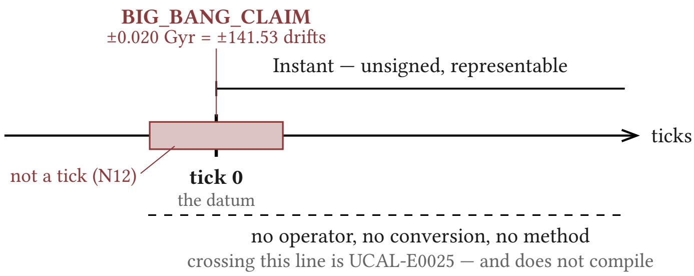

Figure 5 — `BIG_BANG_CLAIM` against the timeline. The claim straddles the datum and extends where no tick exists; the timeline is unsigned. Nothing converts between them.

The provenance chain

The second move — declaring the claim separately — would be worth little if the declaration were prose. It is data, and it re-executes.

ucal datum — the chain

```
datum_provenance:
  input     13.787 Gyr +/- 0.020 Gyr (age_of_universe)
  citation  Planck 2018 results VI, A&A 641, A6 (2020)
  unit_defs:
    Gyr = 10^9 x 31 557 600 s (Julian years, exact)
  chain:
    AGE_s     = 13 787 000 000 x 31 557 600
              = 435 084 631 200 000 000 s        (exact)
    AGE_ticks = AGE_s x SECOND
              = 80702040028955965162632000000...  (exact)
    beats     = round_half_even(AGE_ticks / BEAT)
              = 9 304 311 741 502 590 385
    ORIGIN_OFFSET
              = beats x BEAT
  residual_ticks     -31885691436436281946953338...
  residual_rendered  -0.017190364 s
  rationale  a whole-beat datum makes all sub-beat
             digits of the bridge epoch zero (2.4)
```

Every step is there: the cited input, the unit definitions, each exact multiplication, the rounding to a whole beat, and — the part that matters most — the **residual that the rounding discarded**.

Seventeen milliseconds. The datum is not the published age; it is the published age rounded to a whole beat, and it differs from the input by 0.017190364 seconds. The system prints that number rather than absorbing it.

> **INTERPRETATION**
>
> A provenance chain that reports its own rounding error is doing something a citation cannot. A citation says **where the number came from**. This says where the number came from **and what happened to it on the way in** — which is the part that a reader checking the work would otherwise have to reconstruct.
>
> The chain is also re-executable, which means the claim "this constant follows from that measurement" is testable rather than assertable. The constants harness does exactly that on every run, along two independent integer routes.

Kant policed it by discipline; the crate polices it by the compiler

Chapter 11 borrowed Kant's First Antinomy. Here is the borrowing that goes deeper.

> Kant distinguishes **constitutive** principles, which tell you what objects are, from **regulative** ones, which tell you how to go on investigating. The idea of the world-whole is regulative: it directs reason to keep extending the series and never licenses the assertion that the completed series exists.
>
> Transcendental illusion, for Kant, is precisely the slide from the second to the first — taking a rule for how to proceed as a description of an object.

The datum is a regulative posit. It says **count from here**; it does not say **here is where time began**. The physical claim is the constitutive-looking statement, and it is the one that has been rendered inert.

> **INTERPRETATION**
>
> Kant had no mechanism for this. He had argument, and vigilance, and the expectation that a careful reader would maintain the distinction under pressure. He says himself that the illusion does not go away once diagnosed.
>
> What this project has is not a better argument. It is a worse philosopher with a compiler: the distinction is maintained by something that does not get tired, does not skim, and does not want the flattering answer.
>
> Whether that counts as philosophical work or merely as engineering hygiene is the question Part VIII exists to argue. The claim made here is narrower and, I think, hard to dispute — that a distinction which previously depended on the reader's attention now does not.

UC-Θ, unbuilt

One thing this part must not do is imply the design is finished.

There is a second profile that has been specified and not built. Call it UC-Θ. In it, the datum is not the FLRW limit but the beginning of time **at organization** — a cosmology in which matter is organised rather than created from nothing, so that a physical origin lies later than the reckoning's zero, and there is a "before" that is not a **when**.

INTERPRETATION

Two things make UC-Θ interesting rather than idle.

The unsigned domain, which under UC-1 is a limitation to be defended, becomes **room** — the interval between the datum and the physical origin is representable rather than refused.

And Rule X becomes inapplicable. The cosmology module derives ages under a declared ΛCDM model anchored on the FLRW limit; move the datum off that limit and the certified enclosures no longer mean what they meant. A whole subsystem would have to be re-argued rather than re-parameterised.

That is what makes the profile mechanism more than versioning. Two profiles here are not two configurations of one system. They are two cosmologies, and Part VI, chapter 21 has to say plainly that UC-Θ is heterodox by the standard of the tradition that supplied chapter 13′s cautionary tale.

It is listed as unbuilt because it is unbuilt. If it is ever implemented, this section becomes a worked second datum and the book will have to say what changed.

WHAT THIS CHAPTER ESTABLISHED

- `BIG_BANG_CLAIM` is a `SignedWindow`: two fields, and no arithmetic operators, conversions, or methods returning a computable type.
- Three compile-fail tests encode the absence — each is a natural attempt to use the claim as a number, and each must fail to build.
- The type is signed because the claim straddles the datum; `Instant` is unsigned. Splitting them satisfies both requirements exactly rather than compromising between them.
- `datum_provenance` re-executes: cited input, exact chain, and the −0.017190364 s residual that rounding to a whole beat discarded.
- Kant maintained the constitutive/regulative distinction by vigilance and said the illusion persists after diagnosis. Here it is maintained by something that does not get tired.
- UC-Θ is specified and unbuilt. Under it the unsigned domain becomes room, and Rule X becomes inapplicable — which is why profiles are cosmologies, not configurations.

CHAPTER 13

13

Florensky's radius

Chapter 12 described a rule against letting a formal artifact be read as a physical fact. Rules against things exist because the things happen.

This chapter is about the clearest case of it happening, in a book by a mathematician who was better at this than the present author, published in 1922, which contains both the strongest precedent for what this project attempts and the exact failure it guards against — separated by a few pages.

Мнимости в геометрии

Imaginaries in Geometry, by Pavel Florensky: mathematician, electrical engineer, Orthodox priest, and the author of a study of geometry on the projective plane.

The book's project is to give geometric meaning to imaginary quantities, and most of it is exactly what it says. Then, at the end, it turns to Dante.

What Florensky got right

The achievement first, because it is real and because the failure is only interesting if the author was capable.

> Consider the **Comedy**'s trajectory. Dante and Virgil descend through the earth to its centre, pass the point of gravity, and continue — and emerge at the Mount of Purgatory on the far side. Dante then ascends through the spheres, past the fixed stars, to the Empyrean, and the poem ends.
>
> In a Euclidean cosmos the trajectory does not close. You go down, keep going, come out somewhere else, go up a long way, and stop. Standard diagrams of Dante's universe handle this by not quite drawing it.
>
> Florensky's observation is that the journey closes — that it returns coherently to its start — if the cosmos is **finite and non-orientable**. The traveller who passes through the centre and continues arrives back, reoriented, without ever turning around.

That is a genuine piece of mathematical reading. It takes a structural feature of a poem seriously enough to ask what geometry the poem presupposes, and the answer is neither trivial nor forced. Whatever one thinks of the rest, this part is good work.

What Florensky did next

In §9 of the same book, he turns to relativity.

The argument runs: consider a cosmos rotating rigidly. At a sufficient radius the co-rotating velocity would exceed c. In the Lorentz factor $\sqrt{1 - v^2/c^2}$, that makes the quantity under the root negative, and lengths and durations become imaginary. He computes the radius at which this happens. It comes out near the distance to the sphere of the fixed stars in the Ptolemaic scheme.

And then he identifies that surface with the Empyrean.

INTERPRETATION

The distance between the Dante geometry and this is exactly one move, and it is worth naming precisely, because from the outside they look like the same kind of reasoning.

In the first, a formal structure is used to **interpret** a text: the poem's trajectory has a property, and a geometry in which that property holds is exhibited. The claim is about the poem.

In the second, a formal artifact — the imaginary value of an expression outside its domain of validity — is treated as **designating a place**. The claim is about the world, and it is licensed by nothing except that the number came out at a suggestive magnitude.

One move. There is no third thing between them, and no bright line in the prose where the register changes.

What was missing

Florensky had no rule against the second move.

That is not a moral observation. He had a formal apparatus, a serious theological project, and enormous mathematical facility, and no principle in his method that said: **here is where the mathematics stops describing and starts being read**. Nothing in his toolkit distinguished a structure that carries meaning from a structure that is a physical fact.

> **INTERPRETATION**
>
> This project has such a rule, and chapter 12 is what it looks like implemented: a claim about the world, declared with its citation and its magnitude, and made incapable of entering a computation.
>
> It would be comfortable to conclude that the rule makes this project safe from Florensky's error. It does not, and saying so is the point of putting this chapter in Part IV rather than in Part VI.
>
> What the rule protects is the **arithmetic**. `BIG_BANG_CLAIM` cannot become an operand. But nothing prevents a reader — or the author, on a bad day — from looking at a 61-digit integer and reading it as a fact about being rather than a count under a stipulation. That is a failure of interpretation, not of arithmetic, and no type system reaches it.
>
> The last chapter of this book is about exactly that residue.

Basil, fifteen centuries earlier

The insight Florensky's §9 lacked had been available for a very long time.

> Basil of Caesarea, in the **Hexaemeron**, addresses what "in the beginning" means and answers with an image: the beginning of a road is not itself the road, and the beginning of a house is not the house. A beginning is not a member of the series it begins.
>
> Applied to time: the beginning of time is not a time. It is not a very short duration, not the first instant of a sequence, not a moment you could in principle point at. It is of a different kind entirely.

That is Rule Q's content, stated in the fourth century with no arithmetic whatsoever. The datum is not the first tick of the universe; it is where counting starts, and those are different sorts of thing.

WHAT THIS CONVERGENCE IS AND IS NOT

Basil is not being credited with anticipating anything, and Rule Q is not being validated by his agreement. Chapter 21 handles the Patristic material properly, including where it conflicts with this project.

The point here is narrower and it is about Florensky. The distinction he needed in §9 was not obscure, not recent, and not outside his tradition — he was an Orthodox priest, and Basil is not a marginal figure in it. He had every resource required to see the problem, and the problem was not visible from where he stood.

That is the useful version of the cautionary tale. Not **he lacked something**, but **he had it and it did not help**, because nothing in his working method forced him to reach for it at the moment it was needed.

What it cost

The chapter should not end on a methodological note, because the people in it did not.

Florensky was persecuted from the early 1920s onward. He continued working — on electrical engineering, on materials science, producing real technical results — under conditions that grew steadily worse. He was arrested, sent to Solovki, and executed in
1937.

Alexei Losev, whose work on number and name appears in chapter 25 and whose position Rule N declines, was imprisoned and sent to the White Sea Canal construction. He went nearly blind there. He survived, and spent decades publishing under constraints that required him to disguise what he was arguing.

INTERPRETATION

It matters that this book's clearest cautionary tale is a man who was killed for the intellectual tradition the tale is drawn from.

The failure identified in §9 is a real failure and this chapter does not soften it. But there is a way of writing about historical mistakes that treats their authors as material, and it is available here in a particularly cheap form: the mathematician who confused a formal artifact with a place, held up as a lesson.

He was a serious thinker working with the resources he had, on a question this project also finds worth asking, and he paid a price for the asking that no one writing today is being asked to pay. The rule in chapter 12 exists because his error is easy, not because he was careless.

WHAT THIS CHAPTER ESTABLISHED

- Florensky's reading of the **Comedy** — that its trajectory closes only in a finite, non-orientable cosmos — is genuine mathematical work on a real structural feature of the poem.
- In §9 of the same book he computes a radius from the Lorentz factor outside its domain of validity and identifies the resulting surface with the Empyrean.
- The distance between the two is exactly one move: interpreting a structure, versus asserting that a formal artifact designates a place.
- He had no rule against the second move, and Basil's road-and-house image — the distinction he needed — was fifteen centuries old and inside his own tradition.
- Chapter 12′s rule protects the arithmetic. It does not protect the reading, and the last chapter of this book is about that residue.
- Florensky was executed in 1937; Losev went nearly blind at Belomorkanal. The error is easy, which is why the rule exists — not because its author was careless.

PART V

Any celestial body

How far the approach generalises, and precisely where it fails.

CHAPTER 14

14

Universal baselines

The claim this part examines is that the approach works for any body, not just Earth. That is a strong claim, and the next three chapters test it: this one says why it should be true, chapter 15 shows it working, and chapter 16 says where it fails — at no less length than chapter 15, because a capability section longer than its limits section is a sales document.

Why it generalises

Three properties, each established in Part I or II, and together they are the whole argument.

The tick is not an Earth artifact. It is composed from G, $\hbar$ and c. A Martian, a Titanian, or anyone else running the same physics arrives at the same unit — with the caveat from chapter 2 that its **length in seconds** is a metrological convention, and the caveat matters less here than it did there, because nobody counting ticks needs the conversion.

The tier ladder has no body content. It is the powers 5^{5k}. There is no rung called "day" and none called "year", and the spacing was chosen for digit packing. Nothing on the ladder would have been different if Earth had a thirty-hour rotation.

The arithmetic never references a rotation, an orbit, or a civil calendar. This is the one that can be checked rather than argued, and chapter 7 described the check: a workspace lint fails the build if any identifier in the core crate names a foreign unit outside the single bridge declaration.

> INTERPRETATION
>
> The third property is doing more work than the first two, and it is worth separating them.
>
> The first two are claims about **choices**: the tick and the ladder were selected to be body-independent, and one could take that on trust or verify it by reading the definitions.
>
> The third is a claim about **what the code cannot contain**, enforced mechanically. Body independence in this system is not a design intention that survives as long as contributors remember it. It is a property the build checks.

One mechanism, and Earth as an instance

Chapter 8 gave the shape: every calendar is (Body, Anchor, LeapRule, Cycles). What matters for this part is that there is exactly **one** implementation of that shape.

There is no Earth path and no Mars path. There is no `if body == earth`. There is no crate named after a body — the crate is `ucal-body`, and Earth, Mars and Titan are entries in a data table it reads.

ucal cal list — three calendars, one mechanism

```
earth-d:  kind  derived — Rule K
          body  earth
mars-d:   kind  derived — Rule K
          body  mars
titan-d:  kind  derived — Rule K
```

The test that matters here constructs `earth-d` and `mars-d` through the identical generic path, from data alone, and asserts that neither required a special case. If someone adds a branch that treats Earth differently, that test is what notices.

WHY THIS NEEDED ENFORCING AT ALL

Because the failure it prevents is not a bug, it is a **drift**.

Nobody sets out to make Earth the template. What happens is that Earth is implemented first, because it is the one you can check against a wall calendar. A convenience gets added for it. A constant gets a default that happens to be Earth's. A test asserts something true only of bodies with one large moon. None of these breaks anything, and after enough of them the mechanism is an Earth calendar with parameters.

Failure mode F9 is that drift, and Rule K is the response: one mechanism, Earth as an ordinary instance, checked by a test that builds two bodies the same way.

The Copernican point, stated at its actual size

There is a temptation here that should be named and declined.

It would be easy to write that this system dethrones Earth the way Copernicus dethroned it — that a calendar with no privileged body is a philosophical achievement of the same kind. The sentence writes itself, and it would be an overclaim.

> **INTERPRETATION**
>
> What was actually done is narrower and it is worth stating precisely, because the narrow version is defensible and the grand one is not.
>
> Earth was moved from being the **template** of the mechanism to being an **instance** of it. That is a change in software architecture with a philosophical shape, and its entire content is: the structure that generates calendars does not contain a planet.
>
> It is not a claim about the universe. It does not show that Earth is unimportant, or that a body-independent reckoning is more true than a local one. Chapter 22 will argue, from a tradition that has thought about it longer than this project has, that a calendar is by definition an instrument of the created order — which cuts against reading any of this as transcendence.
>
> And the privilege was not abolished. It was **relocated**, from a body to a power-of-5 grid, and a grid is not nothing. Chapter 23 is where that gets pressed properly, by a tradition that puts a governing body exactly where this system puts an abstract ladder.

What generalising does not include

One boundary, stated here so chapter 15 cannot be read as claiming more than it shows.

The mechanism generalises over **bodies**. It does not generalise over **physics**. There is no relativistic model here: no time dilation, no worldline, no frame transformation beyond the single comoving frame the profile declares. Two clocks in different gravitational potentials are not modelled as running at different rates, because nothing in this system models rates at all — it counts.

That is a real limit and chapter 16 lists it among the others. It is mentioned here because “works for any celestial body” is the kind of phrase that expands while nobody is watching.

WHAT THIS CHAPTER ESTABLISHED

- The tick is built from constants, the tier ladder is a power ladder, and neither contains a body.
- The arithmetic cannot reference a rotation, an orbit, or a civil calendar — checked by a lint, not by intention.
- One mechanism, no per-body path, no crate named after a body; a test builds Earth and Mars identically from data alone.
- The failure this prevents is drift rather than a bug: Earth becomes the template one convenience at a time.
- Earth moved from template to instance. That is the claim, at its actual size — the privilege was relocated to a grid, not abolished.
- The mechanism generalises over bodies, not over physics. There is no relativistic model.

CHAPTER 15

15

Deriving a calendar

Three bodies, one mechanism, end to end. Earth because it can be checked against a wall calendar; Mars because it is a different planet with a different answer; Titan because it is strange enough to break a mechanism that only looked general.

Earth

The full derivation appeared in chapter 8 and the finding it produced was chapter 10. Here it is as one line of a comparison:

body	days/year	rule chosen	at the default bound
Earth	365.242190	31/128 — conv. 4	1 day slips in 400,000 yr
Mars	668.599	45/76 — conv. 6	1 sol slips in 16,566 Mars yr
Titan	—	derived	no anchor: phase unavailable

The Julian rule, 1/4, is Earth’s first convergent. The Gregorian rule is not a convergent at any depth. Both facts came out of the mechanism with no knowledge of Rome or of 1582.

Earth's months, and the Metonic cycle

Earth names the Moon as its grouping satellite, so the mechanism derives cycles the same way it derives intercalation — continued-fraction expansion, this time of the ratio between the year and the synodic month.

ucal cal show earth-d — cycles

```
cycles:
  satellite        moon
  cycles_per_year  12.368266523
  convergents:
    12/1
    25/2
    37/3
    99/8
    136/11
    235/19        <- the Metonic cycle
    4131/334
    8497/687
```

The sixth convergent is **235/19**: 235 lunar months in 19 years.

That is the Metonic cycle, named for an Athenian astronomer of the fifth century BC, used by the Babylonians before him, and still the basis of the Hebrew calendar’s intercalation and of the computus that fixes Easter. It falls out of the ratio of Earth’s two periods, with nothing supplied but those periods.

INTERPRETATION

This is the most satisfying single output the mechanism produces, and it is worth being careful about what it demonstrates.

It does not show that Meton was doing continued fractions. He was not. It does not show that the mechanism has rediscovered a truth about the cosmos.

What it shows is that 235/19 is **the good approximation at that denominator** — that anyone, in any century, with an accurate enough ratio and a motive to find a repeating cycle, is under pressure toward the same fraction. The convergents are where the good approximations are; Meton found one by observation and patience, and the algorithm finds the same one because there was nowhere else for either to land.

That is a claim about the arithmetic of approximation, not about anticipation.

Mars

A different planet, run through the identical code path.

● ● ● *ucal cal show mars-d — intercalation*

```
intercalation:
  whole_days_per_year  668
  rule                 45/76
  bound                1 local day per 10000 local years
  walked:
    1: 1/1     — 1 day slips in 2 local years
    2: 1/2     — 1 day slips in 11 local years
    3: 3/5     — 1 day slips in 128 local years
    4: 13/22   — 1 day slips in 796 local years
    5: 16/27   — 1 day slips in 2342 local years
    6: 45/76   — 1 day slips in 16566 local years   <- chosen
    7: 106/179 — 1 day slips in 76079 local years
   ...
   13: 486756/821993 — 1 day slips in never (exact)
```

A Mars year is 668.599 sols, so the fraction to absorb is 0.599 — much larger than Earth's 0.242, and a different shape of problem. The convergent ladder is correspondingly different: it starts at 1/1, meaning "every Mars year is a leap year", which is nearly right and not right enough.

At the default drift bound the mechanism selects **45/76**: forty-five intercalary sols every seventy-six Mars years, drifting one sol in 16,566 Mars years.

> **A NUMBER THE ARTICLE RFC STATES DIFFERENTLY**
>
> This book's own specification says Mars selects **16/27** at a bound of one sol per 2,342 years. Both figures are in the walk above, and they are consistent: 16/27 is convergent 5 and it does slip one sol in 2,342 Mars years.
>
> But 2,342 is not the default bound. At the default — one local day per ten thousand local years, the same bound Earth is given — 16/27 fails and the mechanism goes one rung further to 45/76.
>
> Rule S again. The specification quoted a real convergent at a bound it did not state, and the crate quotes what the default actually produces.

> **INTERPRETATION**
>
> Notice what the drift bound is **stated in**: one local day per ten thousand local years. Not seconds. Not Earth days.
>
> That is delta D-A13, and chapter 9 recorded it as a cost. Here is the benefit. The identical bound, applied to two planets, means the same **thing** on both without meaning the same **duration** on either — and it selects different rules because the planets are different, not because the bound was tuned per planet.
>
> A bound in seconds would have been an Earth constant sitting inside the mechanism Rule K exists to keep Earth-free. The awkwardness of local units is the price of not having one.

Titan

Titan is where a mechanism that merely looked general would fail, and it is in the book for that reason.

Titan is tidally locked to Saturn. Its rotation period equals its orbital period about Saturn, so its **solar day** — the interval between successive noons — is its month, in any ordinary sense of the word. And its **year** is not its orbit about Saturn at all; it is Saturn's orbit about the Sun, roughly 29.5 Earth years.

So the four components come apart in a way they do not for a planet: the body's rotation and its primary-orbit are the same number, and the period that gives the year belongs to a different body entirely.

> **INTERPRETATION**
>
> The mechanism handles this with no special case, and the absence of the special case is the result.
>
> Nothing in the derivation knows that Titan is unusual. It takes a rotation period, a solar day, and an orbital period as exact rationals of ticks, and it expands a continued fraction. That two of those numbers coincide, and that the third comes from Saturn's motion rather than Titan's, are facts about the **data** — and the data is where facts about bodies are supposed to live.
>
> A mechanism that needed to be told about tidal locking would be a mechanism with a list of the cases its author had thought of.

Titan also demonstrates the mechanism's limit, immediately and unambiguously:

ucal cal show titan-d

```
UCAL-E0062: calendar has no anchor; local fields cannot be
produced
```

That is chapter 16's subject, and it is not a defect in the derivation. The units, the intercalation and the cycles are all computed. What is missing is phase, and phase is not derivable from any of them.

One instant, three calendars

The mechanism's output, seen from outside:

ucal show — one instant, three calendars

```
ticks
8070205189128471254993117657693008777530466139316558837890625

earth-d:      earth-d/1: 0027-213.7987 c328    derived (Rule K)
mars-d:       mars-d/1:  0082-086.1665         derived (Rule K)
earth-civil:  2026-07-31T19:11:12 TT           legacy (§8.6)
                                               UCAL-W0005
```

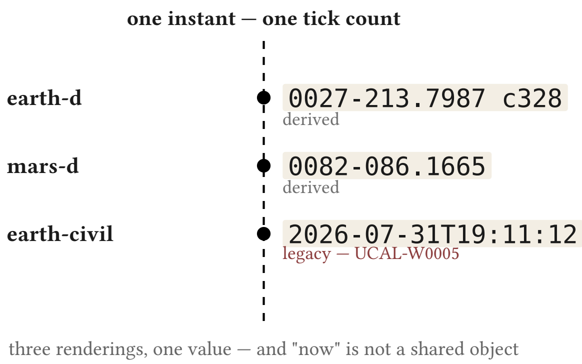

Figure 7 — Cross-body simultaneity. The same tick count rendered in three calendars, each carrying its kind and its anchor revision.

Three things to notice. Each rendering carries its **kind**, so a derived value is never mistaken for a declared one. Each derived rendering carries its **anchor revision**, so values computed under different anchor determinations are never silently compared. And the legacy rendering carries a warning, every time.

WHAT THIS CHAPTER ESTABLISHED

- Earth: 31/128 at the default bound; the Julian rule is convergent 1 and the Gregorian is absent.
- Earth's Moon yields 235/19 — the Metonic cycle — as the sixth convergent, from the two periods alone. That is a fact about where good approximations live, not about anticipation.
- Mars: 45/76 at the same default bound, not the 16/27 the article RFC quotes — which is convergent 5, correct at a bound the RFC did not state.
- The bound is stated in **local** days and years, so the same bound means the same thing on both planets without meaning the same duration.
- Titan is tidally locked and takes its year from Saturn; the mechanism handles it with no special case, which is the result.
- One instant renders in three calendars, each carrying its kind and its anchor revision.

CHAPTER 16

16

Where it breaks

Chapter 15 showed the mechanism working on three bodies. This chapter is the same length, and that is a rule rather than a coincidence: a book whose capability section outruns its limits section is selling something.

Six limits. Each is stated as a limit, not as a caveat — the difference being that a caveat is a qualification you are meant to read past.

1. The anchor is empirical and cannot be derived

This is the largest one, and everything else on the list is smaller.

The mechanism derives units, intercalation, and cycles from a body's periods. It cannot derive **phase**. Knowing exactly how long Earth takes to rotate tells you nothing about whether it is currently noon, and no amount of additional tick counting will supply the missing fact.

Getting phase requires knowing where a body actually was at some actual moment, which means ephemerides — observation, tabulated. That is a different kind of input and a different discipline, and it is out of scope for this system.

So every calendar carries one declared, cited, interval-valued constant:

ucal cal show earth-d — the anchor

```
anchor:
  phase          mean solar midnight
  revision       1
  method         mean solar midnight at the prime meridian on
2000-01-01,
                 i.e. 00:00:00 UT1, converted through TT = UT1 +
Delta-T
                 with Delta-T(2000.0) = 63.8285 s
  uncertainty    dominated by the resolution of the published
Delta-T
                 series, quoted to 0.0001 s near 2000.0; widened to
1 ms
  citation       IERS Conventions (2010), IERS Earth Orientation
Centre
```

INTERPRETATION

It is tempting to present the anchor as a minor input, and the honest framing is less comfortable: **the mechanism is not self-sufficient, and one component of every calendar it produces is a measurement it did not make.**

What can be said in its defence is structural. The anchor is no more privileged than the rotation period beside it — both are cited measurements with uncertainty windows — and it is no less necessary. The design does not pretend the anchor is derived, does not hide it among the computed fields, and does not default it. Its absence is an error rather than a fallback, which is the whole of chapter 16′s second half.

2. A body with no qualifying satellite has no month

Months come from a satellite. A body without a suitable one has none, and the mechanism must return nothing rather than synthesise something.

Mars is the worked case, and it is worth going through carefully because the article RFC for this book gets it wrong and the error is instructive.

Mars has two satellites. Under §9.6′s formula as originally written — synodic period measured against the primary's solar day — Phobos comes out at **0.450 sols** and Deimos at **5.363 sols**.

The specification admitted a satellite as month-giving if its synodic period fell in the bracket 5 to 100 solar days.

DEIMOS IS INSIDE THE BRACKET

This book's specification says both moons "fail the bounds". Phobos does, at 0.45. Deimos does not: 5.363 is inside [5, 100], comfortably.

So the RFC's own algorithm **admits** Deimos and hands Mars a month of 124.67 cycles per Mars year — while the same document states elsewhere that Mars has no month. The specification contradicted itself, and the implementation is what surfaced it.

There is a second layer to this, and it is the one that matters.

Delta D-A12 established that §9.6′s formula computes the wrong quantity: it measures against the primary's **solar day** where Appendix I.2′s own worked example measures against the **year**. Under the corrected, year-relative formula Deimos's synodic period is **1.2315 sols**, which is outside the bracket.

So the bracket's verdict on Deimos depends entirely on which formula you use — 5.363 and admitted, or 1.2315 and rejected.

> **INTERPRETATION**
>
> Two ways to fix a rule that admits a satellite you did not want. Tune the bracket until it excludes Deimos. Or notice that the bracket was calibrated on Earth's Moon — synodic period 29.5 days, comfortably interior — and is therefore an Earth constant sitting inside the one mechanism built to keep Earth out.
>
> The first is what a system built to make its conclusions provable would do, and it is precisely Maimonides' charge from chapter 10. The second is delta D-A5: the bracket was removed entirely, and a calendar must now **name** its grouping satellite with a citation for why.
>
> `mars-d` names none. So Mars has years and sols and no months — not because 5.363 fell on the convenient side of a constant chosen for the Moon, but because nobody has declared a Martian month and the mechanism will not invent one.
>
> Whether a satellite is "month-like" is not derivable, because **month-like** is an Earth predicate.

3. A rogue planet has no year

A planet not bound to a star has no orbital period. There is no fraction to absorb, no continued fraction to expand, and no intercalation to derive.

The mechanism has nothing to compute, and it returns nothing. A day-count is still available — rotation is intrinsic — but a calendar in the ordinary sense, with a repeating annual structure, does not exist for such a body and cannot be manufactured.

This is a small limit in practice and a clarifying one in principle: it shows that three of the four components depend on a **relationship** between the body and something else, and only rotation is the body's own.

4. A tidally locked body's day equals its orbit

Chapter 15 treated this as a success, and it is one — Titan is handled with no special case. It is also a limit, and both are true.

When a body is tidally locked to its primary, two of the four calendar components collapse into one number. The solar day and the primary-orbit are the same period, so a structure that assumes they are independent has one fewer degree of freedom than it expects.

Nothing breaks. But the calendar that results is thinner than a planet's: there is no relationship between rotation and primary-orbit to derive anything from, because they are the same relationship.

5. Relativistic environments are out of scope

There is no relativistic model here at all.

No time dilation. No worldline. No transformation between frames. The profile declares a single comoving frame and every quantity is counted within it.

> **INTERPRETATION**
>
> For the uses this system is aimed at — deep time, cosmological scales, comparing epochs — that is defensible, because the frame is declared and the differences it ignores are small against the intervals it measures.
>
> For a body deep in a gravity well, or for two clocks that need to agree to a precision where dilation matters, this system is simply the wrong instrument. It would give an answer, and the answer would be wrong in a way it could not detect, which is the worst kind.
>
> Rule F is what makes this a limit rather than a hazard: the frame is **declared**, so the assumption is visible rather than implicit. That does not make the system usable outside the frame. It makes the boundary findable.

6. Body parameters are wrong outside their validity windows

The final limit is the one most likely to bite in practice, and it bites hardest at exactly the scales this project exists for.

Every body parameter carries an epoch, a secular rate, and a window of validity. They are not constants. Earth's rotation is lengthening by roughly 1.8 milliseconds per century; its tropical year is shortening by about half a second per century.

Evaluate a parameter outside its stated window and the system warns rather than extrapolating confidently.

> **INTERPRETATION**
>
> Consider what that means for a system whose domain reaches 2.29×10^{103} years.
>
> The parameters are good near J2000 and degrade as you move away. A calendar derived for a body a billion years hence is derived from **today's** rotation rate, and Earth's rotation a billion years from now will not be today's — the Moon is receding and the tidal braking that drives the change will itself have changed.
>
> So the mechanism's reach and its accuracy point in opposite directions. It can address any tick in a 10^{103}-year domain, and it can derive a trustworthy calendar only near the epochs where somebody measured something.
>
> The warning is the honest response and it is not a solution. A drift bound guaranteed over 400,000 years is guaranteed **under parameters that will not hold for 400,000 years**, and the system says so rather than quietly implying otherwise.

What the six have in common

Read together, the limits fall into two kinds, and the distinction is worth having.

Limits 3 and 4 are **structural**: the mechanism gives less because the body has less to give. A rogue planet genuinely has no year. Those are not deficiencies; correct output for an unusual body is unusual output.

Limits 1, 2, 5 and 6 are **epistemic**: the mechanism needs something it cannot compute. Phase requires observation. A month requires a declaration nobody has made. A relativistic answer requires a model this system does not have. A parameter far from its epoch requires a measurement nobody took.

> **INTERPRETATION**
>
> In every one of the four epistemic cases the system's response is the same: return an error or a warning, and never a default.
>
> `UCAL-E0062` for a missing anchor. An empty cycle list for an undeclared satellite. `UCAL-W0003` for a parameter out of its window. A declared frame that says what it does not model.
>
> That consistency is the actual claim of Part V — not that the approach works everywhere, because it does not, but that where it stops working it **says so** rather than producing a confident number nobody can audit.
>
> A calendar that silently defaulted Titan's anchor to Earth's would work. Every function would return a value, no test would fail, and every Titanian date it produced would be wrong by an unknown amount, with nothing anywhere to indicate it.

WHAT THIS CHAPTER ESTABLISHED

- The anchor is empirical and cannot be derived: phase is not a consequence of period, and getting it needs ephemerides.
- A body with no declared grouping satellite has no month. Mars is the case — and the article RFC's claim that both moons fail the bracket is wrong, which is how the bracket's Earth-calibration came to light.
- A rogue planet has no year; three of the four components depend on a relationship, and only rotation is the body's own.
- A tidally locked body collapses two components into one — handled without a special case, and thinner for it.
- Relativistic environments are out of scope entirely. The declared frame makes that boundary findable, not passable.
- Parameters are wrong outside their validity windows, and the reach of the domain and the accuracy of the derivation point in opposite directions.
- Four of the six limits are epistemic, and in all four the system errors or warns rather than defaulting. That is Part V's actual claim.

CHAPTER 17

17

The one qualification

The preface said the practical utility of this system is questionable and that there is exactly one qualification to that. This is it. It is one page, deliberately, and it does not grow.

The qualification

For a single planet, this is a curiosity. Everything in Parts I through IV stands: no task you have today needs a Planck-tick count since a stipulated datum, and the system is actively awkward for ordinary work because nothing on its ladder is near an hour.

For timekeeping across **two or more bodies**, the picture changes — not because this system becomes useful, but because the alternatives become incoherent.

The alternative to a common absolute reckoning is pairwise conversion: Earth time to Mars time, Mars time to Titan time, each pair with its own leap rules, its own epoch, its own accumulated conventions. With n bodies that is $n(n-1)/2$ conversions to define, agree, implement and keep correct, and no fact that all of them are about.

A universal ladder with local overlays replaces that with n conversions to one referent. Chapter 15′s cross-body rendering is what it looks like: one tick count, three calendars, each carrying its kind and its anchor revision so that values from different determinations are never silently compared.

INTERPRETATION

The argument is about **coherence**, not efficiency. Pairwise conversion is not merely more work; it has no common object. "Now" under it is a web of agreements about translation, and there is nothing the agreements are agreements about.

With a common referent there is something — a tick count — that every local calendar is a rendering of. Whether anyone needs that is a separate question, and the answer today is no.

What this is not

Three things this paragraph must not be read as saying, since it is the one place in the book where the practicality question is even partly reopened.

It is **not** a recommendation for adoption. Whether a system should be used depends on migration cost, tooling, training, and the enormous inertia of working conventions, and none of those favour this. The claim is that a universal ladder **could** be coherent, not that it **ought** to be adopted.

It is **not** a prediction. Multi-body civil timekeeping is not a live problem. If it becomes one, the people facing it will have constraints nobody here can anticipate, and they will very likely choose something else for reasons that are good.

It is **not** a retraction of the preface. The three facts stand: the artifact is real, its practical utility is questionable, and it is therefore research of another kind. A conditional about a situation that does not exist is not a use case.

WHY THIS CHAPTER IS ONE PAGE

Because the temptation is to let it be twenty.

There is a version of this book where the multi-body argument grows — where each chapter finds another way the approach would help, and the honest assessment of Part V's limits gets balanced against a mounting case for eventual usefulness. That book would be more comfortable to write and would have quietly abandoned its thesis.

The thesis is that this is research conducted in the medium of a working program, and that the rigor is the medium rather than a preliminary to a payoff. A rescue of the practicality question would not strengthen that. It would replace it.

WHAT THIS CHAPTER ESTABLISHED

- For one planet the system is a curiosity, and the preface's second fact stands unretracted.
- For two or more bodies, pairwise conversion has no common referent: $n(n-1)/2$ agreements about translation with nothing they are about.
- A universal ladder with local overlays replaces that with n renderings of one value. That is a claim about coherence, not efficiency.
- It is not a recommendation, not a prediction, and not a retraction — and it is one page because the temptation is to let it be twenty.

PART VI

Nine readings

The traditions as readers of the artifact — neither validating it nor validated by it. Every chapter names its conflict.

CHAPTER 18

18

Greek

HOW TO READ PART VI

Nine chapters, one shape. Each says what a direction holds, which rule of the artifact it illuminates, where the two converge, **where they conflict**, and what the conflict changes about how to read the instrument.

The conflicts are fixed in advance and none is optional. A tradition that only agrees has not been read — it has been quarried. Where a conflict cuts against this project, the chapter says so and does not resolve it in the artifact's favour.

No tradition here is argued true. The code is evidence for none of them.

What the direction holds

Three positions, from three centuries, that between them set the terms this book has been using without acknowledging where they came from.

> **Aristotle**, **Physics** IV.11: time is "the number of motion with respect to before and after." Not motion itself, and not a container motion happens in — the **countable aspect** of change.
>
> In IV.14 he presses further and asks whether time could exist if there were no soul to count. His answer is careful: there would be the substrate of time, the changing thing, but not time as number, since number requires a numberer.
>
> In **Physics** VI he establishes that time is a continuum. Between any two instants lies another; the instant is a **limit** of an interval, as a point is a limit of a line, and not a very short piece of one.
>
> **Plotinus**, **Enneads** VI.6, distinguishes two senses of number: οὐσιώδης ἀριθμός, substantial number, which is constitutive of what a thing is, and μοναδικὸς ἀριθμός, monadic number, the counting-number we use to tally. The first is prior to being; the second is our instrument.
>
> Behind both stands Plato's doctrine of ideal numbers as ἀσύμβλητοι — non-addible. Two of them do not make four, because they are not units of a common kind.

Which rule it illuminates

Rule Q, on the datum and the counter. Rule G, on the tier grid and what a number is. N1, the refusal to call the tick a quantum of time. And `derive_leap_rule`, which turns out to be older than it looks.

The convergences

Euclid X.2 is the intercalation algorithm

This is the strongest of them, and it is not a resemblance.

> **Elements** X.2 gives the procedure Greek mathematics calls ἀνθυφαίρεσις — reciprocal subtraction. Given two magnitudes, subtract the smaller from the larger repeatedly; take the remainder and repeat against the smaller; continue. If the process never terminates, the magnitudes are incommensurable.
>
> Euclid's purpose is a criterion for incommensurability. The procedure is the Euclidean algorithm run on magnitudes rather than numbers.

Continued-fraction expansion **is** anthyphairesis. The quotients at each step are the partial quotients; the convergents are what you get by truncating. Chapter 8′s `derive_leap_rule`, which takes a year and a day and returns 1/4, 7/29, 8/33, 31/128, is running Euclid X.2 on two orbital periods.

> INTERPRETATION
>
> So Appendix I of this project's specification is a Greek text with different vocabulary. That is not a coincidence and it is not anticipation. There is one good algorithm for this problem and it was found early, because the problem is old and the algorithm is short.
>
> What the convergence is evidence **for** is narrower and more useful: that the mechanism is not an invention of the author's. It is the standard procedure, and its results are therefore not tuneable in the way chapter 10′s charge worried about.

Aristotle's two numbers, and the ladder

Physics IV.11 distinguishes the number **counted** from the number **by which we count** — the twenty sheep from the twenty.

That distinction maps onto the artifact with unusual precision. The tick count is the number counted: how many ticks have elapsed. The tier ladder is the number by which we count: the system of units in which the count is expressed.

Archimedes

> **RESONANCE**
>
> The **Sand Reckoner** is addressed to King Gelon and its purpose is to refute the claim that the number of sand grains filling the cosmos is infinite — or if not infinite, at least unnameable. Archimedes' response is to **build a notation**: a positional hierarchy of orders and periods, expressly constructed so that a cosmic-scale magnitude becomes expressible.
>
> He arrives at something around 10^{63}.
>
> The present tick count is about 8×10^{60}.
>
> Three orders of magnitude apart, from a problem about sand and a problem about time, twenty-two centuries apart. This is a coincidence. It is recorded because the **method** is the same — invent a positional hierarchy so that the unnameable becomes nameable — and because the numerical closeness is striking and means nothing.
>
> It is not a premise for anything in this book.

Two boundary markers

Epicurus held that time, like magnitude, has minima — indivisible least parts. So temporal atomism is native to Greek thought, and N1′s refusal to call the tick a quantum declines a Greek position rather than only a later one.

The Stoics held ἐκπύρωσις: the cosmos periodically consumed by fire and reconstituted, identically, without end. That is a cosmology in which **no datum is possible**. There is no first cycle to count from, and any origin you stipulate is arbitrary in a way that UC-1′s is not — because UC-1′s is at least stipulated **against** a model that has a limit to point at.

THE CONFLICT

Two, and the second is worse.

Time is a continuum; the instrument is discrete. Aristotle's **Physics** VI is not a casual position — it is argued at length, and the argument that the instant is a limit rather than a part is one of the more durable things in the corpus. This system's finest addressable quantity is a tick, and between tick n and tick $n + 1$ there is nothing.

N1 declines to claim that reality is discrete, and that helps. But it does not dissolve the tension: an instrument whose resolution is a smallest unit is not neutral about a continuum. It can only ever address a countable subset of it.

Physics IV.14 is the harder one. If time as number requires a counter, then the datum's objectivity depends on there being someone to count from it — and Rule Q, which makes the datum stipulated rather than found, **concedes this rather than answering it**. The datum is where a counter decided to start. Aristotle would recognise that immediately, and he would not regard it as a solution to the problem he raised.

What it changes

Here is the thing the Greek material makes visible, and it is uncomfortable.

The tick count is the number counted. The tier ladder is the number by which we count. And in this system **they are the same integer**: the ladder is not a separate apparatus applied to the count — a timestamp is the count itself, written in base 5 and grouped in fives. Chapter 4 presented that identity as the design's central elegance.

INTERPRETATION

Neither Aristotle nor Plotinus would grant it.

For Aristotle the two numbers are distinct in kind. The twenty sheep and the twenty are not the same thing; one is a feature of the flock and the other is what the mind supplies. Collapsing them is not simplification but conflation.

For Plotinus it is worse. Substantial number is prior to being and monadic number is our tally, and to identify them is to mistake the instrument for the constitution — which is close to the error chapter 13 spent a chapter on.

So the design's most elegant property is the one this direction rejects most firmly, and the rejection is not a misunderstanding. It is a considered position from people who thought about number harder than this project has.

I do not think the identity should be withdrawn — the count and its notation really are one object here, and pretending otherwise would add a layer that does no work. But it should be **defended** rather than presented as obviously good, and this book has until now presented it as obviously good. That is what the chapter changes.

WHAT THIS CHAPTER ESTABLISHED

- Time as the number of motion; the instant as a limit, not a part; two senses of number, substantial and monadic.
- Euclid X.2′s anthyphairesis **is** continued-fraction expansion — the intercalation mechanism is a Greek algorithm, which is why it is not tuneable.
- Archimedes built a positional hierarchy to make a cosmic magnitude expressible and landed within three orders of the present tick count. Recorded as resonance; a premise for nothing.
- **Conflict:** the instrument is discrete where Aristotle holds a continuum — and **Physics** IV.14 makes the datum's objectivity depend on a counter, which Rule Q concedes rather than answers.
- **What changes:** the design's central identity — that the number counted and the number by which we count are one integer — is precisely what this direction denies. It needs defending, not displaying.

CHAPTER 19

19

Jewish

This chapter contains the hardest sentence in the book, and it arrives in the conflict section. Everything before it is preparation for being able to write it honestly.

What the direction holds

> The Hebrew calendar's computational epoch is **molad tohu** — "the new moon of chaos" — fixed by the mnemonic BaHaRaD: day 2, hour 5, 204 **ḥalakim**. By rabbinic reckoning it falls roughly a year **before** the creation it anchors, because

> the calendar's arithmetic needs a molad to count from and the first actual molad is not it.
>
> Time is subdivided by the **ḥelek**: 1/1080 of an hour, about 3⅓ seconds. The number 1080 was not chosen for convenience of size. It was chosen because it divides — by 2, 3, 4, 5, 6, 8, 9, 10, 12 and more — so that the fractions the calendar needs come out whole.
>
> Intercalation follows a nineteen-year cycle: seven leap years in nineteen, fixing the lunar year against the solar.
>
> And chronology is not held to be independent of interpretation. **Seder Olam Rabbah**, the second-century chronology that underlies the traditional Anno Mundi count, derives its dates from scripture read as a coherent whole.

Which rule it illuminates

Rule Q, on stipulated epochs. Rule Q.4, on provenance as data. D-3, on divisibility. And Appendix I.2, on the Metonic cycle.

The convergences

Molad tohu is Rule Q, nineteen centuries early

This is the closest precedent for the datum that this book has found anywhere.

The epoch is **computational**. It is placed before the event it anchors. It is not claimed as an observation. And — the part that no modern epoch matches — it is **named for its own emptiness**. Tohu is the word from Genesis 1:2, formlessness, the void before ordering.

INTERPRETATION

Compare the honesty of the two namings.

This project calls its epoch “the datum” and then spends four chapters and a type system explaining that it is stipulated rather than observed. The Hebrew calendar calls its epoch **the new moon of chaos**, and the explanation is in the name.

Whoever fixed BaHaRaD understood exactly what they were doing: an exact arithmetic needs an exact origin, the origin cannot be the thing itself, so you place a computational fiction at a convenient distance and you say in the name that it is one.

Rule Q’s entire content is in that phrase. The project did not know this when the rule was written.

The ḥelek and SECOND were chosen the same way

Chapter 7 noted that `SECOND` is a multiple of 10^{30}, and that the choice makes whole seconds land with thirty trailing base-5 zeros and whole nanoseconds with twenty-one.

The reason is divisibility: pick a constant with many factors and the subdivisions you need come out exact.

The **ḥelek** is the same move with the same reasoning, made around the ninth century. A thousand and eighty parts to the hour, because 1080 has a lot of divisors and lunar arithmetic needs thirds and fifths and eighths of an hour to come out whole.

Both are a designer looking at a unit and asking not **how big should this be** but **what must it divide by**.

The nineteen-year cycle is convergent 6

Chapter 15 showed the mechanism deriving Earth’s grouping cycles and producing 235/19 — the Metonic cycle — as the sixth convergent, from Earth’s two periods with nothing else supplied.

The Hebrew calendar’s nineteen-year intercalation is that cycle. So the mechanism recovers, unaided, the structure this tradition has used for well over a millennium.

INTERPRETATION

As with Meton in chapter 15, this is not anticipation. 235/19 is where the good approximation is, and anyone with an accurate ratio and a motive lands near it.

What it does establish is that the derived/declared distinction of chapter 8 cuts **for** this tradition on this point. The nineteen-year cycle is derivable. The Gregorian leap rule is not. Whatever else is true, the intercalation here is the kind of thing the mechanism finds rather than the kind it has to be told.

Two epochs, and why Rule P exists

The Anno Mundi count is text-dependent. The Masoretic text yields a creation around 3761 BCE. The Septuagint's longer genealogies yield the Byzantine reckoning, 5509 BCE. A gap of 1,748 years, from the same procedure applied to different manuscript traditions.

Those are two profiles in this system's exact sense: same mechanism, different declared constants, and values from one are not comparable to values from the other.

Rule P exists for precisely this — profiles named, versioned, type-bound, and tagged in every serialised form, so that a timestamp from one cannot be silently compared with a timestamp from the other. Failure mode F1 is what happens when they are.

> **THE CONFLICT**
>
> **The instrument judges a tradition, and the author may not.**
>
> **Seder Olam**'s chronology compresses the Persian period. Standard historical reconstruction gives roughly 207 years from the fall of Babylon to Alexander; **Seder Olam** gives roughly 52. Something on the order of 165 years is absent, and the compression is not random — it aligns the timeline with the seventy weeks of Daniel 9.
>
> That is provenance overruled by doctrine. And this system's provenance chain is **machine-readable and re-executable**: the whole point of Rule Q.4 is that you can run the chain and see whether the constant follows from the cited inputs.
>
> Encode **Seder Olam** as a profile and the chain would not close. The instrument would report, mechanically and without comment, that the declared epoch does not follow from the declared sources.
>
> So the artifact is capable of producing a finding about a religious tradition's chronology — and this book's own rules forbid its author from drawing the conclusion that finding invites.

What it changes

The line this chapter has to hold is: **the instrument may expose; the author may not judge.** It sounds like evasion. Here is why it is not.

INTERPRETATION

Start with what the instrument would actually report. Not "this chronology is false." It would report that a particular arithmetic does not close — that given these declared inputs and this stated procedure, you do not arrive at this stated output. That is a narrow, checkable, and entirely mechanical claim.

Everything interesting lies in what follows from it, and none of that is mechanical. Whether **Seder Olam** was doing chronology in the sense a modern historian means. Whether a second-century text that derives dates from scripture read whole is answerable to the standards of a discipline that did not exist. Whether the compression is an error, a deliberate harmonisation, or a genre distinction the question misunderstands.

The author has views on some of that and they are not evidence. This book's rules say no tradition is argued true and none is argued false, and the rule does not get suspended when the artifact hands over something that looks like ammunition.

The discipline is exactly the one chapter 12 built into the type system, applied to the author instead of to a program. `BIG_BANG_CLAIM` is fully available — you can read it, print it, cite it — and it cannot be used as an operand. The **Seder Olam** finding is fully available: it can be computed, reported, and examined. What it cannot do is enter as a premise in an argument this book makes about whether a tradition is correct.

There is a real cost, and pretending otherwise would be its own dishonesty. A reader who wants to know what the author thinks about the Persian period will not find out. That is the price of a rule that also prevents the artifact being used to validate the traditions its author finds congenial — and a rule that only binds in the uncomfortable direction is not a rule.

WHY THIS IS THE HARDEST SENTENCE

Because the alternative is available, respectable, and would make a better chapter.

A book could reasonably say: here is a mechanism for auditing chronologies, here is what it finds, and the finding is what it is. That book would be more useful to some readers and more honest-seeming to others.

It would also have turned the instrument into an apologetic — one pointed away from the author's own commitments rather than toward them, which is the direction that feels like rigour and is not.

WHAT THIS CHAPTER ESTABLISHED

- **Molad tohu** — "the new moon of chaos" — is a computational epoch placed before what it anchors and named for its own emptiness. Rule Q's content, nineteen centuries early, and better named.
- The **ḥelek** was chosen for divisibility exactly as `SECOND` was: not **how big**, but **what must it divide by**.
- The nineteen-year cycle is the Metonic convergent the mechanism derives unaided — so the derived/declared distinction cuts **for** this tradition here.
- Masoretic and Byzantine Anno Mundi are two profiles in this system's exact sense, and Rule P exists for that case.
- **Conflict:** the re-executable provenance chain would report that **Seder Olam**'s chronology does not close — a finding about a religious tradition that the book's rules forbid its author from concluding from.
- **What changes:** the instrument may expose, the author may not judge — the same discipline chapter 12 put in the type system, applied to the author. It has a real cost, and a rule that only binds in the comfortable direction is not a rule.

CHAPTER 20

20

Islamic

Chapter 8 said that every body parameter carries a validity window, and that evaluating outside it warns rather than extrapolating. That looked like ordinary engineering caution. This chapter is about the position it turns out to encode, and about the fact that the artifact took a side in a live metaphysical dispute without noticing.

What the direction holds

> Kalām atomism holds that the world is composed of indivisible atoms and their accidents, and — the part that matters here — that **accidents do not endure for two moments**. What appears as a persisting quality is God re-creating it instant by instant. Time itself is composed of indivisible moments.
>
> Al-Ghazālī, **Tahāfut al-Falāsifa** XVII, argues that "the connection between what is habitually believed to be a cause and what is habitually believed to be an effect is not necessary." Fire does not burn cotton; God creates burning in the cotton at the moment of contact. The two are conjoined constantly, not necessarily.
>
> His term for the regularity is **ʿāda** — custom, habit. Divine custom is utterly reliable in practice, and it grounds ordinary knowledge, but it carries no necessity and God is not bound by it.
>
> Ibn Rushd replies in **Tahāfut al-Tahāfut** that to deny that things have natures from which their effects follow is to deny the possibility of knowledge. If fire has no nature that burns, the word "fire" names nothing, and demonstration collapses into the observation of sequences.

Which rule it illuminates

N1, the refusal to claim the tick is a quantum of time — which chapter 2 declined an Epicurean position with, and declines a Kalām one with equally. And Rule C, on body parameters carrying `rate` and `valid`.

The convergences

A validity window is ʿāda, compiled

This is the convergence, and it is exact enough to be startling.

Consider what a validity window actually asserts. Earth's rotation is measured near J2000 and found to be lengthening at about 1.8 ms per century. The system records the value, the rate, and a window — and **inside** the window it computes confidently, while **outside** it warns rather than extrapolating.

What is the epistemic content of that?

It is: **the regularity has been observed here, and I will assert it here, and I decline to assert that it holds beyond where it was observed.** Not because a different law is expected outside — no alternative is proposed — but because the warrant runs out where the observation does.

> **INTERPRETATION**
>
> That is Ghazālī's position with the theology removed. Practical certainty inside the range of custom; no claim of necessity; no assertion that the regularity is guaranteed to continue.
>
> `UCAL-W0003` — the warning a parameter emits outside its window — is **ʿāda** compiled. Whatever the author believed about causation while writing it, the epistemic posture the code takes is the one Ghazālī argued for.
>
> I want to be careful about how much this is worth. It does not show Ghazālī was right about causation. What it shows is that a working engineer, forced by the problem to be precise about the scope of an empirical claim, arrives at a distinction — **reliable here, not thereby necessary** — that was worked out in Baghdad in the eleventh century for entirely different reasons.

Atomism, declined twice

Chapter 2 declined to call the tick a quantum of time, and noted that asserting otherwise would have joined Epicurus against Aristotle.

The Kalām position is the more thoroughgoing one: not merely that time has least parts, but that persistence itself is not a fact about things — that endurance is re-creation.

N1 declines this too, and it is worth noting what the declining costs. A system whose finest unit is a tick and whose values are integers is **formally** very close to a world of temporal atoms. If the author had wanted a metaphysical warrant for the design, this tradition offers one, fully worked out, with a serious apparatus behind it.

It is refused for the reason chapter 2 gave: the discreteness is a fact about the instrument's resolution, and taking a metaphysical position as a side effect of choosing an integer type is not a respectable way to hold one.

> **THE CONFLICT**
>
> **A guaranteed drift bound is not something this tradition can grant.**
>
> Chapter 15 reported that Earth's `31/128` intercalation drifts one day in 400,000 years, and chapter 16 qualified that: the guarantee holds under parameters that will not hold for 400,000 years.
>
> Under Ghazālī's seventeenth discussion the qualification is far more severe. If an orbital period is a **habit** rather than a nature, then there is no fact about what Earth's orbit will do that could underwrite a guarantee over four hundred millennia. The regularity holds because God customarily maintains it, and the only honest bound would be **until God wills otherwise**.
>
> Which is not a bound. It has no number in it, and a drift-bound parameter that takes it would not select a convergent.
>
> So the mechanism's central promise — **choose the rule that guarantees this accuracy for this long** — presupposes that periods have natures from which their behaviour follows. **The crate's ontology is Rushdian.** It took a side in a dispute that is still live, and it did so silently, as an implementation consequence.

What it changes

Here is the finding, and it is the most peculiar thing Part VI turned up.

> **INTERPRETATION**
>
> The artifact's **ontology** is Rushdian and its **epistemology** is Ghazālian.
>
> What it computes assumes natures. Continued-fraction expansion over an orbital period yields a guarantee about four hundred thousand years, and that guarantee means nothing unless the period is the kind of thing that has a nature from which its future follows.
>
> What it **claims** assumes custom. The validity window says: I assert this where it was observed and not beyond. `UCAL-W0003` is exactly the refusal to project a regularity past its warrant.
>
> So the system computes as though Ibn Rushd were right and reports as though Ghazālī were.

Is that coherent, or merely convenient?

> **INTERPRETATION**
>
> The case for coherence: the two do different jobs. The ontological assumption is **internal to a conditional** — given that the period is thus, the rule drifts thus — and conditionals do not commit you to their antecedents. The epistemic caution is about **asserting** the antecedent, which the system declines to do outside the window. Read that way there is no contradiction, only a careful separation between what follows from what.
>
> The case for convenience: that reading is available after the fact and was not the reason for either choice. The natures came from wanting a guarantee; the caution came from knowing that parameters drift. Nobody weighed them against each other. A consistency discovered afterwards, which nobody designed and which happens to make the design defensible, should be suspected exactly as chapter 10′s flattering sentence should have been.
>
> I think the first reading is right and I am aware that I would. What can be said without prejudice is that the artifact contains a metaphysical commitment its specification never declares, and that Rule F — which requires the **frame** to be declared — has no analogue requiring this to be.
>
> That is a gap in the rules, found by a tradition, and this chapter is the whole record of it.

WHAT THIS CHAPTER ESTABLISHED

- Kalām atomism: indivisible moments, accidents that do not endure, persistence as re-creation. Ghazālī XVII: causal connection is custom, not necessity. Ibn Rushd: deny natures and you deny knowledge.
- A validity window is ʿ**āda** compiled — reliable where observed, no claim of necessity beyond. `UCAL-W0003` is that posture in code.
- N1 declines Kalām atomism as it declined the Epicurean version, and refuses the metaphysical warrant this tradition would have supplied.
- **Conflict:** a guaranteed drift bound over 400,000 years presupposes that periods have natures. Under Ghazālī the only honest bound is "until God wills otherwise", which is not a bound. The crate's ontology is Rushdian and it never argued for it.
- **What changes:** the artifact computes as though Ibn Rushd were right and reports as though Ghazālī were — and the rules require the **frame** to be declared but contain nothing requiring this commitment to be.

CHAPTER 21

21

Patristic and Latin

Chapter 13 borrowed one image from this direction — Basil's road, which is not its own beginning — and used it against Florensky. This chapter reads the direction properly, and it produces the sharpest conflict in Part VI: the profile mechanism, which was built for versioning, turns out to be doing theological work its author did not design it for.

What the direction holds

Creation ex nihilo, including time. Augustine, **City of God** XI.6: the world was made not **in** time but **with** time. Time is among the things created, so there is no interval before creation for the question “what was God doing then?” to be about. The question is malformed rather than hard.

The beginning of time is not a time. Basil, **Hexaemeron** I.6: the beginning of a road is not the road, and the beginning of a house is not the house. A beginning is not a member of the series it begins.

Aevum. Boethius and later Aquinas distinguish three modes of duration: **tempus**, successive time; **aeternitas**, the wholly simultaneous possession of unlimited life; and **aevum**, the mode belonging to created things that do not change successively.

Act and terminus. Aquinas, **Summa Theologiae** I.13.7: when God is said to become Lord of creatures, the relation is real in the creature and not in God. A timeless act can have a temporal terminus, and dating the terminus does not date the act.

Learned ignorance. Nicholas of Cusa, **De docta ignorantia**: the infinite is not reached by any proportion from the finite. Increasing the precision of a finite measure does not approach it, because approach requires proportion and there is none.

Which rule it illuminates

Rule Z, the unsigned domain. Rule Q, the stipulated datum. And the thesis of this book, which Cusanus states first.

The convergences

`UCAL-E0020` is Augustine's answer

Chapter 3 said that asking for an instant before the datum returns an error rather than a negative number, and that the difference is between refusing a question and answering it.

Augustine's answer to **what was God doing before creating heaven and earth** is exactly that structure. He declines the joke ("preparing hell for those who pry") and gives the serious reply: there was no **then**, because time is among the created things. The question presupposes a container that does not exist.

An error is a malformed query. A negative number is a well-formed query with a value on the other side of a chosen origin. The system returns the first, and the fifth-century argument for why is the better one available.

Basil, and Rule Q's actual content

Chapter 13 used the road-and-house image as a diagnosis of what Florensky's §9 lacked. It is worth restating as a positive claim about the datum.

The datum is not the first tick of the universe. It is where counting starts. Those are different sorts of thing, and the difference is not a technicality — it is the whole reason the physical claim had to be held in a separate, non-computable type.

> INTERPRETATION
>
> Basil is doing with an image what chapter 12 does with a type. Both are keeping a beginning from being treated as a member of the series it begins.
>
> The image is more elegant and reaches more people. The type is enforced.
>
> I do not think that ranks them. It marks a difference in what each medium can do, which is the argument of Part VIII arriving early and from an unexpected direction.

Aquinas makes the project coherent under a timeless God

This convergence matters more than it first appears, because it removes what would otherwise be a fatal objection.

If God is timeless, dating divine action looks incoherent: an act with no temporal location cannot be assigned a date. And a system that offers to timestamp anything at all would seem to be committed to denying divine timelessness.

ST I.13.7 dissolves this. The relation is real in the creature and not in God. The **terminus** of a timeless act is a created thing with a location in the created order, and dating the terminus is dating the creature, not the act.

So a calendar can coherently timestamp events in the created order without any commitment about how the acts that produced them are related to time.

Cusanus states the thesis first

De docta ignorantia: no proportion holds between the finite and the infinite, so no increase in finite precision constitutes approach.

That is this book's thesis, arrived at five and a half centuries earlier and stated more cleanly: **a measuring instrument may legitimately point at what it cannot describe, provided it declares that it is only pointing.** The declaring is what learned ignorance means — knowing the character of what you do not know, rather than merely not knowing it.

> **INTERPRETATION**
>
> The contribution this project can claim is not the thesis. Cusanus has it, and Gregory of Nyssa has a version before him.
>
> It is the second clause: that the declaration can be **enforced mechanically** rather than left to the author's discipline. Cusanus maintained learned ignorance by being Cusanus. Chapter 12 maintains it with a type that has no operators.

THE CONFLICT

UC-Θ is heterodox by this standard, and the book must say so plainly.

Creation **ex nihilo** is not a peripheral commitment in this direction. It is held firmly, argued for repeatedly, and defined against precisely the alternative that chapter 12′s unbuilt profile assumes.

UC-Θ posits a datum at **organization** — matter arranged rather than created from nothing, with an origin of ordering that is not an origin of existence. That is, in the terms Augustine and Aquinas use, the Platonic opinion: pre-existent material shaped by a demiurge. It is the position these authors spent centuries refuting.

So the two profiles are not two configurations of one system. They are two cosmologies, and by this direction's standard one of them is orthodox and the other is not.

The book does not get to soften this by observing that UC-Θ is unbuilt. A profile that has been specified is a position that has been taken seriously.

What it changes

INTERPRETATION

Rule P was written for failure mode F1: timestamps shifting when the age constant is revised. Profiles are named, versioned, type-bound, and tagged in every serialised form so that a value from one cannot be silently compared with a value from another.

That is a versioning mechanism. It was designed by an engineer thinking about constant revision.

What this chapter establishes is that it is **also** the mechanism that keeps two incompatible cosmologies from contaminating each other's timestamps — and that this is not an analogy. UC-1 and UC-Θ differ in what the datum **is**, which means a timestamp under one is not the same kind of statement as a timestamp under the other, which is exactly the condition Rule P exists to detect.

The rule is doing theological work. Nobody designed it to.

Two things follow. First, the specification should say so — a mechanism whose scope is wider than its stated purpose is under-documented, and the gap is the same kind chapter 20 found in Rule F.

Second, and less comfortably: the fact that a versioning rule turned out to be adequate for this does not mean it is **sufficient** for it. Rule P prevents silent comparison. It does not require a profile to declare its cosmological commitments, and UC-Θ's commitments are exactly what a reader would need to see.

WHAT THIS CHAPTER ESTABLISHED

- Creation **ex nihilo** including time; the beginning of time is not a time; **aevum** as a third mode of duration; act and terminus; learned ignorance.
- `UCAL-E0020` is Augustine's move — a malformed question refused, not answered with a value on the far side of an origin.
- Aquinas' **ST** I.13.7 removes what would otherwise be fatal: dating a terminus in the created order commits you to nothing about how a timeless act relates to time.
- Cusanus states this book's thesis first. The contribution is not the thesis but the second clause — that the declaration can be enforced mechanically.
- **Conflict:** UC-Θ assumes organization rather than creation from nothing, which is the position this direction was built to refute. It is heterodox by this standard and the book says so.
- **What changes:** Rule P was written for versioning and turns out to be what keeps two cosmologies apart — doing theological work nobody designed it for, and insufficient for the job, since it never requires a profile to declare what it assumes.

CHAPTER 22

22

Orthodox

This direction gives the book the strongest licence its central sentence receives anywhere, the technical term the research had been circling since Cusanus, and a conflict that puts Rule N on the wrong side of a dispute involving the two thinkers this book most depends on.

What the direction holds

> **Essence and energies.** Gregory Palamas, defending hesychast practice, distinguishes God's **essence** — utterly unknowable, imparticipable — from the **energies**, the uncreated activities by which God is genuinely known and participated. The distinction is real and not merely conceptual: what is known is truly God, and the essence remains beyond knowing.
>
> **Diastema.** Gregory of Nyssa, against Eunomius, makes διάστημα — interval, extension, the spread between before and after — the mark of createdness. Everything created is διαστηματικός; God is ἀδιάστατος, without interval. The gap between creature and Creator is not one of degree along a scale but of **kind**: there is no interval in God for a measure to measure.
>
> **Three modes.** Maximus the Confessor distinguishes ἀϊδιότης, the eternity proper to God; αἰών, the mode of created beings in their fixed existence, a "before" that is not a "when"; and χρόνος, measured successive time.
>
> **The name.** The имяславие controversy of the early twentieth century turned on whether the divine name **participates** in what it names — whether "the Name of God is God" — or is a designation. The Russian Synod condemned the affirmative position in
> 1913. Florensky and Losev defended it.

Which rule it illuminates

The thesis itself. Rule N, on names as display-only. UC-Θ. And §8.6′s `LegacyCalendar`.

The convergences

Palamas is the licence

The book's thesis is that an instrument may point at what it cannot describe, provided it declares that it is only pointing.

Palamas' distinction is the strongest theological warrant that sentence receives from any direction surveyed. Essence unknowable; energies manifest and genuinely known. The structure permits real knowledge of something whose essence is beyond knowing — which is precisely what "pointing without describing" requires in order not to be empty.

> **INTERPRETATION**
>
> I want to be careful here, because this is the place where a book like this most wants to overreach.
>
> Palamas is not talking about calendars, and the essence/energies distinction is a claim about God, not a general licence for instruments that gesture at what they cannot measure. Taking it as one would be quarrying — using a tradition's vocabulary to dignify a much smaller point.
>
> What can honestly be said: the **form** of the thesis has a precedent here in which the two clauses are held together rigorously rather than as a rhetorical balance. That is worth something, and it is less than a licence.

Diastema is the word the research was looking for
This is the most useful single term Part VI produced.

> If διάστημα is the mark of createdness, then anything that measures interval is by definition an instrument of the created order. Not accidentally, not because of a technical limitation, but because interval is what created things have and what God does not.

Everything this system measures is διάστημα. The tick count is an interval. A `Delta` is an interval. A `Window` is an interval with uncertainty. The whole apparatus, top to bottom, measures the spread between before and after.

> **INTERPRETATION**
>
> So the limitation stated in this book's title is not a defect in the instrument. It is a **doctrine** about what the instrument is measuring.
>
> A calendar cannot reach past the created order, and the reason is not that calendars are crude. It is that a calendar measures interval, interval is the mark of the created, and there is no interval in what lies beyond it — nothing for the measure to be a measure **of**.
>
> Cusanus said no proportion holds between finite and infinite. Nyssa says why: they are not two magnitudes of different size but two conditions, one with διάστημα and one without.
>
> Chapter 30 owes this chapter a debt. "An instrument for the immeasurable" reads as a paradox in the title; here it stops being one and becomes a statement about categories.

Maximus supplies what UC-Θ needs

Chapter 12 described UC-Θ as requiring a "before" that is not a "when" — an origin of ordering that is not itself a moment in the ordering.

αἰών is exactly that. It is not eternity and not time; it is the mode of created being that has a beginning without that beginning being a temporal position.

Whether UC-Θ can actually be built on it is another question, and chapter 21′s conflict stands regardless. But the concept UC-Θ was reaching for exists, is technical, and is seventh-century.

The Julian retention defends `LegacyCalendar`

This is the convergence with the most practical force in the book.

Chapter 8 classified `earth-civil` as legacy — declared tables, four items of arbitrary content, a leap rule that is not a convergent — and chapter 10 established just how far `97/400` sits from the derived ladder.

The Orthodox churches that retain the Julian calendar retain a rule, `1/4`, that is the **crudest** convergent available: one day of drift in 128 years. Every church using it knows this. It has been debated for a century, and the calendar has been retained anyway, for reasons of communion and continuity that have nothing to do with accuracy against the tropical year.

INTERPRETATION

That is the strongest available argument that **derivation is not a calendar's only criterion**, and it is an argument from practice rather than from theory.

A calendar is a thing a community keeps together. Accuracy is one property of it. Not breaking communion is another, and there is no arithmetic that ranks them.

So `LegacyCalendar` is not a holding pen for calendars that failed to be derived. It is a category for calendars whose authority comes from somewhere the mechanism cannot reach — and chapter 8′s insistence that "legacy" is a classification rather than a judgement gets its clearest vindication here.

THE CONFLICT

Two, and the second is the one that hurts.

The **ex nihilo** conflict of chapter 21 applies here in full. UC-Θ is heterodox by this direction's standard too, and for the same reasons.

The second is Rule N. It says: **the canonical identity of a tier is its exponent; names are display-only.** Nothing in the system decides behaviour from a name; names live in a locale table alongside their Russian translations, and the tier is the exponent.

That is a position on the relationship between a name and what it names. It says the name is a label attached to an identity that exists independently of it.

In the имяславие dispute, that is the Synodal position of 1913 — the one Florensky and Losev opposed. Rule N sides against them.

Chapter 13′s cautionary tale is built on Florensky. Chapter 25 depends on Losev. These are the two thinkers this book most needs, and on the question of what a name **is**, the artifact holds the view they were persecuted for opposing.

What it changes

INTERPRETATION

Two things, and the second is unusual because it is falsifiable.

First, the title stops being a paradox. Διάστημα makes "an instrument for the immeasurable" a statement about what kind of thing an interval-measure is. Chapter 30 should be written from here rather than from the engineering side.

Second: **the Rule N conflict can be settled in code rather than on the page.**

Rule N holds that a tier's identity is its exponent and its name is display. If that is true, then no behaviour anywhere in the system should ever depend on a name. The lint and the tests currently assume this.

But it is an empirical claim about the software, and it can fail. If some future feature makes a tier name load-bearing — if a name ever determines behaviour rather than merely rendering it — then Florensky and Losev win the argument **on the machine** rather than in the commentary, and Rule N would have to be amended rather than defended.

That is the strangest thing Part VI produced: a hundred-year-old theological dispute with a test condition in a Rust workspace. I do not think it is trivial. Rule N was written to prevent a display concern from acquiring semantics, and the имяславие position is precisely that display concerns **do** acquire semantics when what is named is the sort of thing a name can participate in.

A tier is not that sort of thing, and the analogy should not be pressed further than the shared structure warrants. But the test condition is real, and if it ever fires this chapter has to be rewritten.

WHAT THIS CHAPTER ESTABLISHED

- Essence and energies; διάστημα as the mark of createdness; ἀϊδιότης / αἰών / χρόνος; the name as participating or designating.
- Palamas gives the thesis its strongest precedent — with the caution that it is a claim about God, not a general licence for instruments.
- Διάστημα is the term the research was circling: everything this system measures is interval, and interval is what created things have. The title's paradox dissolves into a statement about categories.
- The Julian retention — keeping the **crudest** convergent knowingly, for a century of debate — is the strongest available defence of `LegacyCalendar` as a classification rather than a judgement.
- **Conflict:** Rule N sides with the 1913 Synodal position against Florensky and Losev, the two thinkers this book most depends on.
- **What changes:** that conflict is falsifiable in code. If a tier name ever becomes load-bearing, they win on the machine and Rule N is amended rather than defended.

CHAPTER 23

23

Latter-day Saint

Rule K holds that reckoning is body-relative and that Earth is an instance rather than a template. Chapter 14 presented that as a piece of software architecture with a philosophical shape.

It is also a doctrinal proposition, stated in 1843, in a text that says it more directly than this project's own specification manages.

What the direction holds

> **Reckoning is body-relative.** Doctrine and Covenants 130:4–5: “In answer to the question — Is not God’s time, angel’s time, prophet’s time, and man’s time, according to the planet on which they reside? I answer, Yes.”
>
> **A declared bridge, with a tag on each side.** Abraham 3:4, of Kolob: “one revolution was a day unto the Lord, after his manner of reckoning, it being one thousand years according to the time appointed unto that whereon thou standest.”
>
> **Law with bounds and conditions.** D&C 88:38: “unto every kingdom is given a law; and unto every law there are certain bounds also and conditions.”
>
> **Organization rather than creation from nothing.** The King Follett discourse of 1844 argues that the word rendered “created” means to organise — that matter is co-eternal with God and is arranged rather than brought from nothing.
>
> **What is numbered, and to whom.** Moses 1:37: “the heavens, they are many, and they cannot be numbered unto man; but they are numbered unto me.” Alma 40:8: “all is as one day with God, and time only is measured unto men.”

Which rule it illuminates

Rule K, one derivation mechanism with Earth as an instance. Rule Y, the declared bridge. Rule C, validity windows. And the datum.

The convergences

D&C 130:4–5 is Rule K's thesis, stated in 1843

Rule K exists to prevent failure mode F9 — Earth becoming the template rather than an instance. Chapter 14 described the drift it guards against and the test that catches it.

The 1843 text states the underlying claim without the software: time is **according to the planet on which they reside**. Not that Earth’s reckoning is primary and others derive from it, and not that there is a true reckoning somewhere that local ones approximate. Reckoning is indexed to a body, and the indexing is the whole story.

> **INTERPRETATION**
>
> This is the most direct statement of Rule K's premise found in any direction surveyed, and it predates the software by 183 years.
>
> What it does not supply is the mechanism. "According to the planet on which they reside" is a proposition; `derive_leap_rule` is a procedure. The convergence is at the level of the claim, not the method, and the book should not inflate it past that.

Abraham 3:4 is a bridge constant with a profile tag on each side

This one is structurally precise enough to be worth laying out.

Read the verse as a conversion statement:

element	what it corresponds to
one revolution = one day unto the Lord	the source quantity, in the source body's units
"after his manner of reckoning"	the source profile tag
one thousand years	the target quantity
"according to the time appointed unto that whereon thou standest"	the target profile tag

That is a declared bridge constant with both sides tagged — which is exactly Rule Y's requirement and exactly Rule P's. The verse does not say "a day is a thousand years" and leave you to guess whose day and whose years. It names both frames.

> **INTERPRETATION**
>
> Compare how often ordinary technical writing fails at this. "The timeout is 30" — thirty what, measured by whom? A ratio without both frames named is not a conversion; it is a number waiting to be misread.
>
> The verse names both. Whatever one thinks of its source, its **form** is the form this project's Rule Y exists to require.

D&C 88:38 is a validity window

“Unto every law there are certain bounds also and conditions.”

Chapter 8 said every body parameter carries an epoch, a rate, and a window of validity, and that evaluating outside the window warns rather than extrapolating. Chapter 20 found that posture to be Ghazālī’s ‘**āda** with the theology removed.

Here it is again, in a different tradition, as a general proposition about law: a regularity holds **within bounds and conditions**, and the bounds are part of the law rather than a caveat on it.

What cannot be numbered, and what can

Moses 1:37 and Alma 40:8 together do something useful for this book’s central worry.

Moses 1:37 says the **heavens** cannot be numbered unto man. It does not say durations cannot. Alma 40:8 says time “is measured unto men” — measuring time is presented as what creatures do, in contrast to the divine mode where all is as one day.

> **INTERPRETATION**
>
> So this direction, like chapter 22′s διάστημα, locates time-measurement firmly on the creaturely side and treats that as its proper place rather than as a limitation to be overcome.
>
> A calendar that counts ticks is doing the thing given to men to do. That is a modest claim and it is the one the book has been circling since the preface.

THE CONFLICT

Two, and the second relocates rather than resolves.

Revelation has no vocabulary in `Determination`. The system records how a parameter was determined — measured, derived, cited to a published source with an uncertainty. A revealed ratio does not fit any of those categories.

Abraham 3:4 can enter as a **declared bridge constant**, because Rule Y's requirement is that a bridge be declared and tagged, and the verse is. It cannot enter as a Rule J anchor, because anchors must be cited to a determination with a stated uncertainty window, and "revealed" is not a determination method the schema knows.

That is a real exclusion and it is not neutral. The system is hospitable to this tradition's **ratios** and closed to its **epistemology**.

Kolob is a body where Rule K.5 permits only a grid. Abraham 3 describes Kolob as the governing body — "nearest unto the throne of God", the one after whose reckoning the others are set. The hierarchy is real and it is centred on a **place**.

Rule K.5 permits no privileged body. The universal ladder is the powers 5^{5k}, and chapter 14 was explicit that nothing on it is any body's.

What it changes

INTERPRETATION

The Kolob conflict is better stated as **the privilege is located differently**, and the restatement is not a softening — it is more accurate and less flattering.

Chapter 14 declined the grand Copernican reading and said the privilege was **relocated, not abolished**: from a body to a power-of-5 grid. At the time that was a caution against overclaiming.

This chapter shows what it means. Abraham 3 has a governing body at the centre of a hierarchy of reckonings. UC-1 has an abstract ladder. Both are hierarchies with a privileged referent; they differ in **what sort of thing** sits at the privileged point.

And a grid is not neutral merely by being abstract. It was chosen — base five, anchored at 5^{60}, with the beat placed where a human can notice a duration. Chapter 4 admitted that last part as "a concession to the reader". Someone reading Abraham 3 could fairly observe that a ladder whose reference rung is set to human perception is not obviously less parochial than a ladder whose reference is a named star; it is parochial about a different thing.

I do not think that is right, and I think the reason is that a power ladder can be re-anchored without changing any arithmetic while a governing body cannot be replaced without changing the cosmology. But the objection is good enough that the book has to make the argument rather than assume it, and until this chapter it had assumed it.

WHAT THIS CHAPTER ESTABLISHED

- D&C 130:4–5 states Rule K's premise in 1843: reckoning is according to the planet on which one resides.
- Abraham 3:4 is a declared bridge constant with a profile tag on **both** sides — the form Rule Y requires and ordinary technical writing routinely omits.
- D&C 88:38′s "bounds also and conditions" is a validity window as a general proposition about law.
- Moses 1:37 and Alma 40:8 locate time-measurement on the creaturely side and treat that as its proper place — the same move as chapter 22′s διάστημα.
- **Conflict:** revelation has no vocabulary in `Determination`, so a revealed ratio may be a bridge constant and never an anchor. And Kolob is a governing **body** where Rule K.5 permits only a grid.
- **What changes:** the privilege was relocated, not abolished — and a ladder anchored where humans notice durations is parochial about something too. The book has to argue that, not assume it.

CHAPTER 24

24

Modern philosophy

Chapter 11 borrowed the First Antinomy and chapter 12 borrowed the constitutive/regulative distinction. This chapter reads Kant properly, which means reading the parts that do not help.

What the direction holds

> **Absolute against relational.** Newton's **Principia** declares absolute time, "of itself, and from its own nature, flowing equably without relation to anything

> external." Leibniz replies in the correspondence with Clarke that time is an order of successions and nothing more — that a universe shifted uniformly in time would be indiscernible from this one, and indiscernibles are identical.
>
> **A- and B-series.** McTaggart distinguishes the A-series — past, present, future, which shift as time passes — from the B-series — earlier-than and later-than, which do not. He argues the A-series is contradictory and the B-series alone cannot constitute time.
>
> **Kant's forms.** Time is a pure form of sensible intuition: not something perceived but the form under which anything is perceived. It is **empirically real** — every object of experience is in time — and **transcendentally ideal** — it is not a property of things as they are in themselves.
>
> **The First Antinomy.** Thesis: the world has a beginning in time. Antithesis: it does not. Both proofs valid; both conclusions false, because both assume the world-series is a completed given totality.
>
> **Number as schema.** In the Schematism, number is the schema of magnitude: the successive addition of homogeneous units, the representation of a procedure for synthesising a quantity.
>
> **Transcendental illusion.** The illusion is not a mistake to be corrected. It is natural and unavoidable, and it persists after diagnosis — "as the astronomer cannot prevent the moon from appearing larger at its rising, although he is not deceived by this illusion."

Which rule it illuminates

Rule Q and `UCAL-E0025`. N1. Rule F, the declared frame. Rule G, on number.

The convergences

The First Antinomy is Rule Q's fourth reason

Chapter 11 gave this in full: the specification offers three reasons the datum cannot be measured, and Kant supplies a fourth it does not state — the question presupposes a completed totality that is not a possible object of knowledge.

What chapter 11 did not say is where the antithesis argument lands.

> Kant’s argument against the thesis runs: suppose the world began. Then there was an empty time before it. But no part of empty time has any distinguishing condition of existence rather than non-existence — one moment of nothing is indistinguishable from another — so no reason could determine the world to begin **then** rather than at some other moment. Therefore it did not begin.

That argument is why Rule Z is right to treat a “before” as malformed rather than as a region with values in it. An empty time before the datum would have no distinguishable parts, so a tick count within it would be a coordinate on a structure with no distinctions to index. `UCAL-E0020` refuses to produce one.

Quanta continua gives N1 a better defence

Chapter 2 declined to claim the tick is a quantum of time and gave a procedural reason: the discreteness is a fact about the instrument, and taking a metaphysical position as a side effect of choosing an integer type is not respectable.

Kant supplies a stronger reason. Space and time are **quanta continua**: magnitudes in which no part is the smallest, and the parts are possible only through limitation. The instant is a limit, not a part — the same position chapter 18 found in Aristotle, now argued from the form of intuition rather than from the analysis of motion.

So N1′s refusal is not merely modest. It declines a claim that two of the most careful treatments of time in the Western tradition independently argue is false.

Constitutive and regulative, enforced

Chapter 12 made this its closing move and it is worth restating precisely.

The datum is a regulative posit: **count from here**. The physical claim is constitutive-looking: **this is where time began**. Kant’s illusion is the slide from the first to the second.

`UCAL-E0025` and the compile-fail tests make the slide impossible in the arithmetic. Kant policed the boundary by argument and vigilance; here it is policed by a type with no operators.

THE CONFLICT

Two, and the first cuts at chapter 18′s finding from the other side.

Kant's number is monadic. In the Schematism, number is the schema of magnitude — the successive addition of homogeneous units. That is exactly Plotinus' μοναδικὸς ἀριθμός, the counting-number, and it is exactly what Plotinus says is **not** the number that is constitutive of being.

Chapter 18 found that Rule G asserts an identity between the number counted and the number by which we count, and that the Greeks would deny it. Kant denies it too, from the opposite direction: for him there is only the monadic sense, and the substantial sense is not a further kind of number but a confusion.

So Rule G is caught between two rejections. Plotinus says the identity collapses a distinction that matters; Kant says one of the two terms does not exist. The rule survives by declining to be a claim about number at all — but it is stated as one.

UC-Θ asserts a first term. The unbuilt profile posits that time begins to be at organization. That is the thesis of the First Antinomy, and Kant's whole point is that it cannot be asserted — not that it is false in a way the antithesis is true, but that asserting either is a transcendental error.

Chapter 21 found UC-Θ heterodox by the Patristic standard. Kant finds it **ill-formed**, which is worse, and he finds UC-1′s stipulation acceptable for exactly the reason UC-Θ's assertion is not: a stipulated origin claims nothing about the series.

What it changes

INTERPRETATION

Transcendental illusion is incurable, and that is what chapter 12′s achievement is bounded by.

The astronomer knows the moon is not larger at the horizon. He sees it larger anyway. Diagnosis does not dissolve the appearance; it only stops him being deceived.

`UCAL-E0025` is diagnosis with teeth. It stops the claim entering the arithmetic — which is a real thing to have stopped, and it is not the same as stopping the illusion. Every reader who sees a 61-digit integer will read it as a fact about being, because that is what a very precise number looks like. No type prevents that. Nothing prevents that.

So the honest form of the book's claim is narrower than Part VIII would like: the instrument contains the illusion rather than curing it. It refuses to compute with a claim it cannot help but suggest.

Chapter 32 is that, and this chapter is what earns it.

WHAT THIS CHAPTER ESTABLISHED

- Newton's absolute time against Leibniz's order of successions; McTaggart's two series; Kant's time as a form of intuition, empirically real and transcendentally ideal.
- The First Antinomy supplies Rule Q's unstated fourth reason, and its antithesis argument is why an empty "before" has no distinguishable parts to index.
- **Quanta continua** gives N1 a stronger defence than modesty: the instant is a limit, not a part.
- **Conflict:** Kant's number is monadic, so Rule G is rejected from both sides — Plotinus says it collapses a distinction, Kant says one of its terms does not exist.
- **Conflict:** UC-Θ asserts a first term of the series, which the First Antinomy holds cannot be asserted at all. Ill-formed, not merely heterodox.
- **What changes:** the illusion is incurable, so `UCAL-E0025` contains rather than cures. The instrument refuses to compute with a claim it cannot help but suggest.

CHAPTER 25

25

Russian

Chapter 13 used Florensky as a cautionary tale. That was one text and one failure, and it would be a poor account of a tradition to leave it there.

This is the direction the project turns out to be closest to, and the closeness is uncomfortable: it supplies the best frame for what the artifact does, and it holds two positions the artifact declines — one of them argued by the man chapter 13 is built on.

What the direction holds

> **Arithmology.** Nikolai Bugaev proposed discontinuity as a worldview: that alongside analysis, which studies continuous functions, there is **arithmology**, which studies discontinuous ones — and that the two together constitute the completeness of mathematical knowledge. Not a rival to analysis; a complement.
>
> **Number as constitutive, and the name as лик.** Alexei Losev, in **Философия имени** and **Диалектические основы математики**, treats number as the constructive basis of every thing, and the name not as a label but as the **лик** — the countenance — through which a thing is present.
>
> **The unknowable as ground.** Semyon Frank, **Непостижимое**: the unknowable is not a residue left over after knowledge, not the part we have failed to reach. It is the ground on which knowing stands.
>
> **Total addressability of the past.** Nikolai Fyodorov's **Философия общего дела** makes the recovery of the past a moral obligation — the common task being, literally, the restoration of all who have lived.
>
> **Process-relative time.** Vladimir Vernadsky argues that biological, geological and physical processes have their own times and are not adequately described by one external measure. Alexander Chizhevsky dates terrestrial history by the solar cycle.

Which rule it illuminates

N1. Rule N. Rule G. Rule K.

The convergences

Fyodorov explains why arbitrary-tick precision matters

Chapter 4 said that writing all the digits pinpoints one tick, and presented it as a property of the notation. It is, and the question it leaves is **why that would matter**.

The ordinary answer is completeness — a system that can address any instant is tidier than one that cannot. That answer is thin.

> **INTERPRETATION**
>
> Fyodorov gives a better one, and it is startling in this context. If the past is something owed recovery rather than merely recorded, then **addressability** is not a convenience. To be unable to name a moment is to have lost it in a way that matters.
>
> Rule G's requirement that a full-precision timestamp pinpoints exactly one tick, with no ambiguity anywhere in a 10^{103}-year domain, is — read through him — a refusal to let any moment be unnameable.
>
> I am not endorsing the Common Task, and the project was not built with it in mind. What Fyodorov supplies is the only frame found in this survey under which arbitrary-precision addressability is a **value** rather than a feature. That is worth recording even by someone who does not share the metaphysics.

Vernadsky and Chizhevsky are Rule K's unacknowledged ancestry

Rule K holds that reckoning is body-relative and that there is no privileged reckoning that others approximate.

Vernadsky's claim that processes have their own times, and Chizhevsky's practice of dating terrestrial history by a solar period rather than a terrestrial one, are the same move made a century earlier and for scientific rather than architectural reasons.

Chizhevsky in particular is doing what chapter 15 does: taking a period that belongs to a **different body** and using it as the frame for events on this one. Titan's year is Saturn's orbit; Chizhevsky's history is indexed to the Sun's cycle.

Frank states the thesis at book length

Непостижимое arrives at this book's central sentence by the same route Cusanus took — and independently, six centuries later.

The unknowable is not what is left over. It is the ground. Which means an instrument that declares what it cannot describe is not confessing a shortfall; it is acknowledging the condition under which it works at all.

THE CONFLICT

Two, and both come from inside the family.

Bugaev proposes discreteness as a worldview, and N1 declines it.

This is the friendliest possible conflict and it is still a conflict. Bugaev is not a crank asserting temporal atoms; he is a number theorist arguing that discontinuity is a legitimate and complementary mode of mathematical description, explicitly **not** exclusive of continuity.

N1 declines even that. The tick is the instrument's resolution floor, and the system makes no claim that discreteness is a mode in which the world is describable. That is a narrower refusal than the one aimed at Epicurus or the Kalām, and it is aimed at the closest kin — at a position that would have cost almost nothing to accept and that offers the artifact a respectable mathematical home.

It is refused because "almost nothing" is not nothing, and because a system that adopts a worldview because the worldview flatters its data structures has done the thing chapter 10 is about.

Losev's имя makes the name constitutive; Rule N makes it a locale-table entry.

Chapter 22 recorded this as the имяславие conflict and located Rule N with the 1913 Synodal position. Here it is in its philosophical rather than its ecclesiastical form: for Losev, number and name are how a thing is constituted and made present. A name is not attached to an identity; it is the identity's countenance.

Rule N says the canonical identity of a tier is its exponent and the name is display. There is no way to hold both.

What it changes

INTERPRETATION

Chapter 13 read **Мнимости** as precedent and cautionary tale in one text, and said the distance between §7 and §9 is exactly one move. This chapter supplies what that reading was missing: the context in which the move was not obviously a move.

For Losev — and Florensky agreed with him, publicly, at cost — a formal structure **can** be constitutive of what it describes. Number is not a description laid over being; it is how being is articulated. If you hold that, then reading a mathematical artifact as designating something real is not a category error. It is the expected case.

So §9 is not carelessness. It is what follows from a position Florensky held on principle and defended against his own church.

That makes the cautionary tale harder and more useful. The lesson is not **do not be careless**. It is: **a rule against reading formal artifacts as physical facts is a substantive metaphysical commitment, and this project made it without argument.**

Rule M — mark every interpretive claim — presupposes that there is a clean line between a structure and a reading of it. Losev denies there is. The artifact assumes there is, the book is built on the assumption, and until this chapter neither had noticed it was an assumption.

I still think the line is there. But chapter 13′s confidence that Florensky simply lacked a rule now looks like the confidence of someone who has not noticed his own rule is contested.

WHAT THIS CHAPTER ESTABLISHED

- Arithmology as a complement to analysis; number and name as constitutive; the unknowable as ground rather than residue; total addressability as obligation; process-relative time.
- Fyodorov supplies the only frame in this survey under which arbitrary-precision addressability is a **value** — a refusal to let any moment be unnameable.
- Vernadsky and Chizhevsky are Rule K's unacknowledged ancestry: process-relative time, and dating one body's history by another's period.
- Frank reaches the book's thesis independently — the unknowable as the ground on which knowing stands, not the residue left after it.
- **Conflict:** Bugaev's discreteness-as-worldview is declined even though it is complementary, cheap, and offered by the closest kin — because a system that adopts a worldview flattering its data structures has done chapter 10′s error.
- **What changes:** Losev makes the name constitutive, so §9 was not carelessness but principle. A rule against reading formal artifacts as physical facts is itself a metaphysical commitment, and this project made it without argument.

CHAPTER 26

26

Method

Eight chapters have read eight directions against one artifact. This one reads the procedure that made that possible, because the procedure is a position and it has not been argued for.

What the direction holds

> **Comparison requires a fixed point.** To compare is to hold something constant and let something else vary. What is held constant is not neutral: it is the frame in which the comparison's results become statements.
>
> **Doxography is not argument.** A list of what thinkers held is not a philosophical work. The ancient doxographers preserved a great deal and settled nothing, and the form's characteristic failure is to present positions as though enumeration were analysis.
>
> **Primary texts do not reduce to summaries.** Jacob Klein's **Greek Mathematical Thought and the Origin of Algebra** turns on a distinction — that **arithmos** means a definite number of definite things, not a number in the modern sense — which is invisible in any summary and which changes what a whole tradition's texts are about.
>
> Richard Sorabji's **Time, Creation and the Continuum** does the same service for the ancient and medieval debates: it makes Greek, Patristic, Jewish and Islamic arguments about time legible as one continuing conversation rather than four separate literatures.

Which rule it illuminates

Rule B, the bibliography. Rule C, the conflict requirement. And this book's own procedure.

The convergences

Klein makes Rule G's problem statable

Chapter 18 found that Rule G asserts an identity between the number counted and the number by which we count, and that Aristotle and Plotinus would deny it. Chapter 24 found Kant denying it from the other side.

That finding is only available because Klein made the distinction visible. Without **arithmos** as "a definite number of definite things," the Greek texts read as though they were talking about numbers in the modern sense, and the eidetic/monadic distinction collapses into a terminological curiosity.

INTERPRETATION

This is worth stating plainly because it is the strongest argument for Rule B, which requires every primary text to be consulted in a named edition and forbids secondary summaries from standing in.

A summary of Plotinus would have said “he distinguishes two kinds of number.” That is accurate and it is useless. What produced chapter 18′s conflict was the specific content of the distinction, which is only in the text and the scholarship that unlocks it.

Rule B is not academic decoration. It is the difference between a book that could find that conflict and one that could not.

Sorabji makes four directions into one field

Chapters 18, 19, 20 and 21 cover Greek, Jewish, Islamic and Patristic material. Read separately they are four literatures. Read through Sorabji they are one argument, with the same questions passing between them across languages and centuries: whether time began, whether it is continuous, whether the instant is a part.

That is why the Kalām atomism of chapter 20 could be set against the Epicurean atomism of chapter 18 as versions of one position rather than as unrelated coincidences.

THE CONFLICT

The method makes the code the invariant and the traditions the variables.

Every chapter of Part VI has the same shape: here is a direction, here is the artifact it illuminates, here is where they converge, here is where they conflict, here is what it changes **about how to read the artifact**.

The artifact is what is held constant. The traditions are what vary against it.

That is a choice, and it is not the obvious one. The reverse book is entirely available: take a tradition as the fixed point and ask what this artifact looks like from inside it — what a Palamite, or a Kalām theologian, or a Kantian would build. That book would have different findings, and some of them would be findings this one structurally cannot reach.

Holding the code constant treats it as the thing with the stable identity and the traditions as perspectives on it. That is a metaphysical position about what sort of object each is, and **the book adopted it without argument** — in the same way chapter 20 found the artifact adopting a Rushdian ontology without argument, and chapter 25 found it assuming a clean line between structure and reading.

Three times now, in one part.

What it changes

INTERPRETATION

The book must disclose its own procedure as a position rather than presenting it as a neutral frame. That is the whole reason this chapter exists rather than being folded into the preface, and it is the last and most self-referential application of Rule M.

Rule M says: mark every interpretive claim, and let no factual claim depend on one. Chapter 12 put that discipline into a type. Chapter 19 applied it to the author, who may compute a finding about a tradition's chronology and may not conclude from it.

Here it turns on the method itself. The comparative frame is an interpretive choice, it is doing work throughout Part VI, and by the book's own rule it should have been marked from the first page rather than disclosed on the last page of the part.

So this chapter is a correction as much as a conclusion.

What the reverse book would find

Worth sketching, since naming a limitation without indicating its shape is cheap.

A book with a tradition as the fixed point would ask whether the artifact is **adequate** to that tradition's account of time — not merely whether the two converge. Chapter 22 would become: does an interval-measure built on διάστημα actually respect the essence/energies distinction, or does it flatten it? Chapter 20 would become: what would a calendar built on ʿ**āda** look like, and would it have drift bounds at all?

Those are better questions than this book asks, and they are unreachable from here. They are unreachable because the artifact is fixed, and asking them would require letting it vary.

WHY THE CHOICE WAS STILL RIGHT

Because the alternative book cannot be written honestly by this author.

A book that takes a tradition as its fixed point is a book written from inside it, or else it is a book that pretends to a standpoint it does not occupy. This author has one artifact he built and understands, and nine directions he has read. Holding the artifact constant is the only frame in which he is competent across the whole survey.

That is a reason. It is not a justification of the frame as neutral, and it should not be read as one.

WHAT THIS CHAPTER ESTABLISHED

- Comparison requires a fixed point, and the fixed point is not neutral; doxography is not argument; primary texts do not reduce to summaries.
- Klein's **arithmos** is what made chapter 18′s conflict findable — the strongest argument for Rule B's insistence on primary texts in named editions.
- Sorabji turns four literatures into one conversation, which is why Kalām and Epicurean atomism could be read as versions of one position.
- **Conflict:** the method makes the code the invariant and the traditions the variables. That is a metaphysical choice, adopted without argument — the third such in this part.
- **What changes:** the book must disclose its procedure as a position rather than a neutral frame. The last and most self-referential application of Rule M, and a correction as much as a conclusion.
- The reverse book — tradition as fixed point — would ask whether the artifact is **adequate** to an account of time, not merely whether it converges. Better questions, unreachable from here.

PART VII

The instrument as research tool

What a program can do that an argument cannot: it can be run against material its author did not choose, and return an answer he did not want.

CHAPTER 27

27

Six samples

What a program can do that an argument cannot

An argument is addressed. It is made by someone, to someone, about something, and its force depends on the person receiving it being willing to follow. That is not a defect — it is what arguments are — but it has two consequences that matter here.

An argument cannot be **run against material its author did not choose**. It can be extended to new cases, but the extending is done by the author, who selects the cases.

And an argument cannot **return an answer its author did not want**. It can fail to persuade; it can be refuted by someone else. It cannot, on its own, produce a conclusion its maker was not looking for.

A program can do both. Chapter 10 is the book's evidence for the second — the mechanism contradicted a claim its author had published twice. This chapter is the evidence for the first: six samples, run against material the project was not built for, each producing an artifact you can open.

> **EVERYTHING HERE IS GENERATED**
>
> Every number in this chapter comes from a file under `assets/output/`, produced by `samples/run-samples.py` against the commit in `PINNED.md`. Nothing is illustrative and nothing was typed by hand.
>
> Where a sample produced a weaker result than hoped, chapter 28 says so.

S1 — comparative chronology on one axis

Four declared epochs, each converted to absolute ticks.

assets/output/S1-chronologies.txt

```
epoch                         civil (Julian)    ticks since datum
--------------------------------------------------------------------------
Seder Olam (Masoretic AM 1)   3761 BC-10-07
807020180242960413984034848...
Byzantine AM 1                5509 BC-09-01
807020077918219588719351401...
Ussher                        4004 BC-10-23
807020166021508493862768038...
Julian Day epoch              4713 BC-01-01
807020136627266382332510054...

Spread between the widest pair:
  1,748.1 years
```

The four differ by 1,748 years, and the difference is not error. Byzantine reckoning follows the Septuagint's longer genealogies; the Masoretic text gives shorter ones. Ussher and Scaliger are working from the same materials with different methods.

> **INTERPRETATION**
>
> What the sample demonstrates is not that one is right. It is that **putting them on one axis at all** requires a mechanism that keeps them apart.
>
> Each row is a different profile. The arithmetic within a row is meaningful; the arithmetic across rows is not, because the rows are not measuring from the same declared origin. Rule P is what makes the table legitimate — every value carries its profile, so the comparison is visible rather than accidental.
>
> Without that, a chronological table is a set of numbers that look comparable and are not. This is the ordinary condition of chronological tables.

S2 — calendar audit by convergent

The sample with a gated experiment attached, and the one worth publishing on its own.

Six historical intercalation rules, tested against Earth's convergent ladder.

assets/output/S2-calendar-audit.txt

```
rule                        value   convergent?    1 day slips in
----------------------------------------------------------------------------
Julian                        1/4      yes — #1           128 yr
Gregorian                  97/400            NO         3,226 yr
Revised Julian            218/900            NO        31,034 yr
Persian (Jalali)             8/33      yes — #3         4,269 yr
Medler                     31/128      yes — #4       400,000 yr
Coptic / Ethiopian            1/4      yes — #1           128 yr

FINDING
  derived (a convergent): Coptic / Ethiopian, Julian, Medler,
Persian
  declared (not one):     Gregorian, Revised Julian
```

INTERPRETATION

The split is real and it does not follow accuracy, which is what makes it interesting.

The Persian calendar of 1079, worked out by a commission including Omar Khayyam, uses 8/33 — the third convergent. It is **derived** in this system's exact sense: it is where the arithmetic says the good approximation is at that denominator.

The Gregorian reform of 1582 uses 97/400, which is not a convergent at any depth and is beaten by 8/33 with a denominator twelve times smaller.

The Revised Julian rule of 1923 is more accurate than the Gregorian — 31,034 years against 3,226 — and is also not a convergent. So accuracy and derivedness come apart in both directions, and neither implies the other.

GE-A2 — THE KILL CRITERION DOES NOT FIRE

The experiment asked whether this audit produces a non-trivial result. Its kill criterion: if every historical calendar turns out to be a convergent, the finding is empty and the sample becomes a footnote.

Four of six are convergents and two are not, and the two that are not include the calendar most of the world uses. The result stands, and the method is reusable on any body and any rule.

S3 — uncertainty audit

For any chronology, separate the **cited** from the **stipulated**.

The worked case is UC-1′s own datum, because that is the one this project is answerable for:

assets/output/S3-uncertainty-audit.txt

```
CITED      13.787 Gyr +/- 0.020 Gyr, Planck 2018 VI
           31 557 600 s per Julian year (definitional)
           SECOND, as the declared bridge constant
STIPULATED that tick 0 is the origin of the count
           that the datum is rounded to a whole beat
DISCARDED  0.017190364 s, reported rather than absorbed
```

Three categories, and the third is the one most systems do not have. A rounding that is **discarded and reported** is different from one that is absorbed, and the difference is auditable.

> **INTERPRETATION**
>
> The same audit applied to **Seder Olam** would separate the scriptural genealogies from the compression of the Persian period. That audit is computable, and chapter 19 is four pages on why this book computes it and does not conclude from it.
>
> Turning the audit on the project's own datum first is not modesty. It is the only order in which the sample is not an accusation.

S4 — cross-body simultaneity

One instant, three calendars — the demonstration that "now" is not a shared object.

assets/output/S4-simultaneity.txt

```
earth-d:      earth-d/1: 0027-213.7987 c328    derived (Rule K)
mars-d:       mars-d/1:  0082-086.1665         derived (Rule K)
earth-civil:  2026-07-31T19:11:12 TT           legacy — UCAL-W0005
```

Each rendering carries its kind. Each derived rendering carries its anchor revision, so values computed under different anchor determinations are never silently compared. The legacy rendering carries a warning, every time.

> **INTERPRETATION**
>
> The Martian date is not a translation of the Earth date. Both are renderings of one tick count, and neither is prior.
>
> That is what Rule K buys, and it is easier to see here than in any amount of prose about Earth being an instance rather than a template. There is no conversion between the two rows. There is a value, and there are three ways of writing it.

S5 — measuring diastema

The datum-to-present interval, stated with no Earth content in the units.

assets/output/S5-diastema.txt

```
ticks since the datum:
  8070205189128471254993117657693008777530466139316558837890625

on the tier ladder:
  T5 deep   31        T2 sweep  3000
  T4 drift  687       T1 arc    1638
  T3 span   2481      T0 beat   3018

in beats — the universe second:
  whole            9304313109135981143
  remainder_ticks  216453223532315359877956032752990722656250
```

No second, no day, no year appears. Every quantity is a count of ticks or of powers of five of them.

INTERPRETATION

Chapter 22 supplied the word for what this measures. Διάστημα — interval, extension, the spread between before and after — is Gregory of Nyssa's mark of createdness, and everything in the display above is an instance of it.

So the sample is not merely a demonstration that Earth units can be avoided. It is the concrete form of the chapter 22 finding: the instrument measures interval, interval is what created things have, and that is both what it can reach and why it reaches no further.

S6 — a revealed ratio, evaluated

Abraham 3:4 states a conversion with a profile tag on each side. The sample takes it seriously as a bridge constant and reports what it does and does not determine.

assets/output/S6-revealed-ratio.txt

```
WHAT IT IMPLIES
  1 Kolob revolution = 5853488070570534936000...000 ticks
                     = 674861227352 beats
                     = 0.007076 drifts

  It is NOT a whole number of beats. The remainder is
    35632189513851911760866641998291015625000 ticks
  which is 5^60-relative, not a rounding artifact.

WHAT IT LEAVES UNDETERMINED
  the phase — Rule J requires an anchor; the text supplies a
period
  the uncertainty — Rule C requires epoch, rate and window; none
given
  the determination method — 'revealed' is not a Determination
value

VERDICT
  ACCEPTED as a declared bridge constant (Rule Y)
  REFUSED  as a Rule J anchor
```

The non-exactness was not anticipated and is the sample's most interesting output.

INTERPRETATION

A thousand Julian years is a whole number of seconds. Seconds carry 5^{30}; the beat carries 5^{60}. So a quantity defined in years cannot land on a whole beat unless it happens to supply the missing thirty factors of five, and a thousand years does not.

That is chapter 7's incommensurability — the two seconds share a measure only at the tick — arriving in a place nobody was looking for it. Any ratio stated in Earth years, from any source, will miss the tier grid the same way.

It says nothing whatever about Abraham 3:4. It is a fact about the arithmetic of expressing year-based quantities on a base-5 ladder, and the sample surfaced it only because it computed the conversion instead of describing it.

INTERPRETATION

The verdict is a statement about the schema, not about the source, and the distinction is the whole value of the sample.

The system does not refuse the ratio because it is revealed. It accepts it — Rule Y requires a bridge to be declared and tagged on both sides, and Abraham 3:4 is tagged more explicitly than most engineering documentation manages.

It refuses it as an **anchor** because anchors carry a determination method and an uncertainty window, and the text supplies neither. The same refusal would fall on a ratio from any source that gave a period without a phase.

This is a template. Any tradition's stated ratio gets the same treatment, and the treatment is legible enough that someone who holds the text as scripture and someone who does not can agree on what the instrument did.

The pattern across the six

S1	chronologies	Rule P keeps four profiles apart
S2	calendar audit	derived and declared come apart from accuracy
S3	uncertainty audit	cited / stipulated / discarded
S4	simultaneity	three renderings, no conversion
S5	diastema	interval, with no Earth in the units
S6	revealed ratio	accepted as bridge, refused as anchor

Figure 8 — The six samples and what each returned. Every row is a file under `assets/output/`*, regenerated by one script.*

INTERPRETATION

Read together, the six do one thing: they take material the project was not built for — rabbinic chronology, a Persian reform, a scriptural ratio — and put it through a mechanism that has no opinion about any of it.

That is the sense in which a program can be run against material its author did not choose. The author chose the six inputs, obviously. What he did not choose was what came out, and S2 in particular came out differently from what a person who likes the Gregorian calendar would have guessed.

WHAT THIS CHAPTER ESTABLISHED

- An argument cannot be run against material its author did not choose, and cannot return an answer he did not want. A program can do both.
- S1: four chronologies on one axis, legitimate only because Rule P keeps their profiles apart.
- S2: four of six historical rules are convergents; the Gregorian and the Revised Julian are not. GE-A2′s kill criterion does not fire, and accuracy and derivedness come apart in both directions.
- S3: cited, stipulated, and **discarded** — the third category is what makes a provenance chain auditable, and the audit is turned on this project's own datum first.
- S4: three renderings of one value with no conversion between them — Rule K, visible.
- S5: the datum-to-present interval with no Earth content, which chapter 22 names as διάστημα.
- S6: a revealed ratio accepted as a bridge constant and refused as an anchor — a statement about the schema, not the source.

CHAPTER 28

28

Null results

Chapter 27 reported what the six samples produced. This chapter reports what they did not, because a demonstration reports only the first and this book is trying not to be one.

It runs eight pages against chapter 27′s ten. That is stated rather than corrected: the remedy for a short chapter is not padding, and a book that criticises the gap between **this ran** and **this established something** should not close a page gap by writing more words about it. The proportion is what it is.

Some of these are weaker versions of results already claimed. Some are things that came out and mean less than they appear to. One is a sample that essentially failed.

S1 — the axis is a picture, not a comparison

The chronology table looks like a comparison and is not one.

Four epochs sit on one axis. Rule P is what makes that legitimate, and Rule P is also what makes it **useless for the obvious purpose**: you cannot subtract Byzantine AM from Masoretic AM and get a meaningful number, because they are not offsets from a shared origin.

> **INTERPRETATION**
>
> What the sample actually produced is a rendering. Four values, each correct in its own profile, displayed together — and the display is exactly as informative as putting four rulers side by side with different zeros marked.
>
> The 1,748-year figure in chapter 27 is the difference between two tick counts, which is a real number about two declared epochs. It is not a disagreement about when anything happened, because the two traditions are not both dating the same event by different methods; they are reading different manuscript families.
>
> I let that figure carry more weight in chapter 27 than it should. It is the width of the picture, not a measured discrepancy.

S2 — what the audit cannot tell you

The strongest sample, and three limits on it.

It only tests solar intercalation. The audit compares leap-day rules against the tropical year. It says nothing about the lunisolar structure of the Hebrew or Islamic calendars, nothing about month lengths, nothing about week structure. A calendar is more than its leap rule, and this tests the leap rule.

"Not a convergent" is not a criticism. Chapter 10 made this point about the Gregorian and it needs making again about the Revised Julian, which is **more accurate** than the Gregorian and equally not a convergent. Derivedness and accuracy are independent, which means neither is a ranking.

Six rules is a small audit. The RFC calls S2 publishable independently. On six data points, with two negatives, that is a method demonstration rather than a survey. A real audit would cover the Islamic, Chinese, Indian, and Mayan systems, and would have to handle calendars whose intercalation is not expressible as a single fraction at all — which several are.

INTERPRETATION

The last is the serious one. The mechanism can only audit a rule that **is** a fraction. An observational calendar — one that intercalates when someone sights the new moon — has no fraction to test, and the audit has nothing to say about it.

That excludes a large share of the historical calendars a survey would want. So the method's reach is narrower than "calendar audit" suggests: it audits **arithmetic** calendars, and a calendar being arithmetic is itself a historical development that not every tradition made.

S3 — the sample that essentially failed

This is the weakest of the six and it should be said plainly.

S3 was specified as an uncertainty audit with **Seder Olam**'s missing years as the worked case. What it produced is an audit of UC-1′s own datum, and a paragraph saying that the **Seder Olam** audit is computable.

INTERPRETATION

Computable is not computed. The sample does not contain the analysis it was specified to contain.

Chapter 19 gives the reason, and the reason is good: performing the audit and publishing the result would put a mechanically-derived finding about a religious tradition's chronology into a book whose rules forbid its author from concluding anything from it. The restraint is deliberate.

But there is a difference between **declining to draw a conclusion** and **not doing the work**, and S3 does the second while chapter 19 argues for the first. Those are not the same position, and the book has been sliding between them.

The honest statement: the audit was not run. Not because it is impossible, and not only because the conclusion would be forbidden, but because running it would have required a level of engagement with rabbinic chronology that the author does not have and that a footnote cannot supply. Chapter 19′s principled restraint is real and it is also, in this instance, standing in for an absence of competence.

S4 — a rendering, not a synchronisation

The cross-body sample shows one instant in three calendars, and chapter 27 called it a demonstration that "now" is not a shared object.

That is right about the **rendering** and says nothing about the physics.

INTERPRETATION

Two observers on Earth and Mars do not share a "now" for reasons this sample does not touch: relativity of simultaneity, light-travel delay of four to twenty-four minutes, differing gravitational potentials.

S4′s point is narrower — that a tick count renders differently in different local calendars, and that the renderings carry their kinds and anchor revisions so they are not confused.

Chapter 16 listed relativistic environments as out of scope. This sample is where a reader is most likely to forget that, because "cross-body simultaneity" sounds like a claim about simultaneity and is a claim about formatting.

S5 — avoiding Earth units is not avoiding Earth

S5 displays the datum-to-present interval with no second, day or year in it, and chapter 22 gives that a name: διάστημα.

The null result is that the display is Earth-free and the **number** is not.

INTERPRETATION

The tick count reached that value through `ORIGIN_OFFSET`, which came from 13.787 Gyr — Julian years — times `SECOND`, the declared bridge constant. Chapter 2 conceded that the tick's length is fixed by convention against the SI second.

So S5 demonstrates that Earth units can be kept out of the **presentation**. It does not demonstrate that they are out of the value, and they are not.

This is the honest form of what chapter 7 called localisation: the dependency was moved to one declared place, not eliminated. S5 is the place where that distinction is easiest to lose, because a display with no Earth units in it looks like a quantity with no Earth in it.

S6 — the sample answered a question nobody asked

S6 accepted Abraham 3:4 as a bridge constant and refused it as an anchor, and produced one unanticipated fact: a Kolob revolution is not a whole number of beats.

INTERPRETATION

That fact is real and it is about arithmetic, not about the text. Any quantity stated in Julian years misses the tier grid, because years reduce to seconds and seconds carry 5^{30} against the beat's 5^{60}.

So the sample's most striking output is a restatement of chapter 7′s incommensurability in a context where it happens to look like a finding about scripture. It is not one, and a reader skimming could easily take it for one.

What the sample did **not** do is anything with the ratio. It converted a number and classified it against a schema. It did not test the ratio against anything, could not have, and the "template for any tradition's stated ratio" framing oversells what a unit conversion plus a schema check amounts to.

The gated experiments, reported

Four of the six §21 experiments in the software's specification hit their kill criteria, and chapter 9 said so. The book's own experiments deserve the same treatment.

id	question	outcome
GE-A1	Does the code diverge from the spec enough to carry Part III?	passed — Part III stands
GE-A2	Does S2 produce a non-trivial result?	passed — 4 of 6, split real
GE-A3	Does the deletion test pass?	passed — green throughout
GE-A4	Can both audiences state the thesis?	**not run**
GE-A5	Does Part VI argue rather than summarise?	passed at the book band
GE-A6	Can anchors be determined usefully?	Earth and Mars yes, Titan no

INTERPRETATION

GE-A4 has not been run and the book is nearly finished. Its criterion is that two readers — one engineer, one not — each state the thesis after a single pass, and its kill condition is that the book picks one audience and demotes Part VI to an appendix.

That experiment requires two people who are not the author. It is the only one of the six that cannot be run by the person writing, and it is therefore the only one that has been deferred at every stage.

Recording it as **not run** rather than quietly dropping it is the minimum. The next thing after the minimum is to make running it cheap, so the protocol is written out in `GE-A4-reader-test.md` — who reads what, the five questions in order, the scoring, and the consequence if it fails. Designing the test was the part the author could do.

The dual-audience claim in the preface is, as of this chapter, untested.

What the null results have in common

INTERPRETATION

Five of the six entries above are the same failure in different clothes: **the sample demonstrated a mechanism and the chapter described it as demonstrating a finding.**

S1 renders four epochs and was described as comparing them. S4 formats one instant three ways and was described as showing something about simultaneity. S5 suppresses Earth units in a display and was described as producing an Earth-free quantity. S6 converts a number and was described as a template for evaluating revealed ratios.

The pattern is not carelessness in any single case. It is that a working mechanism producing real output is genuinely impressive to its author, and the gap between **this ran** and **this established something** is exactly where enthusiasm lives.

Chapter 10 identified the same failure at the level of a specification claim: the error ran in the direction of the thesis, and survived because nobody checked the flattering half. Here it is at the level of six samples, caught only because a chapter was reserved in advance for catching it.

That is the argument for reserving the chapter. Not honesty as a virtue — honesty as a **scheduled step**, because the author who wrote chapter 27 was not in a position to see what chapter 28 says.

WHAT THIS CHAPTER ESTABLISHED

- S1 renders four epochs on one axis; Rule P makes the display legitimate and the subtraction meaningless. The 1,748-year figure is the width of a picture, not a discrepancy.
- S2 tests solar intercalation only, on six rules, and cannot audit an observational calendar at all — which excludes much of what a survey would want.
- S3 essentially failed: the **Seder Olam** audit was specified and not run, and chapter 19′s principled restraint is standing in for an absence of competence.
- S4 is about formatting, not simultaneity; the physics chapter 16 excluded is exactly what a reader will assume it addresses.
- S5 keeps Earth out of the presentation and not out of the value — the dependency was localised, not eliminated.
- S6′s most striking output is chapter 7′s incommensurability wearing scriptural clothes, and the sample tested nothing.
- GE-A4 has not been run. The dual-audience claim is untested, and it is the one experiment the author cannot run alone.
- Five of six failures are one failure: the sample demonstrated a mechanism and the chapter called it a finding.

PART VIII

The claim

CHAPTER 29

29

Why a program

The thesis, restated with everything the book has gathered behind it:

> *A measuring instrument may legitimately point at what it cannot describe, provided it declares that it is only pointing — and that declaration can be enforced mechanically rather than left to the author's discipline.*

The first clause is not new and this book has been careful to say whose it is. Cusanus has it in 1440: no proportion holds between finite and infinite, so precision does not constitute approach. Gregory of Nyssa has the reason a century before: διάστημα is the mark of createdness, and what lies beyond has no interval for a measure to measure. Frank reaches it independently in 1939.

The second clause is the contribution, and this chapter is what it amounts to.

The difference a compiler makes

An essay can assert that a distinction ought to be respected. It can argue at length, and persuade you, and then it ends — and the discipline it recommended survives exactly as long as the next reader's attention does.

Chapter 12′s `SignedWindow` does something an essay cannot. It refuses.

> **INTERPRETATION**
>
> Consider the three ways a rule can be maintained, in ascending order of how much they ask of the person maintaining them.
>
> A **comment** addresses a reader who is paying attention, at the moment they are reading that line, and only if they agree.
>
> A **runtime check** addresses a program that reaches a particular line of execution. It is stronger — it does not depend on agreement — and it is bounded by coverage: a path nobody thought about is a path the check does not guard.
>
> A **type** addresses everyone who ever writes code against the library, at compile time, on every path, including the ones nobody thought about. It does not require the programmer to have read anything. It does not require them to agree. It requires nothing of them at all.
>
> That is not a difference of degree. A comment is a request, a runtime check is a guard, and a type is a **precondition of the code existing**.

Why this is philosophical and not merely technical

The obvious objection: enforcing an invariant with a type is good engineering, and good engineering is not philosophy. Every well-typed program encodes constraints. Why should this one count as anything more?

Three answers, in increasing order of how much they claim.

First: the constraint is about what may be asserted, not about what is well-formed. Most type-level invariants encode structural facts — a list is non-empty, an index is in range. `UCAL-E0025` encodes an epistemological one: that a claim about the world with a published uncertainty may be **recorded** and may not be **computed with**. That is not a statement about data shape. It is a rule about the relationship between measurement and inference, and it happens to be expressible as a missing trait implementation.

Second: the distinction it enforces is one that has needed enforcing for a long time and has never had a mechanism. Kant maintained the constitutive/regulative boundary by argument and vigilance, and said explicitly that the illusion it guards against persists after diagnosis. Florensky, working in a tradition that had the necessary distinction available for fifteen centuries, crossed it in one paragraph of a book he was otherwise right in. This is not a boundary people fail to cross for want of being told about it.

Third — and this is the strong claim — a program can be run by someone who disagrees.

INTERPRETATION

This is where the argument stops being about engineering.

A philosophical argument reaches only those willing to follow it. Its whole mode of operation is voluntary. Someone who rejects the premise is untouched, and there is no version of the essay that reaches them, because reaching them was never what an essay does.

A program does not ask. Someone who thinks the distinction between a stipulated datum and a measured origin is pedantic — who thinks the uncertainty should just be added in — sits down, writes `t.checked_add(&claim)`, and the compiler says no. They have not been persuaded. They have been **stopped**, in under a second, by something with no interest in whether they agree.

I am aware of how this can be misread, so: this is not an argument that force beats reason, and a compiler is not an authority about what is true. If the rule is wrong, the compiler enforces a wrong rule perfectly.

The claim is narrower. It is that for a distinction whose failure mode is **drift** — gradual erosion by people who were never persuaded either way, one convenience at a time — a mechanism that does not depend on persuasion is a different kind of instrument, and philosophy has not had one.

What the medium costs

The argument would be cheap without this section.

A program's argument is only as good as its premises, and it hides them. Chapter 20 found the artifact assuming a Rushdian ontology — that periods have natures from which their behaviour follows — with no declaration anywhere. Chapter 25 found it assuming a clean line between structure and reading, which Losev denies. Chapter 26 found the book assuming the code is the invariant and the traditions the variables. Three metaphysical commitments, none argued, all discovered by reading the artifact against traditions rather than by inspecting it.

An essay wears its premises. A program buries them in type signatures where they function silently and are found only if someone comes looking.

A program's audience is smaller than an essay's. The `SignedWindow` argument is fully available to people who read Rust. To everyone else it is a claim about a piece of code, which is exactly the position an essay is in.

A program can be wrong with more authority. A false claim in an essay is contestable by anyone who reads it. A false claim compiled into a type is enforced against everyone downstream, and the enforcement carries no indication of whether the rule is good.

> INTERPRETATION
>
> So the honest form of the thesis has a clause the preface did not include: mechanical enforcement is a **stronger** way to maintain a distinction and a **worse** way to argue for one.
>
> This book exists because of that asymmetry. The artifact enforces; the book argues. If the enforcing could argue, there would be no need for 200 pages, and there would also be no way to find out that three of the artifact's commitments were never argued at all.

What the book claims, exactly

claim	status
An instrument may point at what it cannot describe	old — Cusanus, Nyssa, Frank. Not this project's
That declaration can be enforced mechanically	this project's contribution
A type addresses those who disagree, as argument cannot	the strong form, argued here
Mechanical enforcement is a better **argument**	**not** claimed — it is a worse one
The distinction enforced here is the right one	not established by the enforcement, and not by this book

The fourth and fifth rows are what keep the third honest. A compiler settles nothing about whether a rule deserves to be enforced. It settles that the rule **is** enforced, which was the whole difficulty.

WHAT THIS CHAPTER ESTABLISHED

- The thesis' first clause is Cusanus', Nyssa's and Frank's. The second — that the declaration can be enforced mechanically — is this project's.
- Comment, runtime check, type: a request, a guard, and a precondition of the code existing. The last is a difference in kind.
- It is philosophical because the constraint governs what may be **asserted**, because the distinction has never had a mechanism, and because a program can be run by someone who disagrees.
- The cost: a program hides its premises — three unargued commitments were found only by reading it against traditions — reaches a smaller audience, and can be wrong with more authority.
- Mechanical enforcement is a stronger way to **maintain** a distinction and a worse way to **argue for** one. That asymmetry is why the book exists alongside the artifact.

CHAPTER 30

30

Uselessness restated

The preface said the practical utility of this system is questionable, and said it as the second of three facts rather than as an apology. Two hundred pages later, it is worth restating — because the intervening chapters have supplied a better reason for it than the one the preface had.

The title stops being a paradox

An Instrument for the Immeasurable. Read as engineering, that is a contradiction with a rueful shrug in it: an instrument that cannot measure its subject is a failed instrument, and the title is admitting as much with good humour.

Chapter 22 makes it something else.

> Gregory of Nyssa, against Eunomius: διάστημα — interval, extension, the spread between before and after — is the mark of createdness. Everything created is διαστηματικός. God is ἀδιάστατος, without interval.
>
> The gap is not one of degree along a scale. It is a difference of **kind**: there is no interval in God for a measure to be a measure of.

Every quantity in this system is an interval. A tick count is the interval from the datum. A `Delta` is an interval. A `Window` is an interval with its uncertainty carried. The whole apparatus, from the 512-bit integer to the tier ladder to the certified cosmological enclosures, measures διάστημα and nothing else.

> INTERPRETATION
>
> So the title is not confessing a shortfall. It is stating a category.
>
> The instrument reaches exactly as far as interval reaches, which is exactly as far as the created order reaches, and the limit is not a failure of precision. Adding digits does not approach what has no interval, in the same way — Cusanus' way — that no finite magnitude approaches the infinite by growing.
>
> An instrument for the immeasurable is therefore not an instrument that failed to measure. It is an instrument that measures interval, pointed at something that is not an interval, and **declaring that this is what it is doing**.
>
> Which is the thesis, arrived at from the theology rather than from the type system.

Uselessness as thesis

With that in hand, the second fact can be restated properly.

The preface put it defensively: no task needs a Planck-tick count, the system does not compete with `chrono`, nothing on its ladder is near an hour. All true, and all framed as things the project is not.

> **INTERPRETATION**
>
> The better statement is positive. This is an instrument built to measure the one quantity that is coextensive with the created order, from the earliest point that order can be said to have, at a resolution no process can exceed.
>
> Nothing about that description mentions utility, and utility is not what it was aimed at. A system built to be useful would have started from a task and worked backwards to a unit. This started from a unit and never acquired a task, and the absence is not a stage it failed to reach.
>
> Chapter 17′s one qualification stands where it was left: for two or more bodies a universal ladder may be the only coherent arrangement. One page, a claim about coherence, not a recommendation. It is not load-bearing here and was never meant to be.

The rigor is the medium

The preface's third fact — that this is research of another kind, conducted in the medium of a working program — has been the least examined of the three. It is worth saying what "the rigor is the medium" means concretely, now that there is evidence.

The tests are the argument's form. That the constants reproduce along two independent integer routes is not a preliminary to chapter 12′s claim. It is what makes chapter 12′s claim about a real object rather than a described one. An essay about a hypothetical type system that would enforce a distinction is a different genre and a much weaker one.

The refusals are the content. `UCAL-E0020` before the datum, `UCAL-E0031` beyond the UCID range, `UCAL-E0043` below the bridge's resolution, `UCAL-E0062` for a missing anchor, `UCAL-W0003` outside a validity window. Chapter 16 counted four epistemic limits and found the same response in all four: error or warn, never default. That consistency is not a coding convention. It is the position the artifact holds, expressed in the only vocabulary an artifact has.

The corrections are the evidence. Chapter 10′s 97/400 finding, chapter 9′s fourteen deltas and one withdrawal, chapter 28′s six null results. A body of work that reported only its successes would be making the same claims with none of the standing.

> **INTERPRETATION**
>
> There is a version of this project that produced a paper arguing that formal systems should mark the boundary between what they compute and what they merely record. That paper would be shorter, easier to write, and read by more people.
>
> It would also have no `SignedWindow` in it, no compile-fail test, no fourteen deltas, and no case where the machinery contradicted its author. The paper would assert that the discipline is possible; the artifact demonstrates that it is, and pays for the demonstration in the only currency that counts — a specification that was wrong in fourteen places and said so.
>
> That is what "the rigor is the medium" means. Not that rigour makes the argument more respectable. That without the rigour there is no argument, only a proposal.

What was actually built

Six crates. 381 tests on two integer backends. A specification vendored, corrected in place, and cited by the source about a thousand times, with a build step that fails if a citation resolves to nothing. A tier grid generated from the library so it cannot drift. A lint that reports every exemption it honours. A book, marked throughout where it stops asserting fact, with a script that checks the marking.

None of it is useful.

INTERPRETATION

I want to leave that sentence sitting there, because the temptation at the end of a long book is to soften it, and softening it would undo the work.

The three facts of the preface are still the three facts. The artifact is real; its practical utility is questionable; therefore it is research of another kind. The third does not rescue the second. It **depends** on it — a useful instrument would have been answerable to its users, and this one is answerable only to whether it is right.

Chapter 31 says what is not claimed, and chapter 32 says what remains after every claim has been withdrawn. Neither of them is a rescue either.

WHAT THIS CHAPTER ESTABLISHED

- Διάστημα makes the title a category rather than a paradox: the instrument measures interval, interval is the mark of createdness, and what lies beyond has none.
- The limit is not imprecision. Adding digits does not approach what has no interval — Cusanus' point, with Nyssa's reason under it.
- The positive statement: an instrument for the one quantity coextensive with the created order, at a resolution no process exceeds. Utility was never the aim, and its absence is not a stage the project failed to reach.
- The rigor is the medium in three concrete senses: the tests make the claim about a real object, the refusals **are** the position, and the corrections are what gives it standing.
- The paper version would be shorter and read more widely, and would have no case in it where the machinery contradicted its author.
- None of it is useful. The third fact depends on the second rather than rescuing it.

CHAPTER 31

31

What is not claimed

A negative inventory. Every item is a sentence this book was in a position to write and did not, and each is listed with what would have made it writable.

It is here because a book that surveys nine traditions alongside a Rust workspace will be quoted, and the quoting will not always be careful. This chapter is what the author can point at.

About time and the datum

Not claimed: that time began. The system stipulates a datum and counts from it. Chapter 11 gave four reasons the origin cannot be measured, and Kant's is that the question presupposes something not available to be true. UC-1 asserts that tick zero is where counting starts. It asserts nothing about whether anything started.

Not claimed: that the Big Bang happened at tick 0. `BIG_BANG_CLAIM` records a published identification with its uncertainty, cited, as metadata. Chapter 12 made it impossible to compute with. A recorded claim is not an endorsed one, and the type is the difference.

Not claimed: that the tick is the smallest unit of time. Chapter 2 declined this explicitly. The tick is the resolution floor of an instrument. Whether the world has a smallest interval is a question this project takes no position on, and would have taken one on by accident had it not said so.

Not claimed: that the uncertainty in the age of the universe makes timestamps uncertain. It does not, because it is not an operand. This is the one negative claim the artifact enforces rather than the author asserting.

About the traditions

Not claimed: that any tradition surveyed is correct. No chapter of Part VI concludes that a direction got something right about the world. Convergences are reported as convergences.

Not claimed: that any tradition anticipated this work. Chapter 18 found Euclid X.2 **is** the intercalation algorithm; chapter 19 found **molad tohu** stating Rule Q's content nineteen centuries early; chapter 23 found D&C 130:4–5 stating Rule K's premise in 1843. None of those is anticipation. There is one good algorithm for continued fractions, one obvious solution to needing an exact origin, and body-relative reckoning is a thought available to anyone who considers another planet.

Not claimed: that the convergences cannot be coincidence. Some of them are structural — Euclid's algorithm and this one are the same algorithm, and that is not luck. Some are coincidence, and chapter 18 labelled Archimedes' 10^{63} as resonance and used it for nothing.

Not claimed: that any tradition is wrong. Chapter 19 is four pages on this. The instrument can compute a finding about **Seder Olam**'s chronology; the author does not conclude from it. Chapter 28 records that the audit was not merely withheld but not run, and that the restraint is partly standing in for an absence of competence.

Not claimed: that this survey is adequate to any of them. Chapter 26 says the method holds the code constant and lets the traditions vary, that this is a position rather than a neutral frame, and that the reverse book would ask better questions this one cannot reach.

About the design

Not claimed: that base 5 is meaningful. $5^5 = 3125$ is five base-5 digits. That is the whole reason, stated in chapter 4 before Part VI could be misread, and Rule N forbids any constant acquiring significance by resembling a number in a tradition.

Not claimed: that the system is useful. Chapter 30. The one qualification in chapter 17 is about coherence across bodies and is not a recommendation for adoption.

Not claimed: that anyone should adopt this calendar. Explicitly not, and the chapter that could have grown into an argument for it was held to one page for that reason.

Not claimed: that the approach works everywhere. Chapter 16 lists six limits at greater length than chapter 15 lists capabilities. Four are epistemic and in all four the system errors or warns rather than defaulting.

Not claimed: that the design has no unargued commitments. Three were found in Part VI alone — a Rushdian ontology in chapter 20, a clean structure/reading line in chapter 25, the comparative frame in chapter 26. All three were discovered by reading the artifact against traditions, and none is declared in the specification.

About the medium

Not claimed: that mechanical enforcement is a better argument. Chapter 29 says the opposite: it is a stronger way to maintain a distinction and a worse way to argue for one, and the asymmetry is why the book exists alongside the artifact.

Not claimed: that a compiler settles anything about truth. If the rule is wrong, the compiler enforces a wrong rule perfectly. That the distinction enforced here is the right one is not established by the enforcement, and is not established by this book either.

Not claimed: that the instrument reaches God. It measures διάστημα, which chapter 22 establishes is the mark of the created order by definition. That is not a limitation being confessed; it is what an interval-measure is.

The one thing that is claimed

Stated once, with the qualifications attached rather than deferred.

> *A measuring instrument may legitimately point at what it cannot describe, provided it declares that it is only pointing — and that declaration can be enforced mechanically rather than left to the author's discipline.*

The first clause belongs to Cusanus, Gregory of Nyssa, and Frank. The second is this project's, and it means one thing: `BIG_BANG_CLAIM` is fully readable, fully cited, carries an exact magnitude, and cannot enter a computation — enforced by a type with no operators and three tests that must fail to build.

That is all. It is a small claim about a narrow mechanism, and the reason it took two hundred pages is that establishing a small claim honestly requires reporting everything that did not establish it.

INTERPRETATION

A reader who takes only one sentence from this book should take this one: **the distinction between what a system computes and what it merely records can be made mechanical, and until it is mechanical it depends on someone remembering.**

Everything else here — the nine traditions, the six samples, the fourteen deltas, the convergent ladders — is either evidence for that or an honest account of what failed to be.

WHAT THIS CHAPTER ESTABLISHED

- Not claimed: that time began, that the Big Bang was at tick 0, that the tick is time's smallest unit, or that the age uncertainty affects timestamps.
- Not claimed: that any tradition is correct, anticipated this, or is wrong — nor that this survey is adequate to any of them.
- Not claimed: that base 5 is meaningful, that the system is useful, that anyone should adopt it, or that the approach works everywhere.
- Not claimed: that the design has no unargued commitments. Three were found in Part VI alone, none declared in the specification.
- Not claimed: that mechanical enforcement is a better argument, that a compiler settles truth, or that the instrument reaches God.
- Claimed: that the declaration can be enforced mechanically. One small claim about one narrow mechanism, and two hundred pages of reporting what did not establish it.

CHAPTER 32

32

Kant's moon

> Transcendental illusion is not a mistake. It is not carelessness, and it is not something a sufficiently careful thinker avoids.
>
> Kant's word is **natural and unavoidable**. It arises from the structure of reason itself, and — the part that matters here — it **persists after diagnosis**. His image is the moon at the horizon: the astronomer knows perfectly well that it is no larger there than overhead, and cannot prevent it from appearing larger. What diagnosis buys is not the disappearance of the appearance. It is only that one is no longer deceived by it.

The 61-digit integer

Here is the present epoch, at full precision:

ucal now — ticks

```
8070205189128471254993117657693008777530466139316558837890625
```

Every reader who sees that number will read it, at least for a moment, as a fact about being.

Not because they are careless. Because that is what a very precise number **looks like**. Sixty-one digits with no rounding anywhere carries an unmistakable impression of having been **found** — of corresponding to something, out there, that is exactly this many ticks old. The impression arrives before any reasoning does and does not wait for permission.

> **INTERPRETATION**
>
> It is false. Chapter 3 said why in one line: the count is exact and the correspondence between tick zero and any physical event is a separate question with a separate answer.
>
> The number is exactly right about how many ticks have elapsed since a stipulated origin. It says nothing whatever about how long the universe has existed, because the origin was declared rather than discovered, and every digit of precision is precision **about the count**, not about the world.
>
> A reader who has finished Part IV knows this. They will still see the number as a fact about being, for the same reason the astronomer still sees the moon.

What `UCAL-E0025` actually does

Chapter 12 presented the compile-fail tests as the book's central exhibit, and they are. It is worth being precise, at the end, about the scope of what they accomplish.

They stop the claim entering the arithmetic. `BIG_BANG_CLAIM` cannot be added to an `Instant`, cannot be converted to a `Delta`, cannot reach an operand position by any route the type system permits. Three programs that try are required to fail to build, and they do.

That is a real thing to have stopped. Chapter 29 argued why: it is a discipline that does not depend on the programmer having read anything, agreeing with anything, or remembering anything.

> **INTERPRETATION**
>
> And it is not the same as stopping the illusion.
>
> The type governs what may be **computed**. It has no reach at all over what may be **read**. A person looking at sixty-one digits and forming an impression about the age of the universe is not performing an operation the compiler can refuse. They are perceiving, and there is no type signature between a number and its reader.
>
> So the honest statement of what chapter 12 achieved is narrower than the chapter made it sound: `UCAL-E0025` **contains** the illusion. It does not cure it.
>
> It ensures that the false impression cannot propagate into a computed value — cannot become a timestamp that inherits an uncertainty it should not have, cannot become an answer someone downstream trusts. The impression itself survives intact, in every reader, permanently.

Florensky, one last time

Chapter 13 said the rule protects the arithmetic and does not protect the reading, and promised the last chapter would be about the residue. This is it.

Florensky's §9 is a case of exactly this failure. He had the formal artifact — the imaginary value of an expression outside its domain — and he read it as designating a place. Chapter 25 established that this was not carelessness: on Losev's account, which Florensky publicly defended, a formal structure **can** be constitutive of what it describes, and so reading a mathematical artifact as designating something real is the expected case rather than an error.

INTERPRETATION

Which means the rule this project holds — that formal artifacts are not physical facts — is not a safeguard against a mistake. It is a **position**, contested by serious people, adopted here without argument, as chapter 25 recorded.

And the position does not protect its holder from the illusion. It only forbids acting on it.

I would like to be able to write that this project is safe from Florensky's error because it has the rule he lacked. It is not. The rule keeps the error out of the computation. The author of this book looks at a 61-digit integer and feels what everyone feels, and has felt it repeatedly while writing, and the type system was no help at all in that.

What a specification can do

So the book ends on a boundary rather than a conclusion.

INTERPRETATION

A specification can say what a system computes with. It can enforce that boundary absolutely, at compile time, against everyone.

It cannot say what a number means to someone reading it. There is no mechanism for that, there has never been one, and the two-and-a-half centuries since the First Critique have not produced one — Kant did not expect them to, which is the whole point of calling the illusion natural and unavoidable rather than remediable.

`UCAL-E0025` is what a specification can do, done as thoroughly as the medium permits. It refuses to compute with a claim the artifact cannot help but suggest.

That is the only thing a specification can do about a transcendental illusion, and doing it is not nothing. The astronomer who knows the moon is not larger still cannot see it correctly — but he does not adjust his instruments for it, and his tables are right.

Tick zero is a stipulated reference point, conventionally identified with the FLRW $t \to 0$ limit. It is not a measurement and not an observed event.

— the datum statement, printed by `ucal datum` on every run

WHAT THIS CHAPTER ESTABLISHED

- Transcendental illusion is natural, unavoidable, and persists after diagnosis. The astronomer knows the moon is not larger at the horizon and sees it larger anyway.
- Every reader of a 61-digit integer will read it as a fact about being, because that is what a very precise number looks like. The impression arrives before reasoning and does not wait for permission.
- `UCAL-E0025` stops the claim entering the arithmetic. It has no reach over what may be read — there is no type signature between a number and its reader.
- So it **contains** the illusion rather than curing it: the false impression cannot propagate into a computed value, and survives intact in every reader.
- The rule against reading formal artifacts as physical facts is a contested position, not a safeguard — and it does not protect its holder from the illusion, only from acting on it.
- That is the only thing a specification can do about a transcendental illusion, and doing it is not nothing.

APPENDIX A

A

Sources

Rule B: every entry is cited at least once with a locus, and uncited entries are removed. The list below was built from the chapters, not from RFC UCAL-A1′s Appendix B — which is a superset, recording what the research consulted rather than what the book uses.

One entry was removed by that rule while this appendix was being assembled. Dershowitz and Reingold's **Calendrical Calculations** is the standard reference for this material and shaped the author's understanding of it; no chapter names it, so it is not here. `samples/check-refs.py` is what found that, and it runs on demand.

WHY LOCI AND NOT FOOTNOTES

The book carries no inline citation marks. That was a decision about voice — the register it inherits does not use them — and it leaves a gap that this appendix fills: for each work, the chapters that use it.

The gap is real and worth naming. A reader who wants to check a specific claim against a specific page has to find the passage from the chapter, which is more work than a footnote would have been.

Greek and Hellenistic

Aristotle, Physics chs. 2, 18, 24. Greek: ed. W. D. Ross, Oxford Classical Texts. Tr. in **The Complete Works of Aristotle**, ed. Jonathan Barnes, Princeton University Press, 1984. IV.10–14 on time as the number of motion and on the counter; VI on the continuum.

Archimedes, The Sand-Reckoner ch. 18. In T. L. Heath, **The Works of Archimedes**, Cambridge University Press, 1897; repr. Dover.

Euclid, Elements chs. 8, 13, 18. Tr. and comm. T. L. Heath, 2nd ed., Dover, 1956. X.2, anthyphairesis.

Plato, Timaeus ch. 18. Tr. Donald J. Zeyl, Hackett, 2000.

Plotinus, Enneads chs. 18, 24. Tr. A. H. Armstrong, Loeb Classical Library. VI.6 on substantial and monadic number.

Long and Sedley, The Hellenistic Philosophers chs. 2, 18. Cambridge University Press, 1987. Epicurean **minima temporis**; Stoic ἐκπύρωσις.

Jewish

Seder Olam Rabbah chs. 8, 19, 27, 28. Eng. tr. Heinrich W. Guggenheimer, **Seder Olam: The Rabbinic View of Biblical Chronology**, Jason Aronson, 1998.

Maimonides, The Guide of the Perplexed chs. 10, 16. Tr. Shlomo Pines, University of Chicago Press, 1963. I.73 on premises assembled to make a conclusion provable.

Islamic

Al-Ghazālī, The Incoherence of the Philosophers ch. 20. Tr. Michael E. Marmura, 2nd ed., Brigham Young University Press, 2000. Discussion XVII on causal connection and ʿ**āda**.

Ibn Rushd, Tahāfut al-Tahāfut ch. 20. Tr. Simon van den Bergh, Gibb Memorial Trust, 1954.

Patristic and Latin Christian

Augustine, The City of God chs. 3, 21. Ed. and tr. R. W. Dyson, Cambridge University Press, 1998. XI.6, the world made with time rather than in it.

Augustine, Confessions chs. 3, 21. Tr. Henry Chadwick, Oxford University Press, 1991. Book XI.

Basil of Caesarea, Hexaemeron chs. 13, 21. **Nicene and Post-Nicene Fathers**, ser. 2, vol. 8; Greek in **Sources Chrétiennes** 26 bis. I.6, the beginning of a road is not the road.

Boethius, The Consolation of Philosophy ch. 21. Tr. P. G. Walsh, Oxford University Press, 1999. Book V, on **aevum**.

Aquinas, Summa Theologiae ch. 21. Blackfriars edition. I.13.7, the relation real in the creature and not in God.

Nicholas of Cusa, De docta ignorantia chs. 21, 22, 29, 30. Tr. Jasper Hopkins, 2nd ed., Banning Press, 1985.

Orthodox

Gregory of Nyssa, Contra Eunomium chs. 22, 27, 28, 30. **Gregorii Nysseni Opera** I–II, ed. W. Jaeger, Brill. On διάστημα as the mark of createdness.

Maximus the Confessor, The Ambigua ch. 22. Ed. and tr. Nicholas Constas, Dumbarton Oaks Medieval Library, Harvard University Press, 2014. ἀϊδιότης / αἰών / χρόνος.

Gregory Palamas, The Triads ch. 22. Ed. John Meyendorff, tr. Nicholas Gendle, Paulist Press, 1983. Essence and energies.

Latter-day Saint

The Doctrine and Covenants ch. 23. Sections 77, 88:38, 130:4–5.

The Pearl of Great Price chs. 23, 27, 28. Moses 1:37; Abraham 3:4.

The Book of Mormon ch. 23. Alma 40:8.

Joseph Smith, King Follett Discourse, 1844 ch. 23. **The Joseph Smith Papers**, Documents series. Creation as organisation.

Modern philosophy

Kant, Critique of Pure Reason chs. 11, 12, 24, 32. Tr. Guyer and Wood, Cambridge University Press, 1998. The First Antinomy; the Schematism on number; **quanta continua**; transcendental illusion and the moon at the horizon.

Newton, The Principia ch. 24. Tr. Cohen and Whitman, University of California Press, 1999.

Leibniz and Clarke, The Leibniz–Clarke Correspondence ch. 24. Ed. H. G. Alexander, Manchester University Press, 1956.

McTaggart, "The Unreality of Time" ch. 24. **Mind** 17, no. 68 (1908): 457–474.

Russian

Бугаев, "Математика и научно-философское миросозерцание" ch. 25. Киев, 1898. Arithmology.

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Лосев, Философия имени chs. 4, 22, 25. Москва, 1927.

Лосев, Диалектические основы математики ch. 25. Academia, Москва, 2013.

Франк, Непостижимое chs. 25, 29. Париж, 1939. Eng. **The Unknowable**, tr. Boris Jakim, Ohio University Press, 1983.

Фёдоров, Философия общего дела ch. 25.

Вернадский, Размышления натуралиста ch. 25. Наука, Москва, 1975.

Чижевский, Физические факторы исторического процесса ch. 25. Калуга, 1924.

Method

Klein, Greek Mathematical Thought and the Origin of Algebra ch. 26. Tr. Eva Brann, MIT Press, 1968. **Arithmos** as a definite number of definite things.

Sorabji, Time, Creation and the Continuum ch. 26. Cornell University Press, 1983. The single most useful secondary source for chapters 18–21.

Technical and scientific

Planck Collaboration, “Planck 2018 results. VI. Cosmological parameters” preface, chs. 3, 11, 12, 27. **Astronomy & Astrophysics** 641 (2020): A6. The cited input to the datum, and the ΛCDM parameter set.

IERS Conventions (2010) chs. 8, 16, 27. IERS Technical Note 36. Earth’s anchor determination and the ΔT series.

Ussher, Annales Veteris Testamenti chs. 27, 28. 1650.

The project's own documents

RFC UCAL-1, and `spec/UCAL-1.1.md` throughout. Cited as historical evidence about intent, never as authority for behaviour — Rule S. Where the two disagree, the source tree is right and the RFC records what was intended.

RFC UCAL-A1 chs. 5, 15, 16, 28. This book’s own specification. Corrected in three places by Rule S: the domain fraction in chapter 5, the Mars convergent in chapter 15, and the Deimos bracket verdict in chapter 16.

The `ucal` source tree at the commit in `PINNED.md` throughout, and the sole authority for every claim about the software.

APPENDIX B

B

Glossary

Terms as this book uses them. Where a word is a technical term of the software, the chapter that introduces it is given; where it is a technical term of a tradition, the chapter that borrows it.

Two words carry a warning. **Legacy** and **declared** are classifications and not judgements, and chapters 8 and 22 both say so, because they read like judgements and will be taken for them.

The software

anchor The phase of a calendar — where its count begins, as opposed to how long its units are. Not derivable from any period, so declared, cited and interval-valued, with its absence an error rather than a default. Chs. 8, 16.

beat 5^{60} ticks, about 46.762 ms. The reference rung of the tier ladder and what §0.5 calls the **universe second**: a unit of human-noticeable size with no Earth content. Ch. 4.

`BIG_BANG_CLAIM` The published identification of tick 0 with the FLRW $t \to 0$ limit, with its ±0.020 Gyr uncertainty, recorded as metadata. A `SignedWindow`, and therefore not computable with. Ch. 12.

bridge The single declared constant through which a foreign unit enters and leaves. `SECOND`, an exact integer of ticks. Conversion in is multiplication and never rounds; conversion out is division and is the only place a rounding mode is chosen. Ch. 7.

convergent A truncation of a continued fraction — the best rational approximation at its denominator. Intercalation rules are derived by walking them. Chs. 8, 10, 15.

datum Tick 0. A **stipulated** reference point, conventionally identified with the FLRW $t \to 0$ limit; not a measurement and not an observed event. Chs. 3, 11.

declared Of a rule or constant: supplied rather than computed. The opposite of **derived**, and not its inferior — chapter 22′s Julian retention is the case where declaring is the better choice. Chs. 8, 22.

derived Of a rule: produced by the mechanism from cited parameters. The Julian leap rule is derived; the Gregorian is not. Chs. 8, 10.

díastēma See the traditions below. The term chapter 22 supplies for what every quantity in this system measures.

digit form The base-5 notation, five digits to a group, anchored at T32. Canonical for parsing and sorting. Ch. 6.

human form The decimal tier-group notation, anchored at T0. Canonical for statements about time. Ch. 6.

instant A point in absolute time — an unsigned integer count of ticks since the datum. Ch. 5.

interval extension The bounding of a function over a panel by evaluating its monotone parts at the panel's ends. Used on every panel of the cosmological quadrature, because the integrand is not monotone. Ch. 5.

legacy A calendar whose authority comes from declared tables rather than from derivation. A classification of **where the authority is**, not a verdict on the calendar. Chs. 8, 22.

profile A named, versioned, type-bound set of declared constants. Values from two profiles cannot be compared, and the text forms carry the tag so the refusal is visible. Chs. 6, 19, 21.

resonance A convergence that is genuine, striking, and proves nothing. Labelled as such, and never a premise. Ch. 18′s Archimedes figure is the only one in the book.

`SignedWindow` The type of `BIG_BANG_CLAIM`. Two fields, no arithmetic operators, no conversion to any computable type, and three compile-fail tests that enforce the absence. Ch. 12.

tick The Planck time, about 5.39×10^{-44} s. The atomic unit — the resolution floor of the instrument, and **not** a claim that time is discrete. Ch. 2.

tier A rung of the ladder, 5^{60+5k} ticks. Each is 3125 of the one below, which is exactly five base-5 digits. A tier's canonical identity is its exponent; its name is display only. Ch. 4.

UCID A 52-character Crockford base-32 identifier, defined below 2^{256}. Sorts chronologically, contains no randomness, and is not a UUID. Ch. 6.

validity window The range over which a body parameter is asserted. Outside it, `UCAL-W0003` rather than confident extrapolation. Chs. 8, 16, 20.

The traditions

ἀνθυφαίρεσις (anthyphairesis) Reciprocal subtraction, Euclid **Elements** X.2. The same procedure as continued-fraction expansion, and therefore the same procedure as the intercalation mechanism. Ch. 18.

ἀριθμός (arithmos) In Greek usage, a definite number of definite things — not a number in the modern abstract sense. Klein's distinction, without which chapter 18′s conflict is invisible. Chs. 18, 26.

αἰών (aion) In Maximus, the mode of created being that has a beginning without that beginning being a temporal position — a "before" that is not a "when". What UC-Θ would require. Chs. 12, 22.

διάστημα (diastema) In Gregory of Nyssa, interval or extension: the mark of createdness. God is ἀδιάστατος, without interval. Everything this system measures is διάστημα, which is why the instrument reaches exactly as far as it does. Chs. 22, 27, 30.

ἐκπύρωσις (ekpyrosis) The Stoic cyclical conflagration — a cosmology in which no datum is possible, because there is no first cycle. Ch. 18.

essence and energies Palamas' distinction between what is unknowable in God and what is genuinely known and participated. The strongest precedent the book's thesis receives. Ch. 22.

ḥelek 1/1080 of an hour, in the Hebrew calendar. Chosen for divisibility, as `SECOND` was chosen to be a multiple of 10^{30}. Ch. 19.

имяславие (imyaslavie) The dispute over whether the divine name participates in what it names. Condemned by the Synod in 1913; defended by Florensky and Losev. Rule N sides with the Synod. Chs. 22, 25.

molad tohu "The new moon of chaos" — the Hebrew calendar's computational epoch, placed about a year before the creation it anchors and named for its own emptiness. Rule Q's content, nineteen centuries early. Ch. 19.

regulative and constitutive Kant's distinction between a principle that directs inquiry and one that describes an object. Transcendental illusion is the slide from the first to the second. Chs. 12, 24.

transcendental illusion In Kant, an appearance that is natural, unavoidable, and persists after diagnosis. The astronomer knows the moon is not larger at the horizon and sees it larger anyway. Ch. 32.

ʿāda In al-Ghazālī, divine custom — the regularity that grounds practical certainty without carrying necessity. What a validity window encodes. Ch. 20.

This book's own apparatus

conflict The section every chapter of Part VI must contain, naming where the direction and the artifact disagree. Fixed in advance so none could be omitted. Ch. 18′s opening callout.

deletion test A-P8. Strip every **interpretation** and **resonance** block, compile, and confirm no surviving prose refers into a deleted one. Run by `deletion-test.py`. Ch. 0, and the book's README.

locus The chapter that uses a source. Given in Appendix A because the prose carries no inline citation marks. App. A.

null result What a sample failed to establish, reported as plainly as what it did. Ch. 28.

APPENDIX C

C

Diagnostic codes

Generated from `ucal-core`'s error module by `samples/gen-diagnostics.py`. The table is derived rather than transcribed, for the same reason §13.5 makes the tier table generated: a hand-copied list drifts silently, and a reference that disagrees with the software is worse than none.

What the codes are for

Chapter 16 counted four epistemic limits and found the same response in all four — the system errors or warns and never defaults. This appendix is that policy enumerated. Every entry below is a place where the artifact declines to produce a plausible number.

Exit codes

The command line maps each family to a process exit status, so a failure is distinguishable by class without parsing the message.

band	subject	exit
`E0001–E0007`	notation and parsing	2
`E0010–E0013`	profile and provenance	6
`E0020–E0025`	domain, ordering, and the claim	3, 9
`E0030–E0032`	identifiers and encoding	2, 3
`E0040–E0043`	the SI bridge and civil time	2, 4
`E0050–E0065`	bodies, anchors, and calendars	5, 7
`E0070–E0080`	numerics and cosmology	3, 8

Errors

code	meaning	exit	ch.
`UCAL-E0001`	malformed timestamp	2	—
`UCAL-E0002`	unknown profile tag	2	—
`UCAL-E0003`	mixed text forms in one string	2	—
`UCAL-E0004`	group value out of range (> 3124)	2	—
`UCAL-E0005`	invalid base-5 digit	2	—
`UCAL-E0006`	non-contiguous tier sequence	2	—
`UCAL-E0007`	calendar rendering without a kind/id qualifier	2	—
`UCAL-E0010`	locale table load failure	6	12
`UCAL-E0011`	duplicate name in the active locale table	6	12
`UCAL-E0012`	unknown key in HJSON data file	6	12
`UCAL-E0013`	profile lacks a datum_provenance record	6	12
`UCAL-E0020`	result precedes the datum	3	3
`UCAL-E0021`	result exceeds DOMAIN	3	5
`UCAL-E0022`	window inversion, lo > hi	3	12
`UCAL-E0023`	comparison indeterminate at stated precision	4	5

`UCAL-E0024`	lossy rendering requested without a rounding mode	4	5
`UCAL-E0025`	BIG_BANG_CLAIM used as a computational operand	9	12
`UCAL-E0030`	binary form is not 64 bytes	3	6
`UCAL-E0031`	instant outside UCID range	3	6
`UCAL-E0032`	invalid Crockford base-32	2	6
`UCAL-E0040`	civil date outside renderable range	2	7
`UCAL-E0041`	invalid civil date for the stated calendar	2	7
`UCAL-E0042`	second = 60 outside a leap-second instant	2	7
`UCAL-E0043`	foreign-unit input finer than the bridge constant permits	4	7
`UCAL-E0050`	profile mismatch	5	5
`UCAL-E0060`	body parameter missing required provenance or as-measured value	7	5
`UCAL-E0061`	leap-rule derivation cannot meet the requested drift bound	7	5
`UCAL-E0062`	calendar has no anchor; local fields cannot be produced	7	8
`UCAL-E0063`	anchor phase definition not evaluable for this body	7	8
`UCAL-E0064`	grouping cycle requested but none derivable from any satellite	7	8
`UCAL-E0065`	legacy calendar supplied where a derived calendar is required	7	8
`UCAL-E0070`	division by zero or by an interval containing zero	8	5
`UCAL-E0071`	requested enclosure width unreachable at the permitted depth	8	5
`UCAL-E0080`	tier index outside the profile grid	3	5

Warnings

A warning is returned alongside a value. It never replaces one, and it is never suppressed by default — chapter 8′s `UCAL-W0003` and chapter 5′s `UCAL-W0004` are both cases where the answer is real and incomplete.

code	meaning	ch.
`UCAL-W0001`	precision loss in the requested rendering	5
`UCAL-W0002`	leap-second table may be stale; bounded error reported	7
`UCAL-W0003`	body parameter evaluated outside its validity window	8
`UCAL-W0004`	cosmology enclosure width exceeds one tick	5
`UCAL-W0005`	value produced by a legacy (non-derived) calendar	8
`UCAL-W0006`	quantity comparable to or smaller than BIG_BANG_CLAIM	12

THE FOUR THIS BOOK TURNS ON

`UCAL-E0020` a result preceding the datum. Chapter 3 — a malformed question refused, not a value on the far side of an origin.

`UCAL-E0025` `BIG_BANG_CLAIM` used as an operand. Chapter 12, and the only code in the set that no program can reach, because the type has no operators for it to reach through.

`UCAL-E0062` a calendar with no anchor. Chapter 16 — an absence reported rather than defaulted.

`UCAL-W0003` a parameter outside its validity window. Chapter 20 found this to be ʿ**āda** compiled: reliable where observed, no claim beyond.

APPENDIX D

D

The rules

All twenty-four, with the chapters that use them. Subject lines are taken from `spec/RULES.md`, which is the normative statement; the glosses are this book's.

The fourteen the book actually cites are marked. The other ten govern parts of the software the book does not discuss, and are listed so that the set is complete rather than curated.

The twenty-four rules

rule	subject	what it requires	ch.
A	atomicity	Everything is counted in ticks. No other unit is fundamental — not the second, not the beat.	—
B	fixed 64-byte canonical binary	Big-endian, never minimal, so byte order is numeric order and the format never has to change.	6
C	body parameter provenance	Epoch, secular rate, validity window, and the as-measured value. Outside the window, a warning rather than extrapolation.	8
D	two text forms, one value	The human form and the digit form encode the same integer, each declaring its own anchor.	—
E	integrality	Not in a signature, a field, an intermediate, or the rendering path. A lint enforces it.	—
F	the frame is declared	The reference frame is stated rather than assumed. What the system does not model, it says it does not model.	16
G	the tier grid	Units are 5^(5k) ticks. A timestamp is the tick count written in base 5 and grouped in fives.	4
I	UCID range and non-uniqueness	52 characters below 2^256, sortable, containing no randomness. Not a UUID.	—
J	the anchor	Phase cannot be derived from period, so it is supplied per body — and its absence is an error, never a default.	8
K	one derivation mechanism	Every calendar is built by the same path from a body's own periods. There is no Earth branch.	8
L	leap seconds at the boundary only	TT is the only pivot. No arithmetic on absolute time ever meets a leap second.	—
M	monotone total order	Of any two instants, exactly one of earlier, same, later holds.	5
N	names are display-only	A tier's identity is its exponent. Nothing decides behaviour from a name.	4
O	overflow is a typed error	Arithmetic never wraps and never saturates, in release builds as well as debug.	—
P	profile binding	Two timestamps from different declared constants cannot be compared, and the text says which is which.	6
Q	the datum is stipulated	Tick 0 is declared, not measured or observed. The physical claim about it is recorded separately and cannot be computed with.	3

R	rounding only on rendering	Values round when displayed, never when constructed, and always under a mode the caller names.	—
S	sort order	Lexicographic order equals chronological order for the fixed-width forms — and not for text.	6
T	truncation is uncertainty	A value printed to a coarser tier **is** an interval, not a point padded with zeros.	—
U	interval arithmetic	Operations on intervals return intervals. A midpoint is a rendering choice, not a measurement.	—
W	one domain across backends	The fixed-width and arbitrary-precision integers enforce one identical domain.	—
X	certified enclosures	An interval that provably contains the answer, with quadrature error and parameter uncertainty reported separately.	5
Y	metrology	Earth units cross one declared boundary, and never appear in the arithmetic.	7
Z	zero and the unsigned domain	Nothing precedes the datum. A result that would be earlier is an error, not a negative number.	3

Non-goals and failure modes

The specification numbers what it refuses (`N`) and what it is built to prevent (`F`). The book cites four.

id	kind	what it says	ch.
`N1`	non-goal	The tick is not claimed to be a quantum of time. It is the resolution floor of an instrument.	2
`N12`	non-goal	No time before the datum. The value is not representable and the request is refused.	3
`F1`	failure mode	Timestamps shifting when the age constant is revised — what Rule P prevents.	11
`F9`	failure mode	Earth becoming the template rather than an instance — what Rule K prevents.	14

Specification corrections

Where verification found the specification wrong, the correction carries a `D-A` number. Chapter 9 is the account; `spec/SPEC-DELTAS.md` is the record.

delta	what changed	ch.
`D-A4`	Appendix C's human forms are truncated at T−5, not tick-exact	6
`D-A5`	grouping cycles are declared per body, not admitted by a global bound	9, 16
`D-A7`	full-width encode is 45 divmod steps, not 44	9
`D-A8`	what a printed form means, and how each form is anchored	6, 9
`D-A11`	obliquity is an angle and cannot be a rated parameter	9
`D-A12`	§9.6′s synodic formula computes the wrong quantity	9, 16
`D-A13`	a drift bound is a rate in local units, not a duration	9, 15
`D-A14`	§10.3′s integral is improper and cannot be quadratured as written	9

THREE THE BOOK FOUND IN ITSELF

Chapter 20 found the artifact assuming that periods have natures from which their behaviour follows. Chapter 25 found it assuming a clean line between a structure and a reading of it. Chapter 26 found the **book** assuming that the code is the invariant and the traditions the variables.

None of the three is a rule. All three are commitments the artifact and the book make without declaring, and there is no `N` or `F` number for them — which is itself a gap in the scheme.

AFTERWORD

About the author

Vladimir Ulogov.

Vladimir Ulogov has spent decades building infrastructure for distributed systems — the kind of software that watches other software. Early in his career he worked on monitoring and telemetry platforms; later years took him into federated observability, telemetry buses, and the architecture of systems that have to make sense of millions of data points without losing the thread.

Observability, in the end, is a discipline of **coherence** — of never reporting a state the system cannot account for, of insisting that every signal follow from something real. It is not an accident that a calendar built to point at an origin it cannot measure carries the same instinct.

It declares what it knows, declares what it has merely stipulated, and refuses to let the second quietly become the first.

What makes him slightly unusual in his corner of the industry is a tendency to write his own tools — not small utilities, but programming languages. The Bund language (its compiler, its VM, its document store, its parser) lives in a long series of Rust crates on crates.io. `rust_dynamic`, `rust_multistackvm`, `bundcore` — each is a building block that exists because the off-the-shelf options didn't fit the shape of the work. `ucal` grew the same way, from an irritation about units that no existing library was going to fix, because no existing library thought it was a problem.

A work of love

`ucal` is open source, under a permissive licence — you can read it, fork it, study it, modify it, and pass it on. It carries no analytics and no telemetry. It was not built to be sold and it was not built to be adopted; it was built because the question would not go away, and because the only way to find out whether a distinction can be enforced by a compiler is to try to enforce one.

A note on cooperation

Vladimir believes firmly in the human capacity for mutual help — that we make better work, and live better lives, when we share what we know and what we build. Open source is one of the most concrete expressions of cooperation our era has produced: code read, improved, and passed forward without payment, without permission, by people who will never meet.

This book is offered in the same spirit. If it helps you see one distinction more clearly — between what an instrument measures and what it merely points at — that is enough.

Where to find more

GitHub `@vulogov` — the source for `ucal`, Bund, and the dozen-plus Rust crates that carry the infrastructure. Issues and pull requests welcome.

LinkedIn `/in/vladimirulogov` — posts on observability, the occasional long-form essay.

YouTube `@vulogov` — talks and walkthroughs from the conference trail.

If you find a claim in this book that the source tree does not support, open an issue. That is the only kind of correction the book is built to receive, and it is the kind it most wants.

END OF THE BOOK